GE Healthcare

LightSpeed 7.X

Installation Manual

(Book 1 of 2)

LightSpeed VCT XT
LightSpeed VCT
LightSpeed VCT Select
LightSpeed Pro32
LightSpeed VCT Limited Access

OPERATING DOCUMENTATION

5116411-100
Rev. 15

Book 1 of 2: Mechanical Installation

Pages 1 - 162

Effectivity

The information in this manual applies to the following LightSpeed 7.X CT Systems:

- LightSpeed VCT (64-Slice)
- LightSpeed VCT Select (32-Slice)
- LightSpeed Pro32 (32-Slice)
- LightSpeed VCT Limited Access

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING
(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

Предупреждение
(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告
(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA
(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL
(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING
(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

HOIATUS
(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS
(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION
(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG
(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch stromschläge, Mechanische oder Sonstige gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εαν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εαν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS
(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN
(IS)

- Þessi þjónustuhandbók er eingöngu fáanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告
(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고
(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS
(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS
(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL
(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE
(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO
(PT-BR)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado ecompreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros.

ATENÇÃO
(PT-PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado ecompreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE
(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării şi înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!
(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE
(SR)

- Ovo servisno uputstvo je dostupno samo na engleskom jeziku.
- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilacke usluge.
- Ne pokušavajte da opravite uredaj ako niste procitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povredivanja servisera, rukovaoca ili pacijenta usled strujnog udara ili mehanickih i drugih opasnosti.

UPOZORNENIE
(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednos″ou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústi″ do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION
(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING
(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT
(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dişinda bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyariya uyulmamasi, elektrik, mekanik veya diğer tehlikelerden dolayi teknisyen, operatör veya hastanin yaralanmasina yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "Damage in Shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://egems.med.ge.com/edq/home.jsp
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment, if not properly used, may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, GE Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION Risk of Explosion **Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.**

ATTENTION Danger d'Explosion **Il y a danger d'explosion s'il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.**

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GE Healthcare PQR Process to report all omissions, errors, and defects in this publication.

Revision History

Rev	Date	Reason for change
15	10/24/08	**Important Precautions:** added new languages **PQR 13195860:** Chapter 6, Section 6–Corrected the tomographic plane specification to ±1.0 mm.
14	3/28/08	**Customer Configuration Preference for Language requirements**: Updated Chapter 5, Section 2.9.2, Step 7c Procedure, and Section 3.5 Warning Labels. Also updated GE form e4879 to record information. **SPRGCTge34610:** Updated Chapter 2, Section 5, Install Options (added Service Cabinet and UPS)
13	8/17/07	**Warnings:** Updated with new languages. **General Updates:** Corrected spelling, grammar, and procedures. **Removed the following Sections (incorporating material into proper location in the manual):** Installer Notices, Appendix F (*Elevation Characterization*), Appendix G (*Additional Characterization Procedures*) **Chapter 1, Section 10:** Updated the Seismic Mounting information **PQR 13096524** **Removed Appendix D** *Bearing Gap Inspection*, updated and incorporated information into Chapter 1, *Position Subsystems*. Re-ordered the numbering of the Appendices.
12	5/22/07	**SPR FCTge29405** **Chapter 7, Section 11.0:** Replaced figure 7-18 with correct picture of VCT console rear bulkhead. **SPR FCTge28968** **Chapter 5, Section 2.4:** Revised System Options **Chapter 4:** Added Section 6.0: FE Workflow, Section 7.0: Checklists for Completed Installation, and Section 8.0: GE and Regulatory Forms. **Chapter 9:** Deleted this chapter. The information has been replaced by Chapter 4, Sections 7.0 and 8.0, and the GE e4879 form.
11	4/27/07	**SPR FCTge28968** **Chapter 5, Section 2.4**: Updated System Options
10	4/23/07	**Chapter 8:** Added the CT System Chassis Leakage Test. **Chapter 9:** Added a notice with instructions for sending completed forms that are available electronically. **PQR 13093409 (et al.)** **Chapter 5:** Updated the Set Time and Date section, changing the information to resolve the Time Jump issue. Replaced screen shots with updated ones. **PQR 13094381** **Chapter 8**:Added the part numbers for the tools needed to perform Patient Touch Leakage Test.

Rev	Date	Reason for change
9	11/30/06	**Title Pages**: Added product names **Chapter 1**:Updated Section 4.0 - Install and Level Table/Gantry **Chapter 3**:Updated Section 4.0 - Mechanical Installation Completion Checklist (Removed the partial list and referred user to Chapter 6 for a printed version or the CD or web site for the electronic version.) **Chapter 7**:Updated Section 1.0 - CT Options Verification List **Chapter 6**: • Added Section 3.0 - Mechanical Installation Completion Checklist • Updated Section 4.0 - System Installation Electrical Calibration Checklist • Updated Section 5.0 - CT Options Installation Checklist • Updated Section 7.0 - GE Form 4879 Installation Data Verification - 7.x Systems • Added Section 8.0 - GE Form 4879M Data Verification - Mobile Van • Added information for North Carolina System Level Ground Test form
8	9/12/06	**PQR13066716** **Appendix A**: Added Section 9.0 - Gantry Auxiliary (Mini) Dolly Installation With Wood Block **Appendix E**: New appendix (Regulatory Clearance Quick Reference Guide) **Chapter 6:** Updated Section 7.0 - Image Series **Chapter 6**: • Updated Section 4.0 - System Installation Electrical Calibration Checklist • Updated Section 7.0 - GE Form 4879 Installation Data Verification - 7.x Systems
7	4/24/06	**Chapter 6:** Replaced “Image Series” with instructions to perform detailed cals.
6	2/16/06	**Chapter 6:** Removed “Table/Gantry Alignment Procedure”

Rev	Date	Reason for change
5	1/24/06	**PQR 13055171**/**FCTge15465**: Corrected PDU power cable terminal identifiers. **General Updates**: Corrected various spelling & grammar errors. **Chapter 1:** • Updated Required Tools List • Added Section 3.2 - Stored Systems • Updated Section 3.11 - Installation Conditions • Updated Section 4.2 - Position the Gantry • Updated Section 4.6 - Table Prep and Set-up • Updated Section 4.10 - Level the Table • Updated Section 4.14.1 - Notes to Mechanical Installers • Updated Section 4.17 - Alignment Recheck **Chapter 2**: • Updated Section 1.0 - Introduction • Added Section 5.0 - Install Options • Updated Section 6.0 - Gantry Cable Connections **Chapter 3**: • Added Section 3.0 - Axial Head Holder Shim Installation • Updated *Mechanical Installation Completion Checklist* **Appendix A:** Added Section 8.0 - CT UMI Tilting Dolly **Appendix D:** Updated Gantry Bearing Gap Inspection procedure **Chapter 4:** • Updated Section 4.0 - VCT Calibration Training Requirements • Updated Section 5.0 - Required FE Common Tools and Supplies **Chapter 5:** • Updated “Modified In Room Start” setting instructions, for Japan sites. • Removed “Adjust Monitor” section • Updated Section 3.7 - Install Service Cabinet **Chapter 6:** • Updated Section 3.0 - Table/Gantry Alignment Procedure • Added Section 3.3 - Gantry Tilt Verification • Updated *System Installation Electrical Calibration Checklist* **Chapter 8:** Updated System-Level Safety Tests
4	9/28/05	**PSR 13050792**: Added Section 3.4 to Chapter 5. **Chapter 1**: • Updated Section 2.5 - Required Common Tools and Supplies • Added Section 3.11 - Installation Conditions **Chapter 2**: • Added Important Cable Notes (Notices) to Section 1.2 • Added Sections 5.3 and 5.5 **Chapter 6**: Updated Section 7.0 (Form 4879)
3	8/17/05	**Chapter 6**: Updates for Pro32.
2	5/19/05	All sections changed for Release 2 for M3.
1	03/08/05	Initial Release, for NPI (M3).

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Table of Contents: Book 1

Preface
Publication Conventions

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0
Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided below. They include paragraph prefixes and modified text styles.

CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**

DANGER
EXCESSIVE VOLTAGE
CRUSH POINT

DANGER IS USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**

WARNING
ROTATING EQUIPMENT
BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

NOTICE Equipment Damage Possible

Notice is used when a hazard is present that can cause property damage but has absolutely <u>no personal injury risk</u>.They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

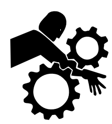
It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

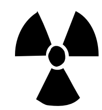
Important information will always be preceded by the exclamation point ⚠ contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL

MECHANICAL

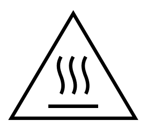


RADIATION

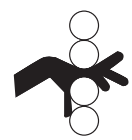


LASER

HEAT

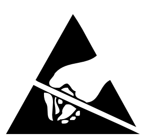


PINCH

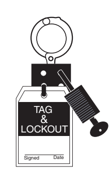


Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION

Section 2.0
Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

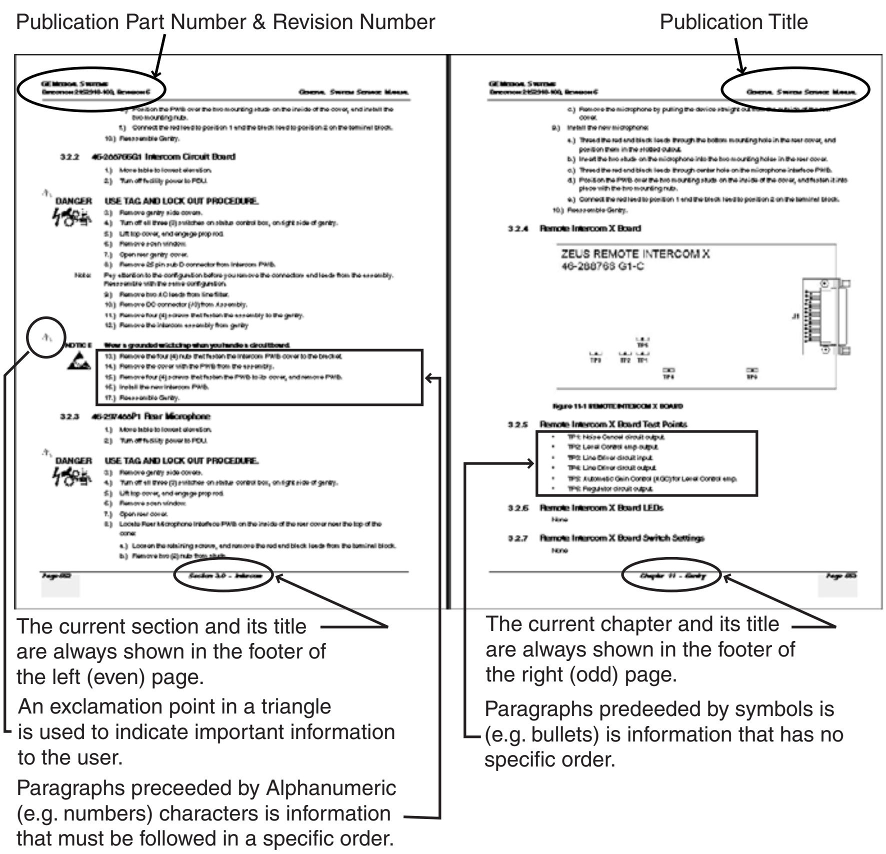
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, such as used at the beginning this chapter for example.

Note: **Conveys information that should be considered important to the reader.**

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout



Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directions. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output
`This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application.It's the most commonly used style used to indicate filenames, paths, and text.`

Example: Variable Output
This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>

Example: Fixed Input
This paragraph denotes fixed input. It's typed input that will not vary from application to application. Fixed text the user is required to supply as input.

Example: Variable Input
This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.

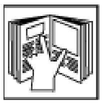
2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys
A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys
Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

Chapter 1
Position Subsystems



NOTICE **Potential for Data Loss and/or Equipment Damage.**

To prevent potential data loss and equipment damage, please do the following:

- **Record data collected from procedures in this chapter into Form 4879 when directed, located in Chapter 6 of this book.**
- **Only use the installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.**

Section 1.0
Installer/FE Notices

1.1 Shipping, Warehouse and Transportation Warning

- This gantry is designed to be moved using the shipping dollies and should not be lifted or moved using a lift truck.
- Do Not Hoist Gantry or Table using Dollies.

1.2 International Shipments

- Dollies must be used to remove the gantry from the shipping skid and to transport the gantry to the customer's site.
- If lifting is required, refer to the Pre-Installation Manual for instructions.

1.3 On Site Warning

This system requires a gantry bearing gap inspection *before* electrical calibration is started. See Gantry Bearing Gap Inspection, on page 43.

1.4 Service Actions

Open a dispatch and record the bearing inspection results first, then close the dispatch and continue with the electrical calibration procedures.

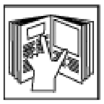
Section 2.0 Introduction

This chapter describes how to mount, position, and level the CT Scanner subsystems.

Note: Before you start the installation, make sure the site preparation complies with conditions and instructions found in the pre-installation manual. Failure to comply will result in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet specific mechanical installation standards that should be reviewed prior to installing this system.

2.1 Skill Set Required

NOTICE **All mechanical installers must meet these minimum requirements.**



MECHANICAL

7.X Installation CBT	**Safety CBTs (6)**	**OJT with Trainer**
Must be trained as FE or Contractor. Must complete all training CBT courses. My Learning verification.	Complete all training CBT courses. My Learning verification.	Complete one VCT training installation with trainer.
Tools	**Access**	**Communication**
Must have all required tools as listed in this manual.	Must have access to some GE Healthcare web sites and tools.	Must be able to complete all required paperwork and complete iTracs as required.

ELECTRICAL

7.X In-Residence Training	**Safety CBT**	**OJT with Trainer**
Must be trained as FE or Contractor Must complete a three-week course in Milwaukee. My Learning verification.	Complete all training CBT courses. My Learning verification.	Complete one VCT training installation with trainer.

Only people trained and experienced in the installation process of a GE CT scanner should perform the work described in this document.

A person doing the installation should have:

- Installed three (3) Lightspeed 5.X series products as a helper prior to leading an installation.
- Completed all of the above training courses and successfully passed all related tests.
- A working understanding of the mechanical installation process.
- Completed Environmental Health and Safety (EHS) training, completed Hazcom training, completed Eye & Personal Protection Safety, completed Blood Borne Pathogens training, and must understand and apply lockout/ tagout processes
- All training and records must be verifiable and available in My Learning.
- Professional communication skills needed to communicate with customer any changes or additional requirements needed to complete the installation project.
- Must be trained to complete site-ready inspections.
- (Installation leader) Must have completed project management, leadership training, and all required training prior to becoming a project leader.
- (Mechanical installation teams) Must have basic measurement skills, system layout skills, and the ability to read and understand blueprints.

Table 1-1 Installation and Training Media

MEDIA	PART NUMBER
LightSpeed 7.X VCT & Pro32 Pre-Installation DVD	5133791-300
LightSpeed 7.X VCT & Pro32 Mechanical Installation DVD	5133792-300
LightSpeed 7.X VCT & Pro32 Safety DVD	5133793-300

2.2 Floor and Room Preparation

2.2.1 Preparation

Consult your local GE Sales and Service representative about your specific needs. It is the purchaser's (buyer) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The LightSpeed 7.X system has a total floor load of approximately 6660 lbs (3021 kg). About 5050 lbs (2291 kg), including patient (500 lbs (227 kg)) is concentrated in the table-gantry assembly. Refer to the *Pre-installation Manual* for more information.

2.2.2 Flooring

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where resilient flooring material should be removed.

Limitations include:

- No part of the floor surface within the table, gantry, or the two interface areas between table and gantry should be higher than the support areas for the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors or through bolts must not reduce the structural strength of the floor.

If you have to remove the gantry covers in order to move the gantry into the room, refer to Appendix A for the cover removal procedure. Please read the caution statement on page 125 before removing the gantry covers.

2.3 Overview

Procedures in this chapter provide detailed instructions to position, level, and anchor the gantry and table securely for operation. The LightSpeed 7.X system uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads located on the gantry base. The table has four (4) pads used for leveling it.

The process you will be following is:

1.) Use the room-layout template to determine the general position of the gantry and table.
2.) Move the gantry into position.
3.) Level gantry.
4.) Use the laser tool to position the table relative to the gantry.
5.) Level the table to the gantry, and anchor the system.

Note: Use the template to position the system; however, use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place. Refer to Section 4.0, on page 35 for an example of this procedure. This CT system installation procedure requires the items listed in Section 2.5, on page 30.

2.4 Pre-Installation Template

Always use the room-layout template (in two pieces), during installation (see Table 1-2, for part number). The gantry and table will not be properly aligned if existing holes are used. The template shows the location of the gantry and table anchor holes.

This template is shipped with the system. It may also be ordered via the Web, from Coakley-Tech.

Table 1-2 LightSpeed 7.X Room Layout Templates

Table	Part Number
GT1700	5111499
GT2000	5111499

2.5 Required Common Tools and Supplies

The following tools and supplies are required for installation of the scanner.

WRENCHES

- Standard and metric combination wrench sets
- Standard and metric hex key (Allen wrench) sets
- ½" and 3/8” drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

SOCKETS AND EXTENSIONS

- 3/8" drive metric and standard socket set
- 1", 1-1/8”, 1-1/4", 1-1/2", 1-5/8" sockets
- 1/2" drive ratchet wrenches
- 3/4” deep well socket
- Metric hex bit set 1/4" or 3/8” drive, including 10mm and 14mm hex bit
- 3/8” drive universal joint
- 21mm socket (optional)

SCREW DRIVERS

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)
- Pozi-drive 0 (S)
- Pozi-drive #1

DRILL BITS

- Complete set of standard (U.S.) drill bits
- Metric tap set (Optional)
- ½" masonry bit, min. 10" long USA; 12" optional (Bit must not be metric.)
- 4" (100mm) hole saw with 1/8" (5mm) masonry bit (to remove flooring)

POWER TOOLS

- 3/8” or ½" drill, cordless or electric
- Reciprocating saw (Sawzall or equivalent) and assorted blades
- Hammer Drill & BIt (8" min, 12" max)
- Sears 17740 Shop Vacuum or equivalent, with “HEPA” or dry wall dust filter (Sears part number 17918 or equivalent)
- 25’ extension power cords

ELECTRICAL TOOLS

- DVM
- Continuity Tester
- FE Electronic Monitor Static Kit
- Dale 600 Leakage Meter
- Temperature/humidity tool: Oregon Scientific Wireless Weather Station Model BAR608HGA

HAND TOOLS

- Ball-peen hammer (1lb or 2lb)
- Tongue & groove pliers (large)
- Diagonal cutting pliers, large (to cut 1/0 ground)
- Diagonal cutting pliers, small and large
- Large pry bar

Table 1-3 Recommended Levels

Johnson Magnetic Level, model 7500M -- 9"
Johnson Professional Box Beam Level, model 9624 -- 24"
Johnson Professional Box Beam Level -- 48" (optional)

PERSONAL SAFETY EQUIPMENT

- Safety shoes*
- Safety glasses*
- Leather gloves
- Knee pads
- LOTO kit* -- MUST have ***approved GE*** LOTO kit
- Hearing protection*
- 6' or 8' step ladders

* Required items

SYSTEM CLEANERS

Purchase Locally:

- Windex Glass Cleaner or equivalent
- Scrubbing Bubbles Bathroom Cleaner or equivalent
- 409 Grease Cutting Cleaner or equivalent
- WD-40

OTHER

Purchase locally: China marker, any color that will be visible on the floor where the system will be installed

GE TOOLS

VCT Alignment Kit (p/n 5148193)
NOTE: This tool may not be available via the tool pool, in some areas.

Section 3.0 Delivery Procedure

3.1 System Transportation - Temperature Extremes

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: 0º to +120º F (-18º to +49º C)

Humidity: 20% to 80%

NOTICE **Component Freezing occurs if CT system is exposed to temperatures below 0º F (-18º C) for a period longer than two days.**

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature prior to installation.

3.2 Stored Systems

If your system have been stored for more than three months, you will need to complete a visual inspection, looking for damage due to improper storage. Check for the latest software revisions, options, and component changes. Contact the OLC or CT support central for additional information.

3.3 Working with the Mover

Follow the instructions provided by your installation specialist regarding working with equipment movers. Help direct movers as to where to place equipment and which items are needed first.

Generally movers should move all equipment into the customer's room. Door removal and other site changes to move equipment should be done only as directed by the install specialist.

For component sizes and weights, refer to the *LightSpeed 7.X Pre-Installation* manual.

Note: Do not place equipment in its final location at this time. Templates must be laid first.

Note: If you have to remove the gantry covers in order to move the gantry into the room, refer to Appendix A, for the cover removal procedure. Please read the caution statement on page 125 before removing the gantry covers.

3.4 Inventory Process

The VCT ships in multiple boxes on pallets and dollies, approximately 26 units.

3.5 Floor Protection

It is suggested that the movers use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

3.6 Equipment Delivery Route

Prior to equipment delivery, review the delivery route with the movers. Refer to the installation specialist for any additional delivery instructions.

3.7 Removing Gantry Dollies and Covers

Refer to Appendix A for the dolly and cover removal procedures. Please read the caution statement on page 125 before removing the gantry covers.

3.8 Damage In Transportation

Check for damage to property that may have occurred at the site during delivery, such as damage to floors, door frames or walls. If damage is found, notify the install specialist.

All packages should be closely examined at time of delivery. If damage is apparent, have notation “damage in shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

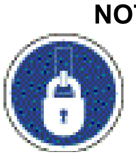
To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm.
- Contact the local service coordinator for more information on this process.

Rev. Jan. 5, 2005

3.9 A1 Breaker

NOTICE **The site MUST have a main disconnect with Lockout/Tagout capability.**



Lock-out and tag-out the A1 breaker now.

A1 BREAKER	UPS
USA 480VAC: **E4502AE**	*VCT 3-Phase Model:* **B7864PP**
Europe and Asia 400VAC: **E4502AF**	*VCT 3-Phase Model* **B7864PP**

Figure 1-1 Sample A1 Breaker

3.10 Installation Support Kits

VCT-5133492 AND PRO32

A new VCT Installation Support Kit is shipped with every system. Locate this box now and open it. All included materials are to be used during the installation process. These items are to be left ON site.

Locate the install packet from the PMI Site Ready Visit. Among other items, it contains:

- Room Print
- Floor Template
- IP Addresses
- Contact names
- Copy of the Site Ready Document

3.11 Installation Conditions

1.) A Final Site Print is REQUIRED.

2.) The room size must match the print.

Measure the room size. If it does not match the stated size, and is smaller, then check all regulatory clearances. If any regulatory clearance is less than the minimum, then **DO NOT** continue. Notify the PMI to set up a site escalation.

3.) **Do Not** start the installation process if the site is under construction:

- In the Room
- In the Scan Area

Refer to the *Pre-Installation Manual* for additional details.

4.) A customer Anchoring Plan is required, if there is anything other than a 4" (minimum) concrete floor. GEH employees shall only install the anchors supplied with this system.

Section 4.0
Install and Level Table/Gantry

4.1 Lay the Floor Template

4.1.1 Tools Required

- Standard Install Tool Kit
- Install Support Kit
- GE Site Print
- VCT Installation Manual
- Floor Template for your system
- 4’ Level
- Chalk Line
- China Marker

4.1.2 Time & Personnel

- 1.0 hour labor on site
- 2 Engineers

4.1.3 Safety

CAUTION **Potential for Injury.**

LightSpeed VCT Gantry Presents a Variety of Mechanical and Electrical Hazards.

Use Appropriate Safety Procedures When Working on the VCT System.

4.1.4 Establish Room Layout

OVERVIEW

Use the GE print developed for your site to establish the room layout. Make sure all the operating and service clearances shown on the print are observed.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

Measure and determine ISO using the GE Site print. Using a china marker, mark ISO on the floor. Use a chalk line to connect the table center line marks on the floor. This is the line on the print that runs down the center of the table through the gantry. This will be used as a reference when positioning the table.

PROCEDURE

1.) Lay out the two (2) pieces of the floor template (GT table: 5111499). Start with the table template, then place the gantry template over the top of the table template. Align per the GE print.

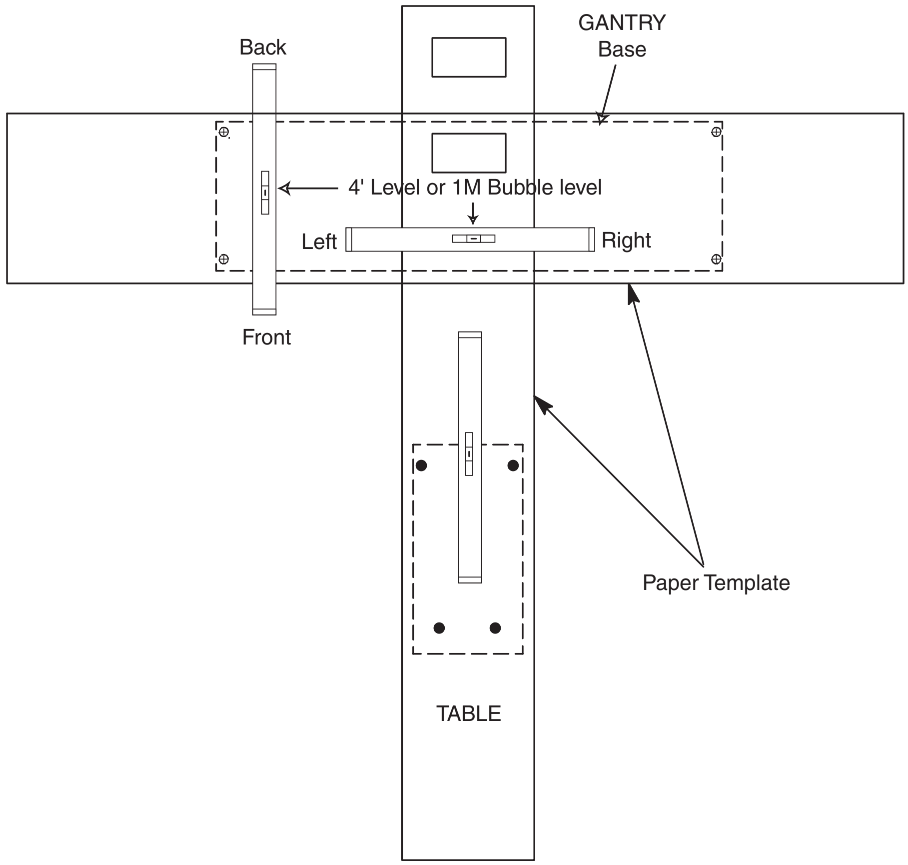
2.) Tape the templates together, making sure that the table and gantry center lines are matched. Then tape the template to the floor.

3.) Recheck the position of the gantry in the room per the GE print. If all is per the GE print, continue. If not, realign per the print.

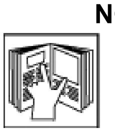
4.) Make sure there are no potential clearance issues. If there are floor obstructions such as conduits or old anchors, be sure to cut them flush to the floor to prevent the gantry from resting on them. Also be sure there is at least 6” (152mm) of clearance between any existing floor penetration and the new gantry position.

5.) Check floor levelness, as shown in Figure 1-2, prior to removing this template. The floor must meet the minimum levelness specification: 1/8” (4mm) over 10 feet.

Figure 1-2 Check Floor Level



NOTICE



Positioning requires cutting eight (8) holes in the floor covering.
Before you drill or cut any flooring, make sure the appropriate hospital personnel have approved the location of the table/gantry.

Any repositioning must meet all regulatory requirements to be completed.

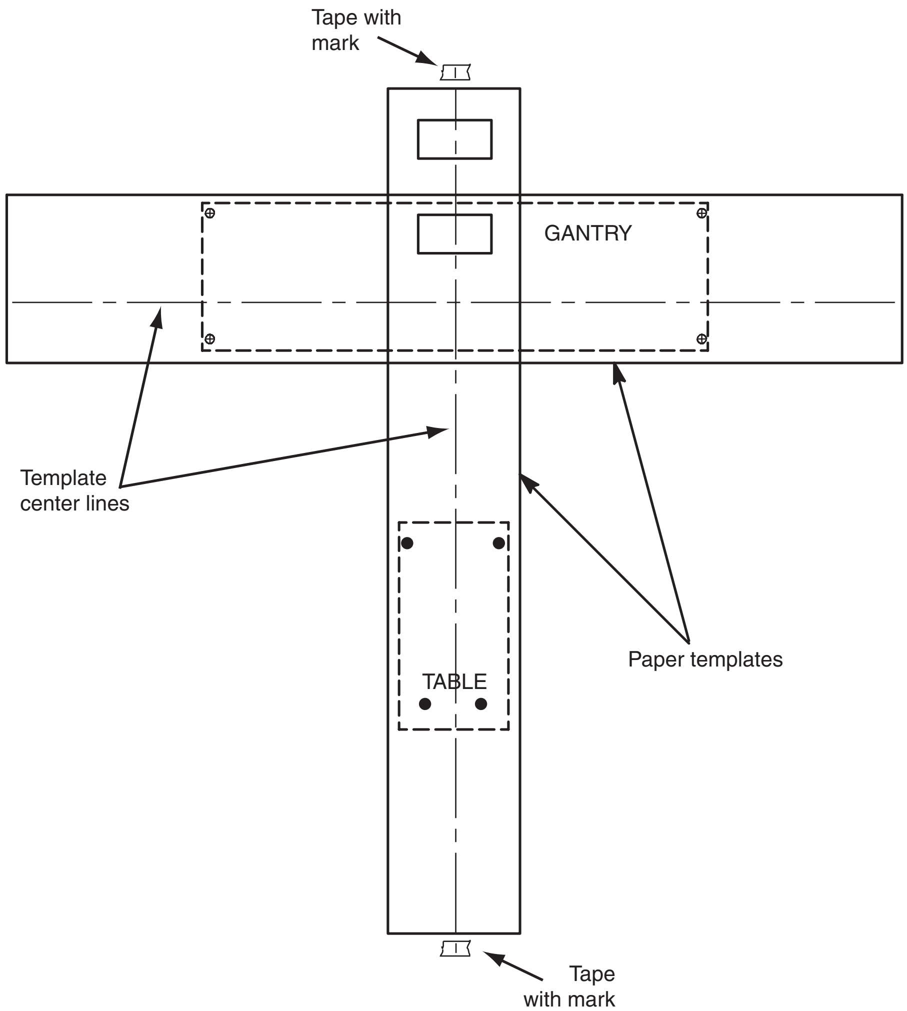
6.) Place the laser at the foot end of the table template and turn it on. Check the straightness of the table template with the laser using the template center line. If necessary, make adjustments to insure the template is straight and accurately positioned. See Figure 1-3.

Note: Do not move the gantry template.

7.) Check with customer for approval of the gantry/table placement.

8.) Place tape on the floor along the table template center line at both ends of the table template (behind the gantry and at the foot of the table.) Mark the tape to indicate the center line. These marks will be used to snap a chalk line in a later step. See Figure 1-3.

Figure 1-3 Center Line with Marks



9.) Use a center punch to mark hole centers for each of the eight (8) leveling pad/anchor locations per Figure 1-4. However, before moving on to the next step, see Step 11 and its note for an alternative method.

CAUTION **Potential for personal injury.**

Use appropriate safety procedures when drilling the floor holes, especially if there is lead under the floor.

Appropriate PPE is required when working with hazardous materials.

10.) Remove the floor template.

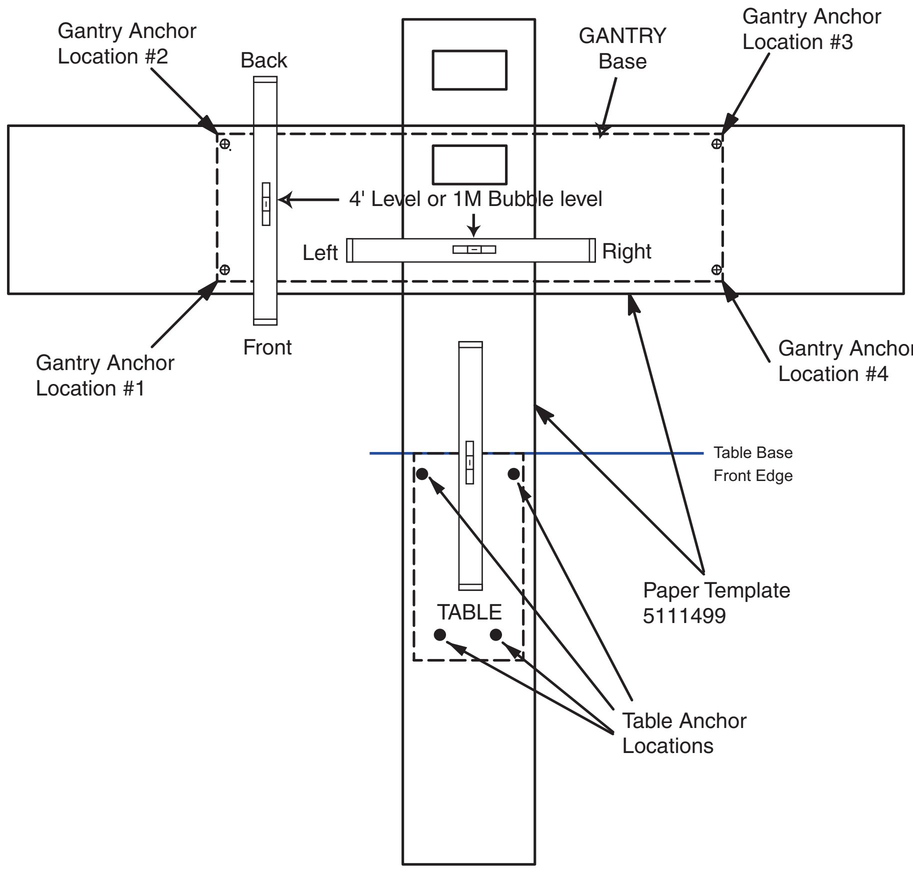
11.) Cut tiles (or other resilient flooring) around all holes punched in the template for the gantry and table.

Note: A fast way to remove flooring is to use a 4” hole saw with a 1/8” masonry bit to cut through the flooring at each leveler pad location.

12.) Some sites require sealing of the floor penetrations after the flooring is removed. If this site does, use RTV or other sealant to seal the floor covering as necessary.

NOTICE **All documentation in this manual is based on mounting the table / gantry on a 4" concrete floor only.**

Figure 1-4 Anchor Hole Locations



13.) Snap a chalk line using the marks that were made on the tape at the ends of the table template.

4.2 Position the Gantry

4.2.1 Additional Tools Required

- Gantry Adjuster Tool, P/N 2107863
- Spanner Wrench, P/N 2110003

4.2.2 Procedure

4.2.2.1 Gantry Prep - For Access Greater Than 28"

Remove all the transportation packaging, except for dollies, from the gantry.

Note: Some sites require floor protection. Locate and install any required floor protection now.

4.2.2.2 Gantry Prep - For Acess Less Than 28"

1.) Remove all the transportation packaging, except for dollies, from the gantry.

Note: Some sites require floor protection. Locate and install any required floor protection now.

2.) Remove the blue dolly from the left side of the gantry so that the gantry can be positioned closer to the left side wall:

Note: Use Floor Protection for this process.

a.) Lower the gantry to the floor so that the gantry is resting on the floor.

b.) Remove the three (3) M14 hex bolts that secure the gantry to the dolly.

c.) Replace the removed dolly with the shipped black gantry-positioning dolly, and reinstall the three (3) M14 hex bolts.

d.) Raise the gantry so that it is once again off of the floor.

4.2.2.3 Gantry Positioning - All Sites

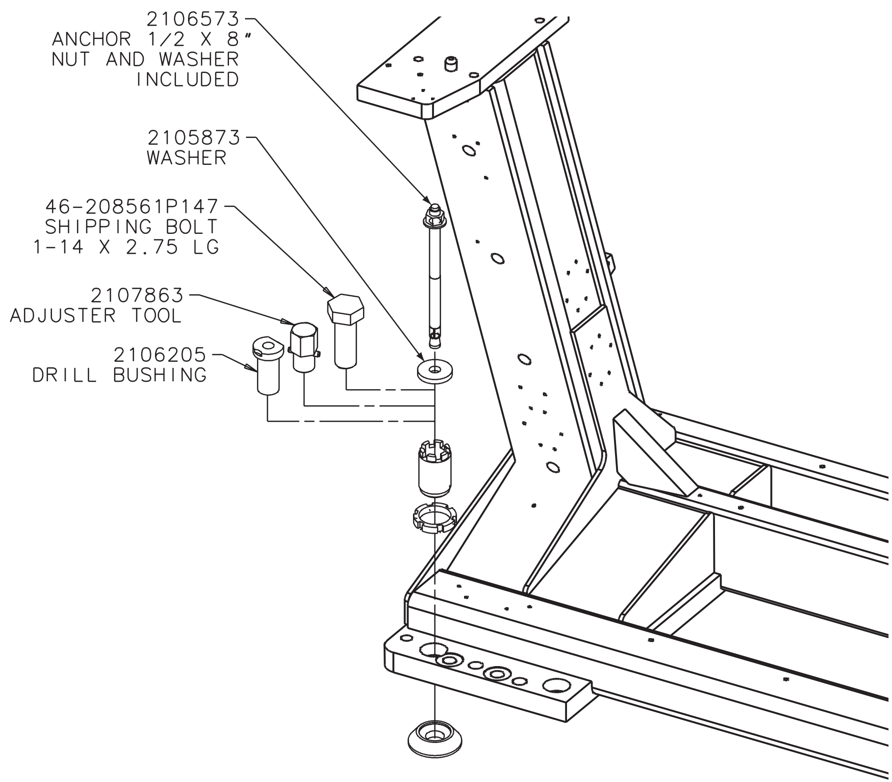
1.) Position the gantry over the floor cutouts appropriately.

a.) Locate the four (4) leveling pads, and position each beneath its associated adjuster. A ½" ratchet and 1-½" socket are required to loosen the gantry shipping bolts.

b.) Use the dollies to evenly lower the gantry, until it is just off of the floor (approximately ¾" or 20 mm). Use a ½" ratchet to raise and lower the dollies.

c.) Carefully rotate the gantry into the correct position over cutouts in the floor.

Figure 1-5 Gantry Base Installation Hardware



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

2.) Loosen the locking rings with the spanner tool. Turn gantry levelers by hand until each leveler is down ¾ in. (or 20 mm) from the bottom of the gantry frame.

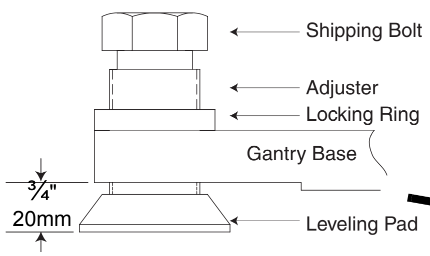
3.) Leave the locking rings loose so you can fine-tune the leveling pads to compensate for slight variations in the floor surface.

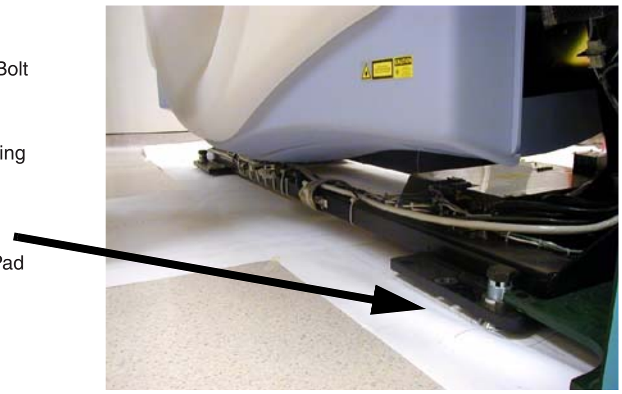
4.) Position the gantry so that the adjusters are centered over their respective holes scribed earlier into the floor cutouts.

Note: Due to the large plug head size, it may be easier to route the gantry 3-phase power cord under the rear gantry rails before lowering the gantry. This cable can be found in the box with the power cables. If not, feed from the PDU side of the cable.

5.) Using a ½ in. ratchet, gently lower the gantry until it rests on the floor over the marked areas. When finished, the gantry should now be resting on all four leveling pads.Gantry and Table Base Leveling Pads (Starting Positions)

Figure 1-6 Gantry Leveling Pads and Adjusters

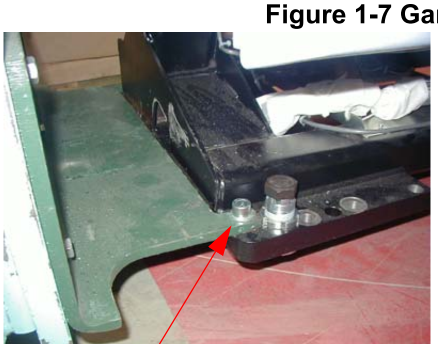


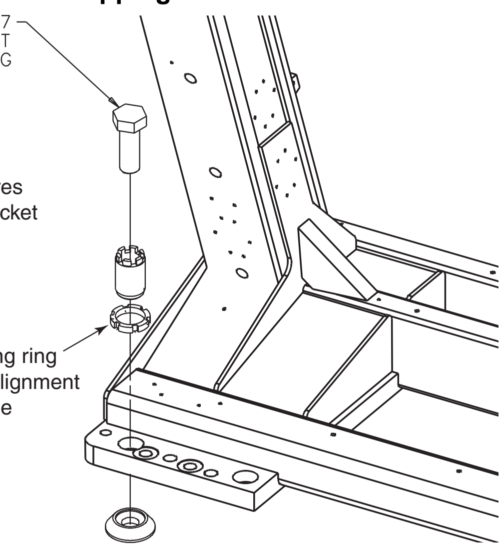


NOTICE **Gantry dollies weigh approximately 250 lbs each. Exercise caution when removing dollies so as to not damage the floor covering.**

6.) Using a 14mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry (Figure 1-7).

Figure 1-7 Gantry Dolly Bolts and Shipping Bolts





7.) If needed, remove the four (4) gantry shipping bolts (Figure 1-7), using a 1½ in. socket.

4.3 Level the Gantry

4.3.1 Additional Tools Required

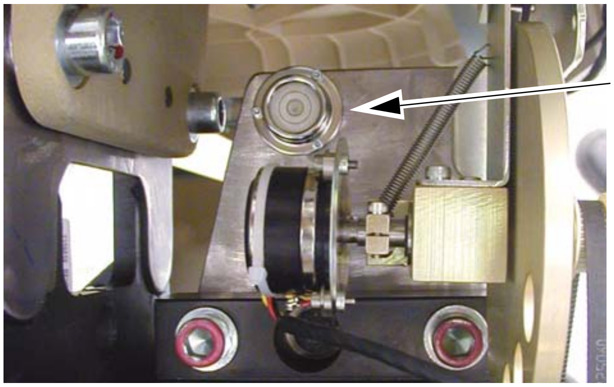
- Gantry Adjuster Tool, P/N 2107863
- Spanner Wrench, P/N 2110003

Note: **This procedure is designed for gantries that are mounted on 4" concrete floors that meet all of the pre-installation flooring requirements.**

The gantry uses two (2) bubble levels that are permanently mounted to machined surfaces on the stationary base, to tell when it is level.

Bubble levels are located on both ends of the gantry stationary base. They are located on the stationary base near a point where the rotating structure pivots mount to the base structure. (See Figure 1-8.)

Figure 1-8 Gantry Bubble Level



4.3.2 Procedure

1.) Loosen all adjuster lock rings (use a spanner wrench or large channel lock pliers).

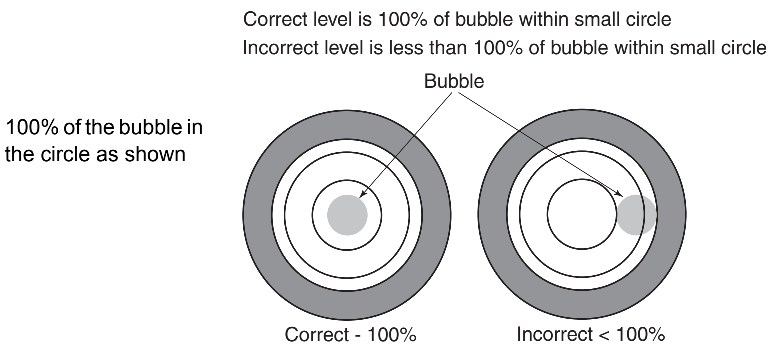
2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3 and 4 in Figure 1-10) until both bubble levels are centered left to right and front to back. (See Figure 1-9.)
   - Begin by turning each adjuster no more than 1 turn at a time.
   - Use the adjuster tool, 1-1/8" socket, and the ½" drive ratchet to turn each adjuster. (Refer to Figure 1-5, on page 39.)

   a.) Level the left side from front to back by turning adjusters #1 and #2.

   b.) Level the right side from front to back by turning adjusters #3 and #4.

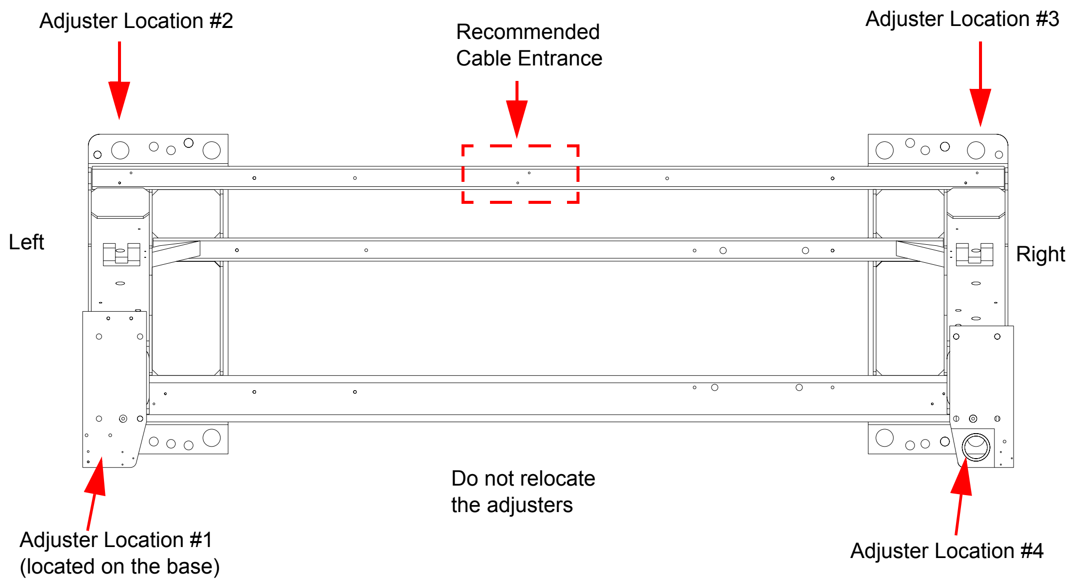
   c.) Level the side (right or left) that is higher with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.

Figure 1-9 Bubble Level Centering



3.) When the bubble levels are centered (Figure 1-9), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation. **DO NOT leave any adjuster un-loaded or floating**.

Figure 1-10 Gantry Base Adjuster Locations — Top View

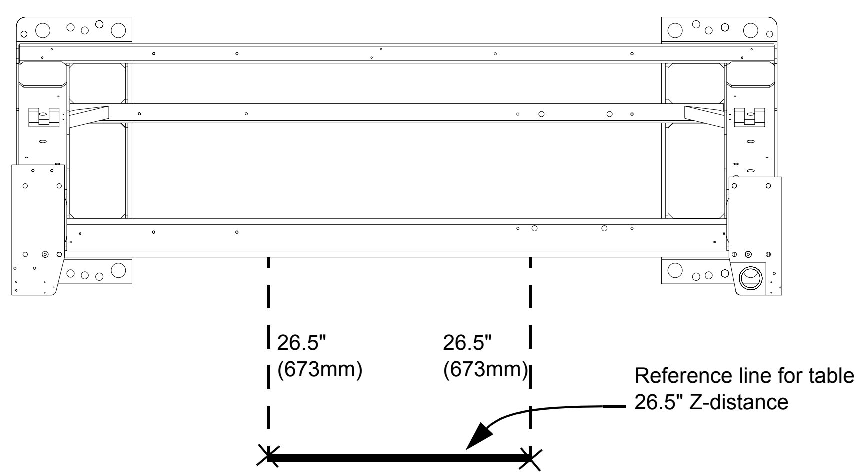


4.) If there are floor obstructions such as conduits or old anchors, be sure to cut them flush to the floor to prevent the gantry from resting on them. Also be sure there is at least 6" (152mm) of distance from any existing floor penetration to the new gantry anchor positions.

5.) Using a measuring tape, measure and mark two spots 26.5" (673mm) from the gantry base frame (under the front cable tray) out toward the table floor cutouts, as illustrated by X marks at the bottom of the dotted lines in Figure 1-11. These spots will be used to snap a chalk line in the next step.

6.) Snap a chalk line using the two 26.5" (673mm) marks. This line should be parallel to the gantry and 26.5" (673mm) from the front of the gantry base. See Figure 1-11. In a later section, you will move the table against the 26.5" (673mm) and center it over a table center line (to be drawn at that time.)

Figure 1-11 Table Base Reference Line and Marks



4.4 Gantry Bearing Gap Inspection

All CT systems require a gantry bearing gap inspection before starting electrical calibration.

All international gantries are shipped in a wooden shipping crate that should not be removed until it arrives at the installation site. This shipping container is designed to reduce the risk of shipping damage.

The back cover needs to be removed to gain access to the gantry bearing.

4.4.1 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
2 (mechanical suppliers or installation team)	15 min	15 min	5 min

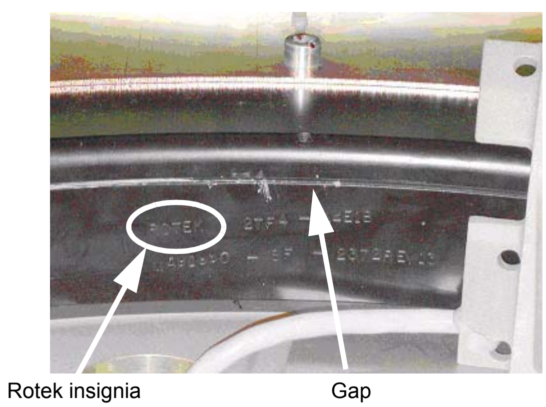
4.4.2 Tools and Test Equipment

- Standard tool kit
- Inspection document
- 2.5 mm Allen wrench
- Rear cover dollies (Qty = 2)
- Flashlight

4.4.3 Damage Indicators

On the inside edge of the black-colored bearing assembly, a mark similar to that shown in Figure 1-12 will be seen, if this is a Rotek bearing.

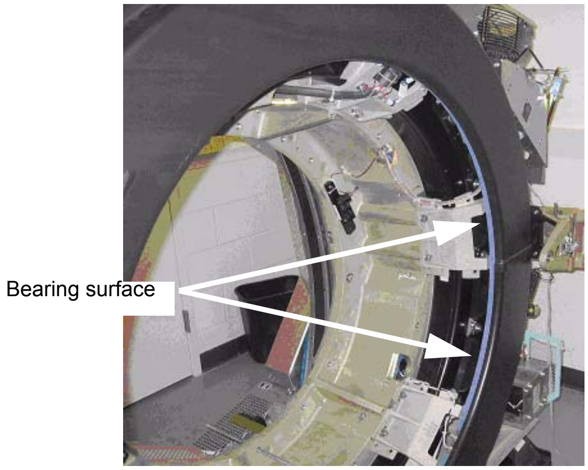
Figure 1-12 Gantry Bearing - Rotek Label



The mark has a serial number in the same format as:
ROTEK 2TF4-44E1B-MA91960-8F-2372-REV13.

The gap to inspect is shown in Figure 1-13 next to the serial number.

Figure 1-13 Gantry Bearing



On most systems, a change in the bearing gap does not cause the gantry to make unusual sounds, unless the gap is severe. If the gantry is badly damaged and the gap is severe, it can cause operation issues. Some systems are shipped with shock indicators that must be returned to Milwaukee.

A severe failure may be seen during installation as a problem rotating the gantry.

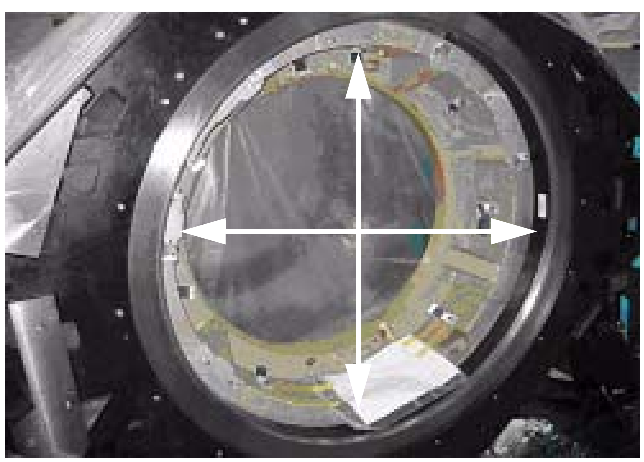
4.4.4 Procedure

1.) Remove the scan window following the procedure in Appendix A, Section 1.0, on page 123.

2.) Remove the top and rear gantry covers, following the procedures in Appendix A, Section 3.0, on page 127 and Section 4.0, on page 129.

1.) Use a 2.5 mm hex wrench as a tool to measure the gap at the positions shown in Figure 1-14. The location of gantry components does not matter. Measure four (4) locations 90 degrees apart from each other.

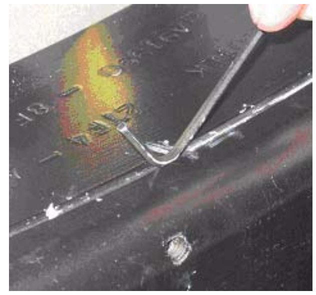
Figure 1-14 Inspection Locations



2.) If the 2.5 mm easily fits without effort in the gap, the gap is out of spec. Figure 1-15 shows a gap that is too large and one that is good. Notice that the hex wrench does not fit in the gap in Figure 1-15 (on the right), but does in on the left. Do not use force when putting the wrench in the gap. Either it slips in or it doesn't.

Figure 1-15 Determining Gap

Gap is too large.



Gap is good.

3.) Replace the top and rear gantry covers, following the procedures in Appendix A, Section 3.0, on page 127 and Section 4.0, on page 129.

4.) Replace the scan window, following the procedures in Appendix A, Section 1.0, on page 123.

4.4.5 Finalization - Mechanical Installers

If the Bearing Gap Inspection passes, complete the sign-off on the GE Form e4879, Installation Data verification form, that this inspection was completed.

If the Bearing Gap Inspection fails, contact your site FE.

4.4.6 FE Service Action Required

If the Bearing Gap Inspection fails, the mechanical installer notifies the site FE that the inspection failed.

The site FE should:

1.) Open a bearing inspection dispatch.

2.) Follow the inspection procedure described in this section.

3.) Record the bearing inspection results.

If no damage is found, close this dispatch and continue with the electrical calibration procedures.

If the system is damaged, go to the Equipment Delivery Quality web site and follow their instructions.

To enter a damaged in shipping claim, go to this web site:
http://egems.med.ge.com/edq/home.jsp

4.4.7 FE Inspection Completion

1.) After the Gantry Bearing Inspection passes, complete the opened service dispatch with the following information:
- Gantry Serial Number
- Gantry Type
- System ID
- Site Name
- Installation date
- Was the gantry transported to the site in the shipping crate? (Yes/No)
- Was the gantry lifted or hoisted, were riggers used, or was the gantry delivered via flatbed wrecker? (Yes/No)
- Number of locations that fail the gap inspection if any: _____

2.) Close the service dispatch.

Should any follow-up be required after this inspection, the site engineer will be contacted directly by CT Engineering.

4.5 Install Gantry Alignment Laser and Bracket

NOTICE **Use caution while removing the gantry scan window.**

1.) Rotate the gantry by hand until the collimator face plate is at the 5 o'clock position.

Note: With power OFF, the gantry movement is tight.
DO NOT pin the gantry during this alignment process.

2.) With the top and back gantry covers removed, locate the two M10 bolt holes as shown in Figure 1-16. These bolt holes will be used to attach the laser tool to the gantry.

- The bolts can be installed using an 8mm Allen wrench. Be careful not to bump the alignment light; the mounting space is tight near the alignment light. Tighten bolts until both are snug.
- Do not drop bolts or the bar on the collimator faceplate. Attach the bar as shown in Figure 1-16.
- Using a 9" (minimum) level placed on the attached bar, level the bar by rotating the gantry.

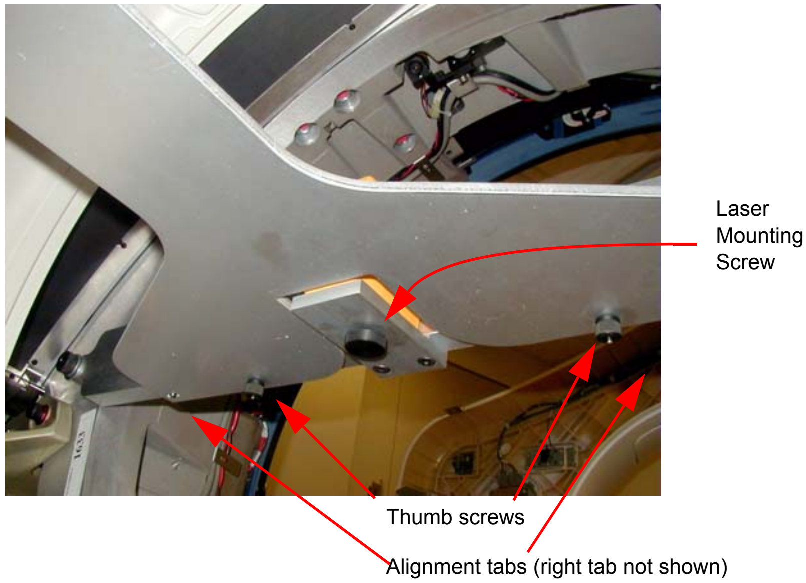
Figure 1-16 Alignment Bar Installation Location

CAUTION

Potential for injury.
DO NOT look into the laser.
Use appropriate safety procedures when working with lasers.

3.) Attach the laser centering plate onto the laser mounting bar as shown in Figure 1-17. The plate is attached from under the alignment bar using two fixed locators and two thumb screws.

Figure 1-17 Attach Laser Center Plate

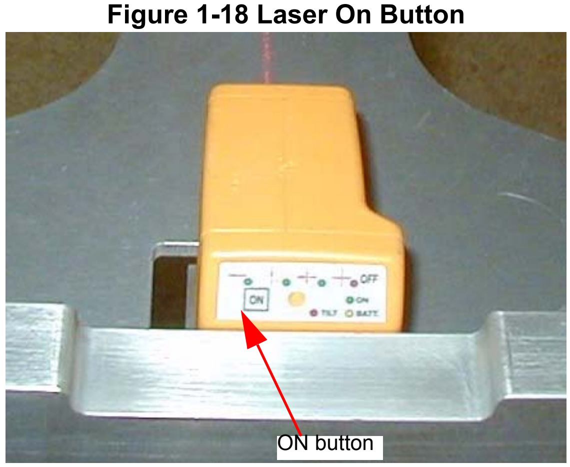


4.) When done, insert the laser and turn on the laser using the controls on the back. If the laser is loose when mounted, use a 2" piece of Velcro loop (fuzzy) section and attach to the alignment plate over the attachment screw. Remount the laser and it should fit snugly without moving.

5.) When pressed, the **ON** button steps through four different beam profiles and "Self-Leveling Off". Press the **ON** button until the "|" beam shows. It will be used for this operation.

Times pressed	Function	Notes
1	—	Self-leveling on
2	|	
3	+	
4	Self-leveling off	Do not use

Figure 1-18 Laser On Button

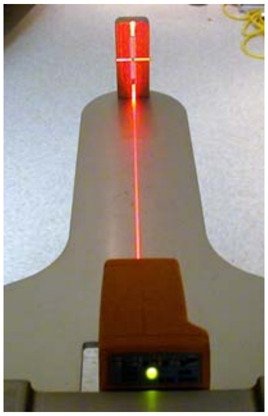


6.) Align the laser by carefully rotating the laser base assembly so that the "I" beam shines through the center of the alignment sight mounted on the end of the alignment plate.

Note: The laser beam may be wider depending on the battery life.

7.) Use the locking screw on the bottom of the alignment bar to secure the laser to the bar, as shown in Figure 1-19. When done the laser should fit snugly without moving on the mounting bracket.

Figure 1-19 Laser Centering



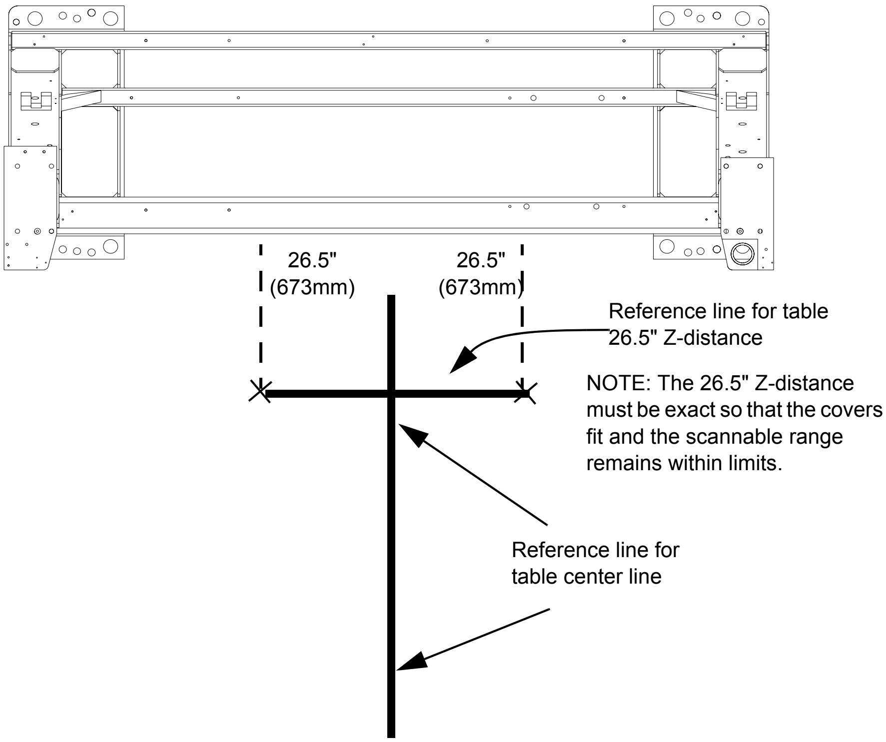
Note: The laser can, but should not, move when tightening. Use caution to prevent any movement. Any movement can result in drilling the table anchor holes in the wrong location.

8.) After the laser is centered, notice that the laser beam also appears on the back wall. Place a piece of masking on the wall at and carefully mark a line on the laser line. This line will later be used in the table alignment. This line is also useful in determining if the laser unit moves during the alignment process.

9.) Remove the alignment centering plate and store in the alignment case.

10.) Using a chalkline, mark a table center line on the floor along the laser light shining on the floor.

Figure 1-20 Table Center Line



11.) Recheck the table-to-gantry reference line for 26.5 in. (± 25 in.) Z-distance. Refer to Figure 1-20.

12.) Turn off the laser but do not remove.

4.6 Table Prep and Set-up

NOTICE **Review the table installation and leveling video for this procedure.**

Tools Required

- Standard Install Tool Kit
- 1-½”, 1-¼”, ¾” sockets
- 1-5/8” (40-41mm) socket
- 10mm and 14mm hex socket bits

Time and Personnel

- 1.5 hour labor on site
- 2 Engineers

SAFETY

CAUTION
Potential for Electric Shock.
Equipment is Energized.
Follow appropriate safety procedures when working with an energized system.

CAUTION
Potential for Injury.
Table will tip if not anchored on the dolly.
Make certain that Table is adequately secured to the dolly.

CAUTION
Potential for Injury.
Table on dolly length is 118" (9’-10”).
Exercise caution when moving the table on the dolly.

PROCEDURE

CAUTION
Potential for Injury.
Table will tip if not anchored on the dolly.
Make certain that Table is adequately secured to the dolly.

1.) Remove all the transportation packaging and boxes, except dollies, from the table. (See Figure 1-21.) Leave a layer of packing material on the cradle to protect the cradle from damage. (It can be removed during laser alignment of the table.)

Figure 1-21 Remove Table Packing

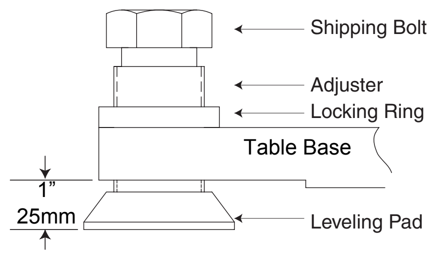
2.) Unpack the items and locate all of the items needed to install the table.

Note: The GT table on dollies is approximately 118” long and may require additional room to maneuver.

3.) Using the table centering and distance locator marks made earlier, wheel the table to its approximate position relative to the gantry.

4.) Locate the table leveling pads inside the table in the back and on the side in the front. Preset leveling pad heights to 3/4” (20mm). (See Figure 1-22.)

Figure 1-22 Table Base Leveling Pads (Starting Positions)



5.) Use a 1-5/8” socket and ½” ratchet to loosen the shipping bolt. Loosen the locking rings if present.

6.) A 1-1/8” socket is used with the adjuster tool if needed to lower the adjuster.

7.) Use the dollies to evenly lower the table until it rests on the leveling pads using a ½”ratchet on each end.

Figure 1-23 Adjusters and Lock Rings

Back Front

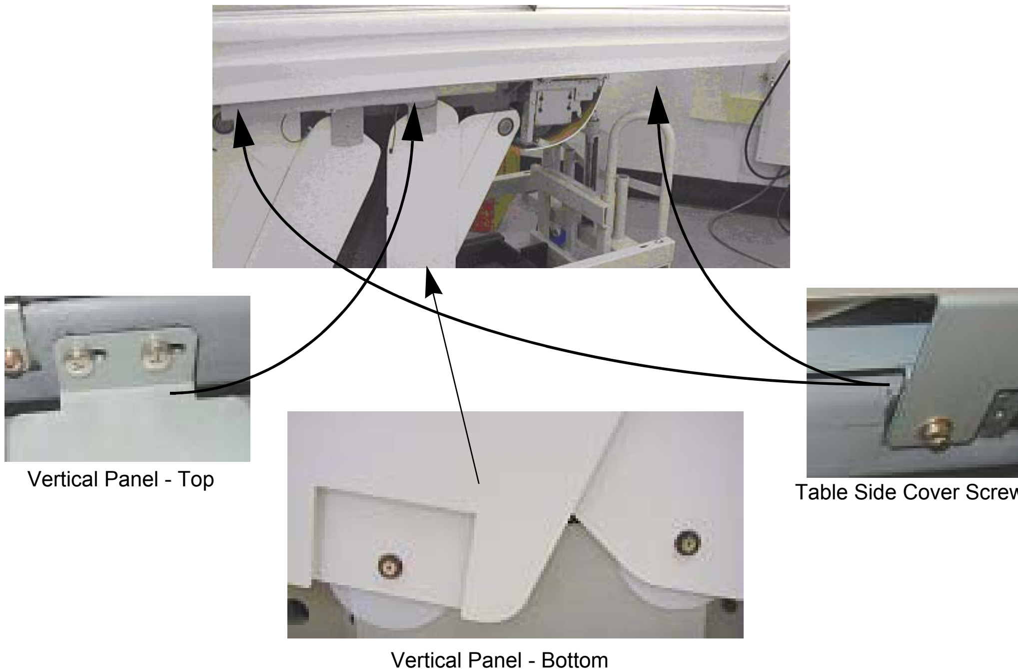
4.7 Table Cover Removal

1.) Remove the table right side cover, as shown in Figure 1-25.

a.) Removing the two screws on each end of the underside of the long side cover of the table.

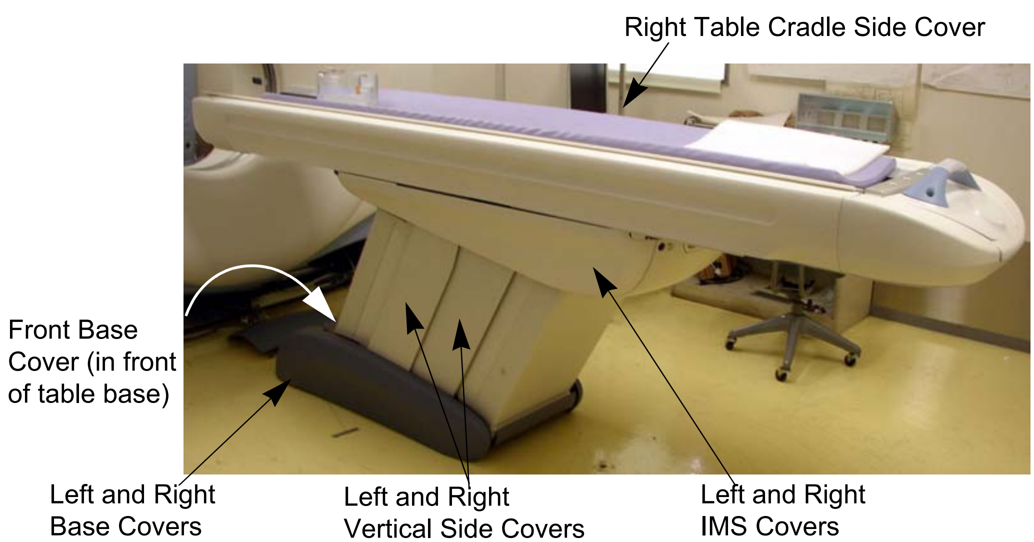
b.) Slide each cover forward to unlatch, lift upward slightly to disengage the latches, and remove the side cover. Doing this procedure will require patience and practice to remove and replace this cover.

Figure 1-24 Table Covers



2.) The table is normally shipped with some of the side/vertical panels removed. If installed, remove the four side panels, using a Pozi drive #1 screwdriver.
3.) Carefully lay the side panels on protective padding out of the way.
4.) Make sure that all four of the table levelers are on the floor. The table should set on the four levelers with the dollies still installed.
5.) Carefully center the four levelers over the 4" (102mm) floor cutouts.
6.) Check that the front table base center line is on the chalk table center line.
7.) If still present, remove all packing materials and the table cradle pad from the table cradle.

Figure 1-25 Table Covers



4.8 Removing the Accessory Rail Strip

1.) Remove the accessory mounting strip attached on each side of the cradle using a small flat blade screw driver. The nylon screws are inserted inside the accessory rail on the cradle.

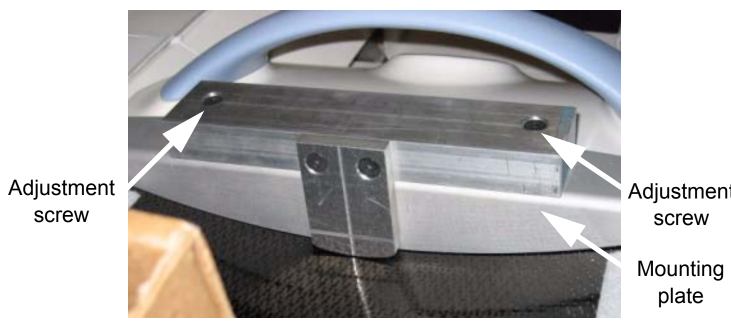
2.) Place the accessory strips on the floor and reinstall the nylon screws into the accessory rail for safe keeping.

Figure 1-26 Accessory Rail Screw

4.9 Install the Table Cradle Laser Alignment Plates

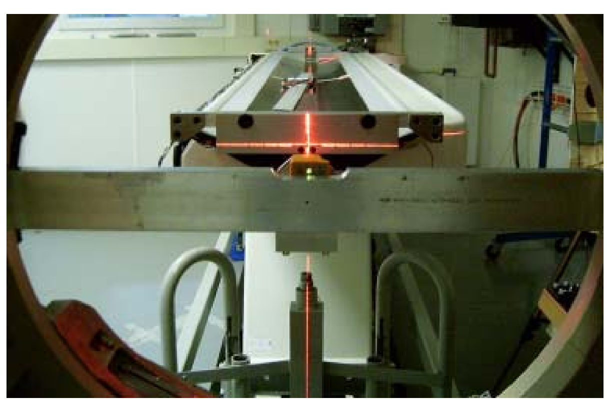
1.) Locate the aluminum accessory tray mounting plate with the three holes on the rear of the cradle. Fit the rear alignment target into the two mounting holes as shown in Figure 1-27. Use the adjustment screw to adjust the fit as needed. See Figure 1-27. The fit should be snug, without play, when you are finished.

Figure 1-27 Cradle Rear Laser Alignment Tool



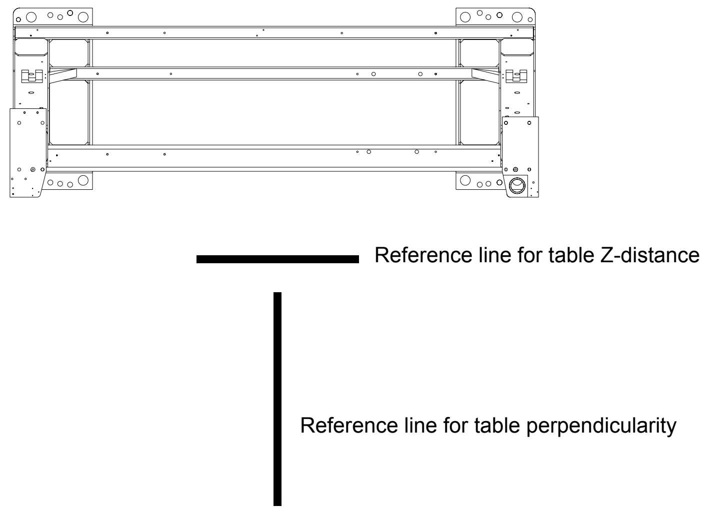
2.) Check that the table base is centered over the table center line, and the base is on the 26.5 in.line (± 0.25 in.) made on the floor.

Figure 1-28 Rear Laser Alignment Tool - Installed



3.) Lower the table to the floor using the dollies, making sure to maintain the 26.5 in. (± 0.25 in.) distance.

Figure 1-29 Two Reference Lines



4.10 Level the Table

4.10.1 Basic Information

4.10.1.1 Tools Required

- Standard Install Tool Kit
- ¾”, 1-¼”, 1-½” and 1-5/8” sockets
- 8mm, 10mm, and 14mm hex socket bits
- Laser Alignment Kit
- Johnson Professional 4’ level
- Johnson Professional 2’ level

4.10.1.2 Time and Personnel

- 0.5 hour labor on site
- 2 Engineers

4.10.1.3 Alignment Conditions

- Before you start, turn on the laser and check that the beam is still on the mark placed on the wall. If not, reset the laser.
- If mark is not present - Using a measuring tape measure and place a 5" piece of masking tape on the cradle at the 1000mm and on the laser line.
- Gantry must be at zero degrees within 0.14 degrees from gantry zero.

4.10.1.4 Alignment Specifications

- Mechanical base alignment must be perpendicular within 0.14 degrees from gantry zero (± 1.5 mm) as measured in this procedure.
- Table cradle travel (X axis) must be perpendicular 0.14 degrees from gantry (Y axis) zero (± 1.5 mm) as measured in this procedure.
- Table cradle must be level in all directions ± 0.0625 in. (centered within the lines on a Johnson Professional level).
- All table adjusters should be preset to a minimum of 3/4 in. (20 mm) down from the table base to make adjustment easier. Based on floor levelness and your experience, a different preset height may work better.

4.10.2 Level and Center the Table to the Gantry

NOTICE **Avoid leaning on the cradle during this procedure.**
DO NOT pin the gantry during this alignment process.
This procedure as described is for systems mounted on 4 in. (102 mm) **concrete floors only!**

Note: If the floor covering was not properly removed with the glue removed or the levelers were not centered over the floor cutouts, the leveler may become trapped against the edge of the floor covering, causing the table to become unleveled. If this happens, move the table and enlarge the 4 in. (102 mm) floor cutout for the table. Glue removal is important and aids in moving the table to its final location.

1.) Have the table side panels removed and have a ratchet, 1-1/8” socket, and a 2-foot level ready to use.

2.) Turn on the laser’s “I” beam (vertical beams) by pressing the **ON** button 2 times.

Note: Step 3 through Step 7 are for perpendicular positioning of the cradle to the gantry.

3.) The table on the dollies should be resting on the floor, and the laser beam visible on the cradle. The laser light should now shine down the cradle onto the rear vertical target. Moving the table on the dollies by raising and lowering makes it easier to center the table right to left.

Note: When using the table dolly to move the table, be sure that the shipping bolts are still attached to the adjuster leveler feet.) This prevents the adjuster levelers from gripping on the floor adhesive, making it difficult to move.

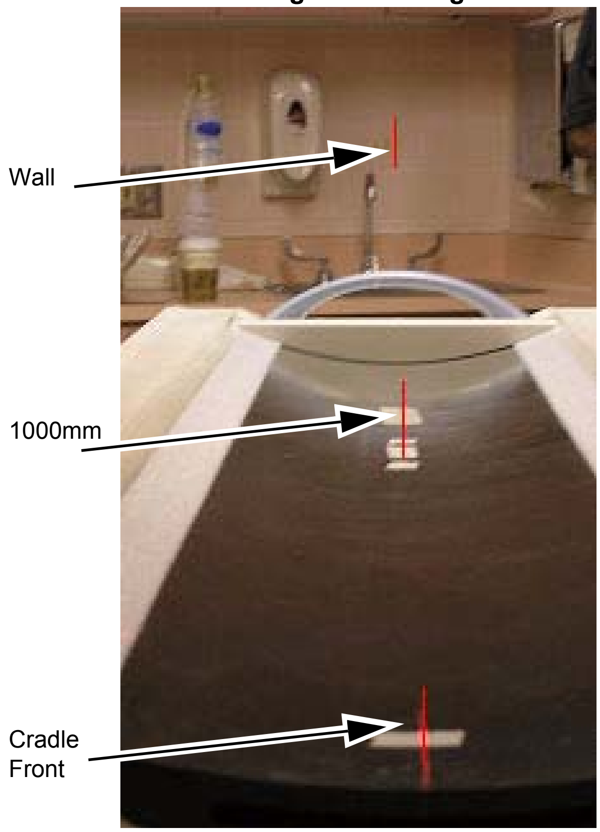
4.) Move the table so that the base is roughly centered over the scan center line, the front edge of the table base is on the 26.5 in. (± 0.25 in.) line, and the table is resting on the floor. Check that the leveling feet are centered in the cutout circles.

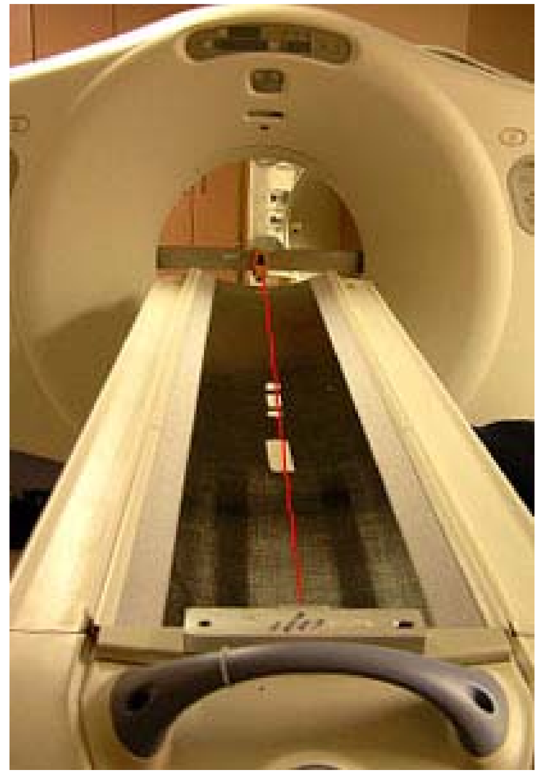
5.) Carefully move the table so that the cradle front center line and the back target are aligned. You may need to raise the table to move the table. When aligned, lower the table to the floor.

6.) If not already done - Measure 1000mm from the front of the cradle, and place a piece of tape under the laser center line. Carefully mark a line along the laser line.

7.) The laser beam should now connect the cradle front centerlines, the 1000 mm cradle center line, onto the rear alignment tool vertical center and finally onto the alignment centering mark placed on the wall. The centering alignment line on the wall is used to be sure the laser is still centered. If the alignment line on the wall is NOT on the original mark, readjust the laser and repeat the above steps. See Figure 1-30.

Figure 1-30 Alignment Laser Marks - Table & Wall





Note: Step 8 through Step 10 are for front-to-back and side-to-side leveling of the cradle.

8.) The table should be completely on the floor and resting on all 4 levelers. Carefully remove one side of the table dolly, taking care not to bump or move the table. Ether side and/or end of the table dolly assembly can be removed.

CAUTION

Potential for Injury.

In the ship position, the table tips easily!

DO NOT lean on the table! The shipping bracket should still be in place!

Figure 1-31 Alignment Laser Marks - Table & Wall

9.) Raise or lower the table as needed using the front and rear levelers and level the cradle horizontally in the front and back locations. Refer to Figure 1-31.

a.) First, horizontally across the cradle at the front of the cradle.

b.) Second, horizontally across the cradle at the 1000mm mark on the cradle.

c.) When the cradle is roughly level in both locations, leave the level on the cradle horizontally.

10.) Level the cradle front to back using a 4' level placed in the center of the cradle. These two leveling actions often conflict and a few iterations may be required.

This process is complete when:

- The cradle is still centered on the front, mid, and rear marks.
- The cradle is leveled horizontally (across table) front and back (with the bubble centered between the lines.)
- The bubble is centered between the lines in the Y direction along the length of the cradle.
- The laser is still centered on the wall center line.
- The table is still on the 26.5" line and the levelers are not resting on the flooring.
- The laser is the same as in Step 7.

Note: The leveling process may take several iterations of Step 1 through Step 10. Patience and accuracy is required to properly complete this process.

11.) When completed, turn off the laser tool.

Note: Do not remove the table dollies.

4.11 Gantry Tilt Zero Alignment Check

4.11.1 Manufacturing Shipping Specification

Must be 90 degrees ± 0.25 degrees

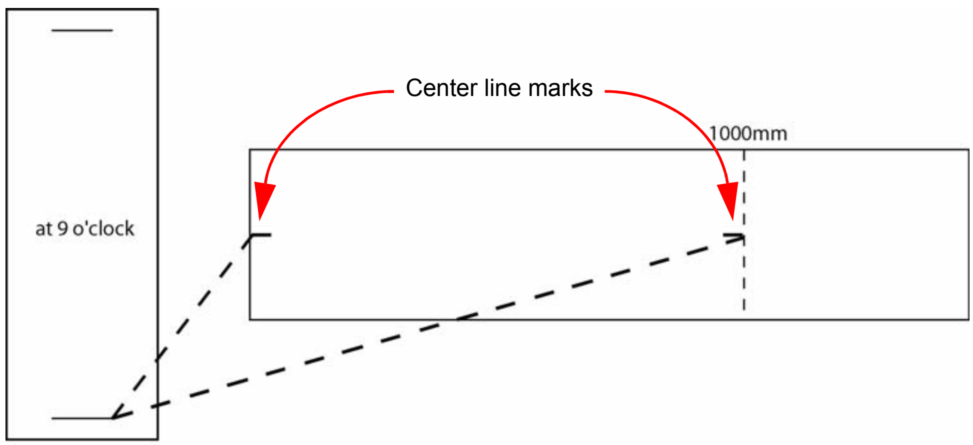
4.11.2 Test

Use an angle indicator mounted on a flat surface of the gantry to determine if the gantry is at 90 degrees (vertical).

4.12 Cradle/Table Parallel Check

1.) With the cradle in the home position, rotate the collimator to the 9 o'clock position. Confirm with a level placed on the collimator face plate.

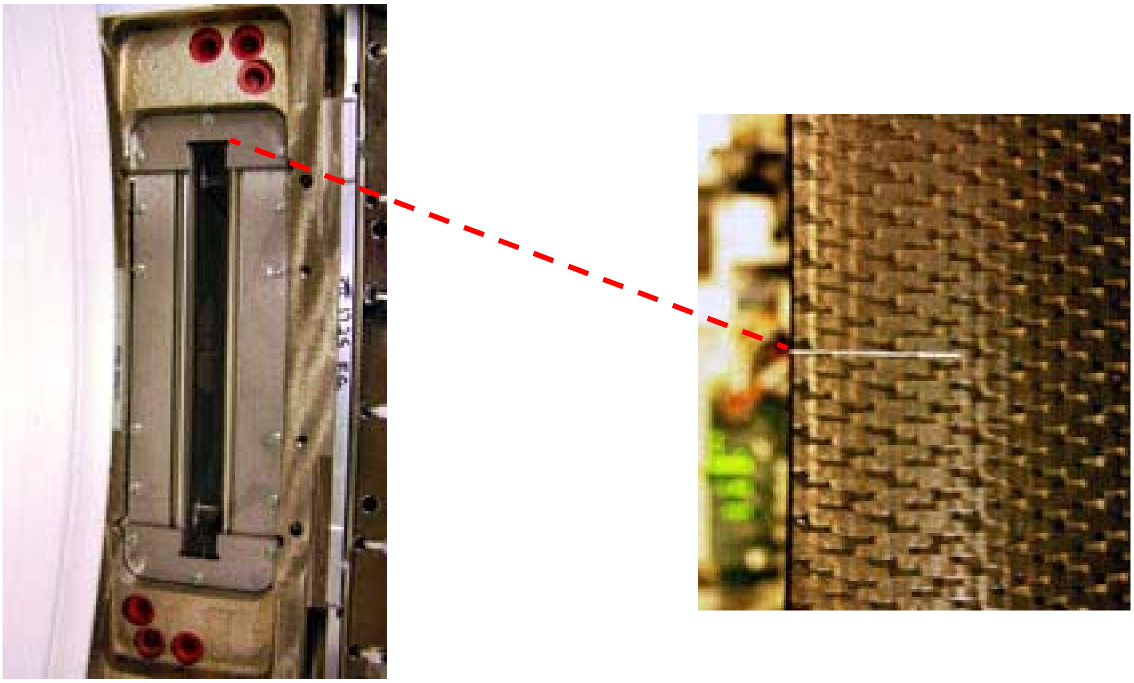
Figure 1-32 First Cradle/Table Parallel Check



2.) Measure the distance from the front top corner of the collimator face plate to the mark at the center front of the cradle. See Figure 1-32, Figure 1-33 and Figure 1-34. Note all measurements in Table 1-4.

Figure 1-33 Collimator Face Plate to Front of Table - Overview

Figure 1-34 Collimator Face Plate to Front of Table - Detail

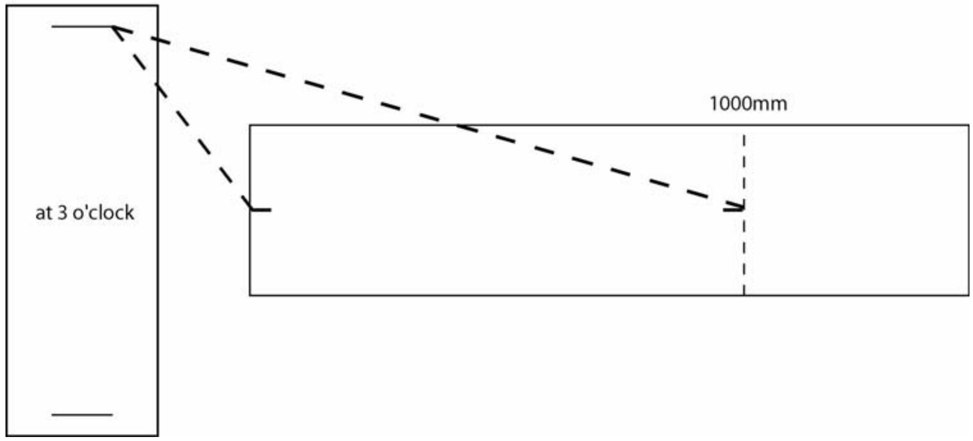


3.) Measure the distance from the same point of the collimator face plate to the center point on the cradle at the 1000mm mark. See Figure 1-32.

4.) Rotate the collimator to the 3 o'clock position. Confirm with a level.

5.) Repeat Step 2 and Step 3. See Figure 1-35.

Figure 1-35 Second Cradle/Table Parallel Check



6.) If needed, move the table using an appropriate tool and re-measure each side until measurements are equal side to side ± 1 mm.

Note: This final adjustment may be slightly different than the placement obtained using the laser.

Table 1-4 Alignment Worksheet

MEASUREMENT	A: 3 O'CLOCK	B: 9 O'CLOCK	DIFFERENCE: | A-B |
Test #1			
To front of cradle			
To 1000mm mark			
Test #2			
To front of cradle			
To 1000mm mark			
Test #3			
To front of cradle			
To 1000mm mark			

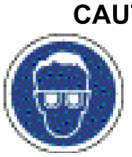
7.) Recheck cradle level (front to back and across) and re-level as required.

4.13 Tighten the Lock Rings

1.) Re-check gantry bubble levels.

2.) Re-check that each of the eight adjuster is loaded by attempting to turn it.

CAUTION **Eye protection is required when using a hammer and chisel.**



3.) Tighten the lock rings at all locations with the spanner, where possible. Use a hammer and chisel to tighten the lock rings only where you can not use the spanner.

4.14 Drill the Table Anchor Holes

WARNING **POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

4.14.1 Notes to Mechanical Installers

4.14.1.1 Note 1: Basic Anchoring Information

GE provided floor anchors are designed for use ONLY on concrete floors that meet the 4-inch concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 3-inches at each anchor point. ANY anchors having more than 1-inch of thread showing above the nut, when torque is set to 55 lb.-ft, shall have a second anchor installed in the closest adjacent hole. This is because the minimum anchor engagement length in the concrete was not met. The second anchor shall be installed to the standard depth and torque specification. **Do not cut anchor bolts that extend longer than the 1-inch limit.**

4.14.1.2 Note 2: Alternate Anchoring

If at least four anchors cannot be set for the gantry, and at least four anchors for the table using the alternate anchor holes, then the installer must inform the PMI that the minimum anchoring cannot be met. Additionally, the customer's structural engineering contractor must be engaged to determine the anchoring method, set the anchors, and certify that their anchoring meets the stated GE minimum load requirement and torque specification.

4.14.1.3 Note 3: Non-Concrete Floors

All other anchoring methods - on floor types other than the concrete minimum - must be determined at the customer's expense by a structural engineering contractor. The anchoring and method must be certified by the customer's contractor to meet the stated GE minimum load requirement and torque specification.

4.14.1.4 Note 4: GE Notification

It is not the role of mechanical contractors or installers (FEs) to determine acceptable methods to install or anchor equipment on non-4-inch concrete floors. The PMI or appropriate GE contact person shall be notified that the facility's floor type DOES NOT MEET the installation mounting requirement for the installation procedure (described in this Installation Manual), and therefore the table-gantry mounting process CANNOT continue.

4.14.2 Requirements

4.14.2.1 Tools Required

- Standard Install Tool Kit
- Hammer Drill
- ½" x 12" Drill Bit (Metric equivalent must not be used)
- ½" Drill Bushing (shipped in install support kit)
- Vacuum with HEPA or drywall dust filter
- Vacuum Hole Attachment – to clean debris from the holes
- PPE

4.14.2.2 Time and Personnel

- .5 hour labor on site
- 2 Engineers

4.14.3 Drilling Procedure

Note: The gantry rear cover should still be removed and the table should still be on the dolly.

1.) Make sure that all table and gantry levelers (four each) are firmly on the concrete floor.

NOTICE Potential for Equipment Damage from Dust

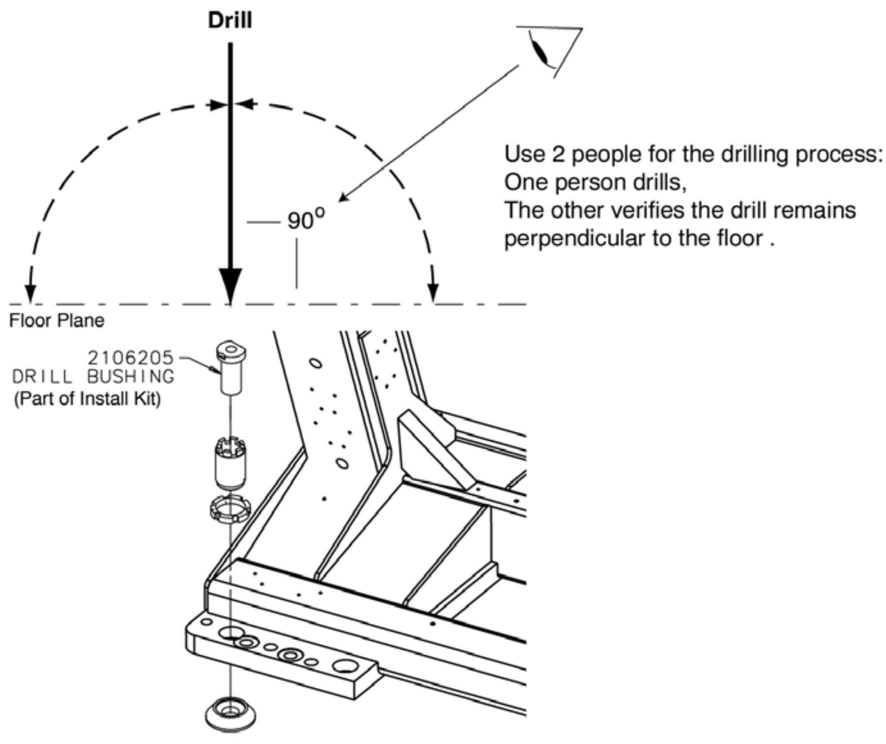
To prevent damage due to the dust created during drilling, you must cover all electronic assemblies in the table base prior to drilling.

2.) Locate the hammer drill and ½" X 12" drill bit. The ½" bit will be used to drill all eight (8) table and gantry anchor holes. You must use the drilling bushing to drill all table and gantry holes. All primary holes can be drilled with the gantry covers installed.

3.) Use a piece of tape to mark the drill bit depth of 7-1/2" (190mm) from the tip of the 1/2" masonry drill bit.

4.) Use the 1/2" bit to drill all eight (8) anchor holes to a depth of 7-1/2" (190mm) as measured from the top of the drill bushing. Review Figure 1-36 and Figure 1-37 prior to drilling.

Figure 1-36 Drilling Position



5.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies.

6.) Place the drill bushing inside each adjuster, to keep the hole vertical and centered within the adjuster.

- Use the drill bushing to center the anchor holes in all adjuster locations, to provide maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
- Take care not to injure yourself on the gantry cover brackets.

7.) Drill the holes perpendicular to the floor.

Important - Follow these guidelines when drilling anchor holes:

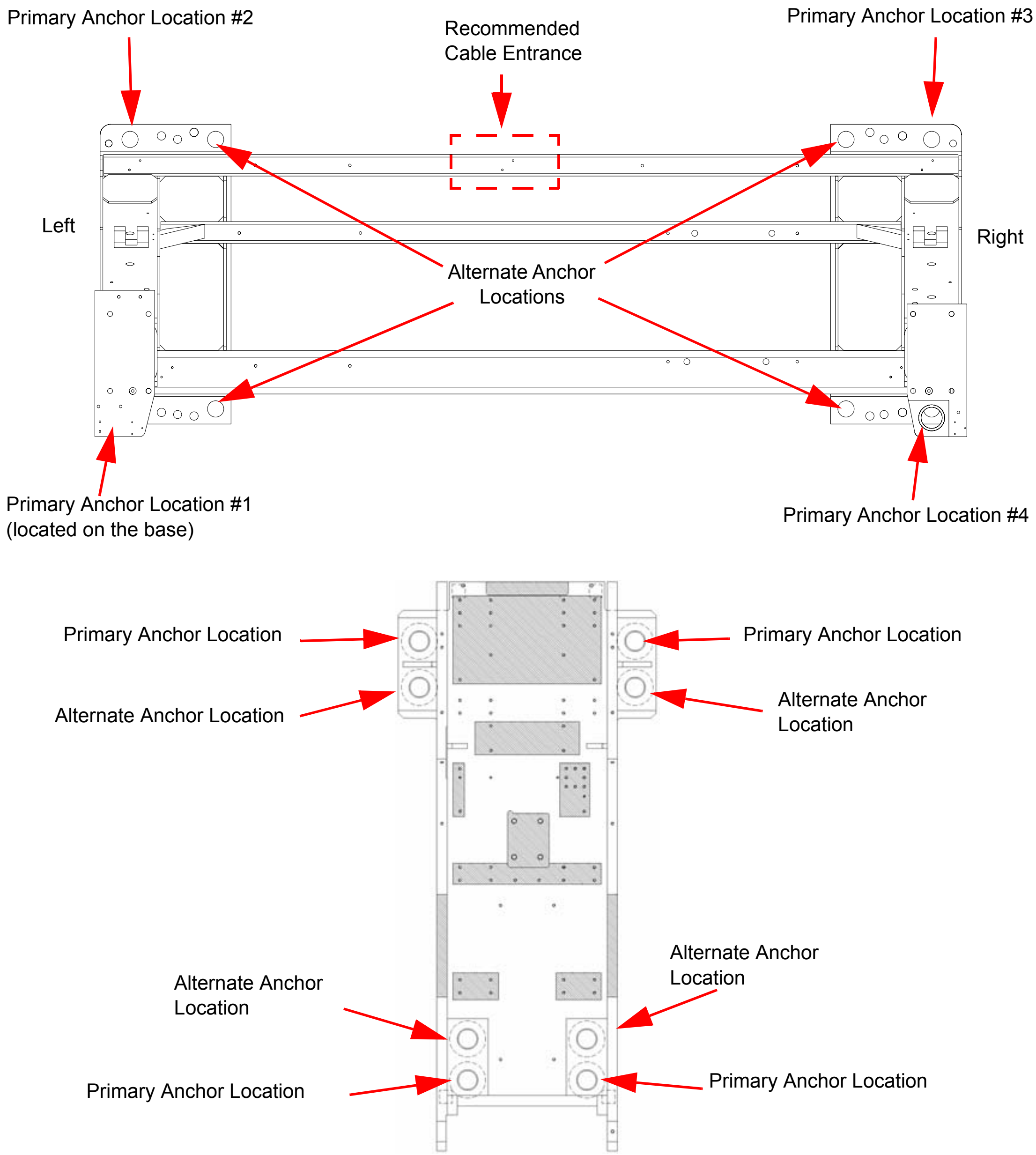
- While one person drills the holes, position a second person to watch the relationship between the drill bit and floor. Make sure the bit remains absolutely perpendicular to the floor throughout the drilling operation.
- Always use the mechanical guide when drilling.
- Stop drilling every 15 or 20 seconds and clear the hole of debris. This lets the drill bit cool and helps to prevent binding of the drill bit.
- Vacuum while drilling to keep gantry and table as free of dust contamination as possible. Place the funnel tip or long extension tip inside the hole.

  A drywall dust filter must be used on the vacuum.
- Drill each hole until the mark on the drill bit is even with the top of the drill bushing. All holes must be a minimum of 7.5" (190 mm) deep, as measured from the top of the adjuster to the bottom of the hole. (See Figure 1-38, on page 65.) Use an upside-down anchor to check the hole depth.

8.) Recheck the depth of all holes by inserting an anchor backward into the hole. A 1/2” (13mm) or less should be showing. Re-drill if needed.

9.) When finished drilling and clearing the anchor holes, vacuum the debris from the inside of each of the holes and from the surrounding (floor) area.

Figure 1-37 Anchor Locations



Note: If alternate location(s) are used to anchor the table or gantry, you must move the respective leveler(s) and pad(s) to the new alternate location(s) and re-drill.

4.15 Gantry & Table Alternate Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, such as reinforcement bars in the concrete, you must use one of the alternate anchor locations, as shown in Figure 1-37. You must also move the respective leveler(s) and pad(s) to the new alternate location(s) and re-drill.

Note: Do not remove the adjuster to move to the alternate anchor hole.

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of four (4) anchors, one (1) at location.

If you must use an alternate anchor hole in the gantry, you must remove the gantry covers to drill the holes. See Appendix A Removal & Installation of Covers, on page 123 for gantry cover removal.

WARNING

POTENTIAL FOR PATIENT INJURY.

IMPROPERLY-SECURED TABLE MAY TIP, DISLODGING PATIENT.

PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements and/or through-bolting.

Note: GE is not responsible for anchoring methods other than those listed in the pre-installation manual.

Provided floor anchors are designed for use ONLY on concrete floors that meet the 4-inch concrete floor requirements.

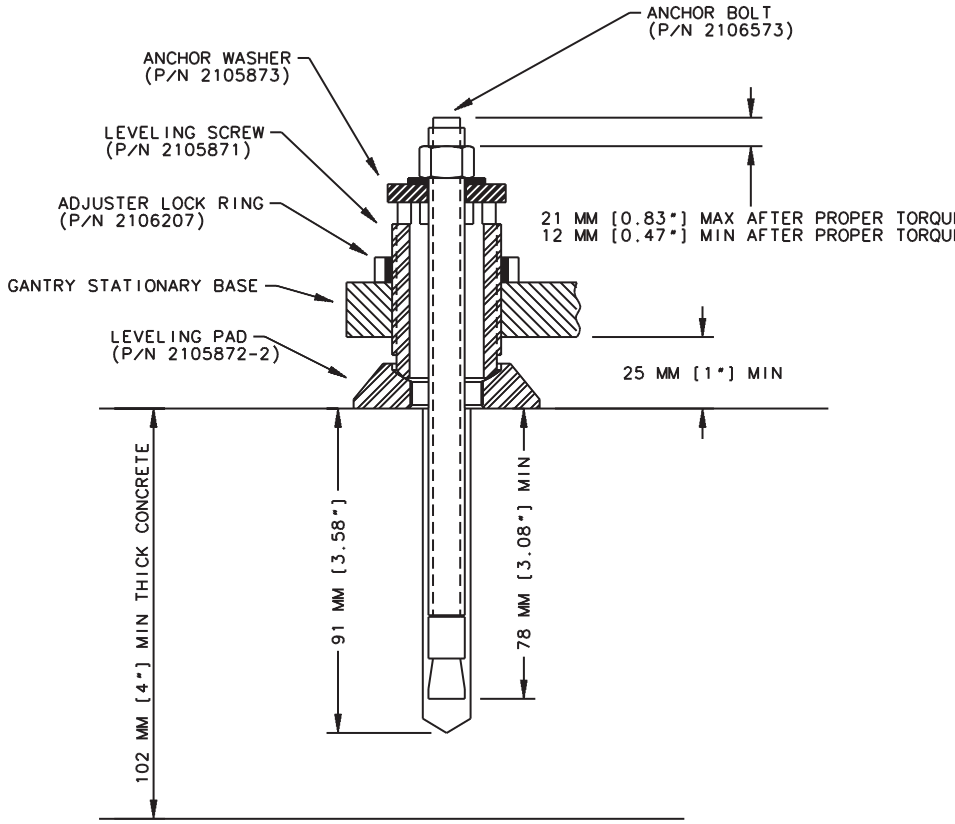
MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum Floor Thickness:	4 inches	4 inches
Recommended Drilling Depth:	3-¾ inches	3-¾ inches
Average Anchor Embedment:	3-½ inches	3-½ inches
Minimum Anchor Embedment:	3 inches	3 inches
Available Alternate Anchor Locations:	Yes	Yes
Shipped Anchor Size:	8 inches	8 inches
Alternate Anchoring Methods:	Yes (see notes, above)	Yes (see notes, above)
FLoor Levelness Requirement:	0.3125 (8mm) over 10 ft	0.3125 (8mm) over 10 ft

Table 1-5 Gantry and Table Mounting Requirements

4.16 Install the Anchors

Recommended - Use “Hilti Kwik-Bolt II” anchors P/N 2106573 (½" dia. by 8" long) as shipped with the VCT system for this procedure.

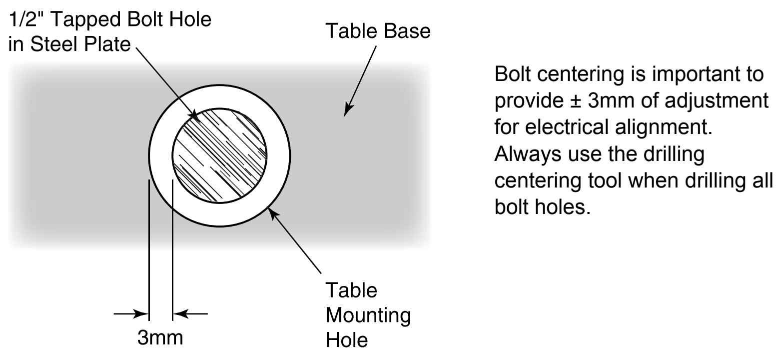
Figure 1-38 Table Base Anchor Assembly



1.) Assemble the anchors before you install them. Refer to Figure 1-38.
   a.) Remove the nut and washer from the anchor.
   b.) Add a ¼" thick washer (PN 2105873) under the regular anchor washer.
   c.) Reassemble the anchor washer and nut and position nut so top is flush with threads of anchor.
2.) Use the anchor seating tool to hammer anchors into the holes.
3.) Adjust all eight (8) anchor bolts until tight.

Figure 1-39 Table Base Anchor Assembly

Figure 1-40 Center tapped holes under mounting holes in table base



4.17 Alignment Recheck

Note: Alignment is critical. Recheck carefully.

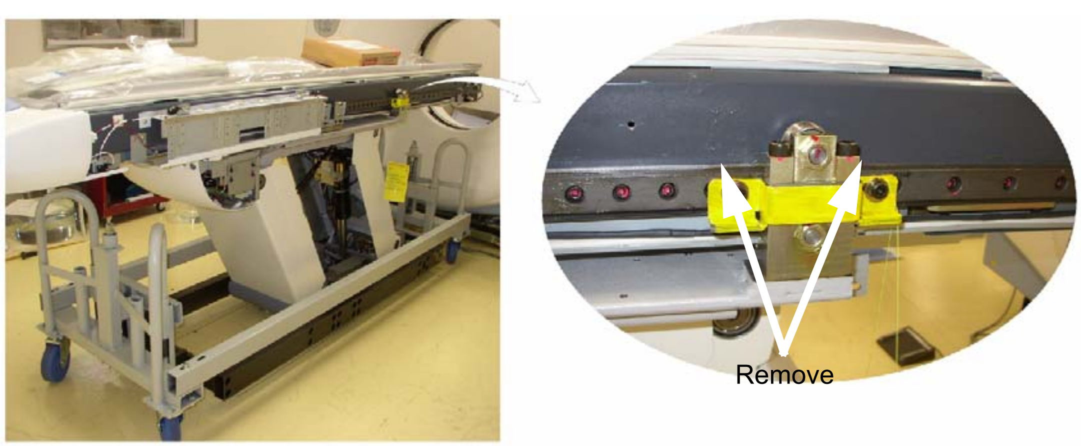
1.) Turn on the alignment tool and recheck alignments. The table alignment must be the same as in Cradle/Table Parallel Check on page 57. If re-leveling is required, repeat this procedure. Using the bubble levels, make adjustments as required to maintain required alignment.
2.) Once alignment has been verified, torque all mounting bolts.
Tighten the location #1 through #7 anchors and torque to 75 ± 6 N-m (55 ± 5 ft.-lb.)
3.) Remove the laser tools.
4.) Reinstall all the removed table panels and hardware.
5.) Reinstall the gantry rear cover.

Note: If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers by half-turn increments to raise the table/gantry until the lower table covers clear the floor. Then return to the alignment sections to level the gantry, level the table, and tighten the locking rings, respectively.

4.18 Remove the IMS Shipping Lock

1.) Remove the right top cover of the table.
2.) A yellow shipping lock is located on the right side of the table.
3.) Unscrew the two bolts and remove the shipping lock.

Figure 1-41 IMS Shipping Lock



4.19 Removing Table Shipping Dollies

4.19.1 Requirements

4.19.1.1 Tools Required

- Standard Install Tool Kit
- 10mm Hex Socket Bit

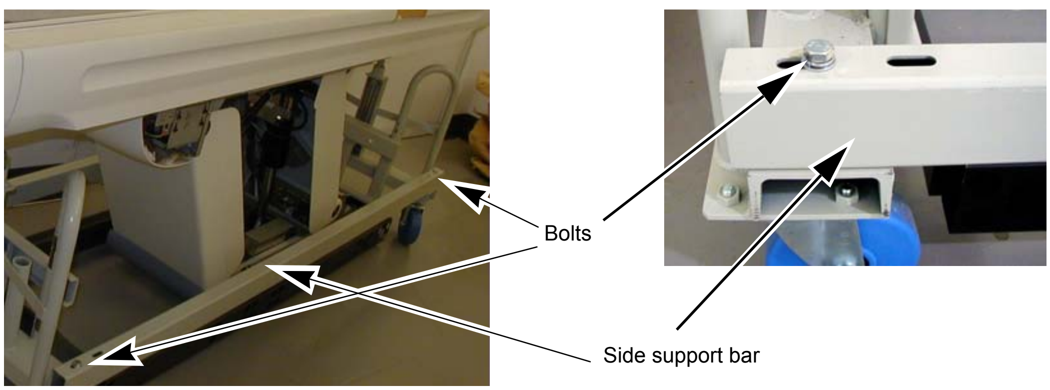
4.19.1.2 Time and Personnel

- .5 hour labor on site
- 1 Engineer

4.19.2 Procedure

1.) Remove the white table dolly support bars on the top of the dolly from each side. Refer to Figure 1-42.

Figure 1-42 Dolly Bolts



2.) Determine which direction is easiest for removing the dolly from the room. We will remove the dolly shipping rail so that the dolly can be rolled out of the room.

3.) There are two table shipping bolts on each end of the black shipping rails that hold the dolly together. Choose the rail on the opposite side of the direction that you plan to use to move the dolly out of the room.

4.) Using a 10mm hex socket, remove the two bolts on each end of the dolly frame. Refer to Figure 1-43.

Figure 1-43 Dolly Bolts

5.) On the long black side rail, there are two 14mm bolts holding the table to the dolly on each side: one near the back of the table base and one in the front of the table base. Refer to Figure 1-44 and Figure 1-45.

Figure 1-44 Dolly Side Panels - Back Bolt

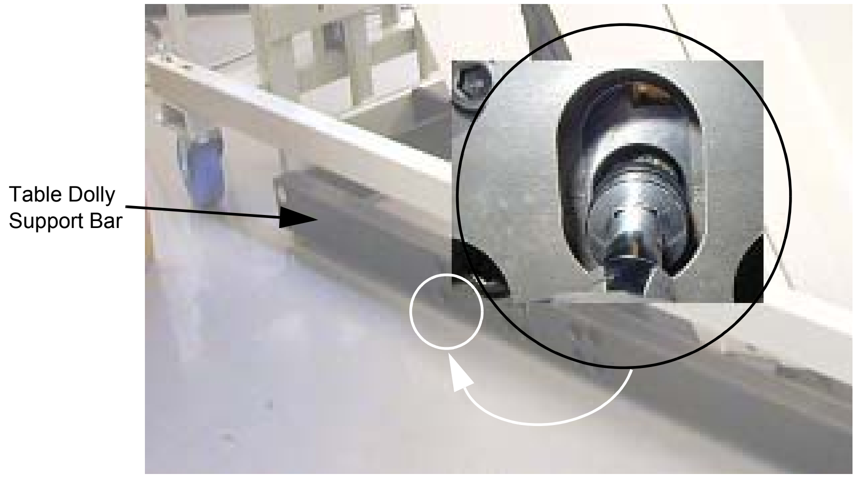
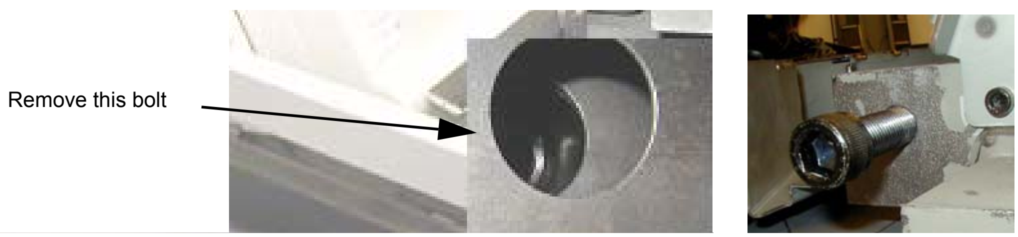


Figure 1-45 Dolly Side Panels - Front Bolt



6.) Roll the dolly away from the table.

7.) Remove the remaining side rail of the dolly from the other side of the table following step 5.

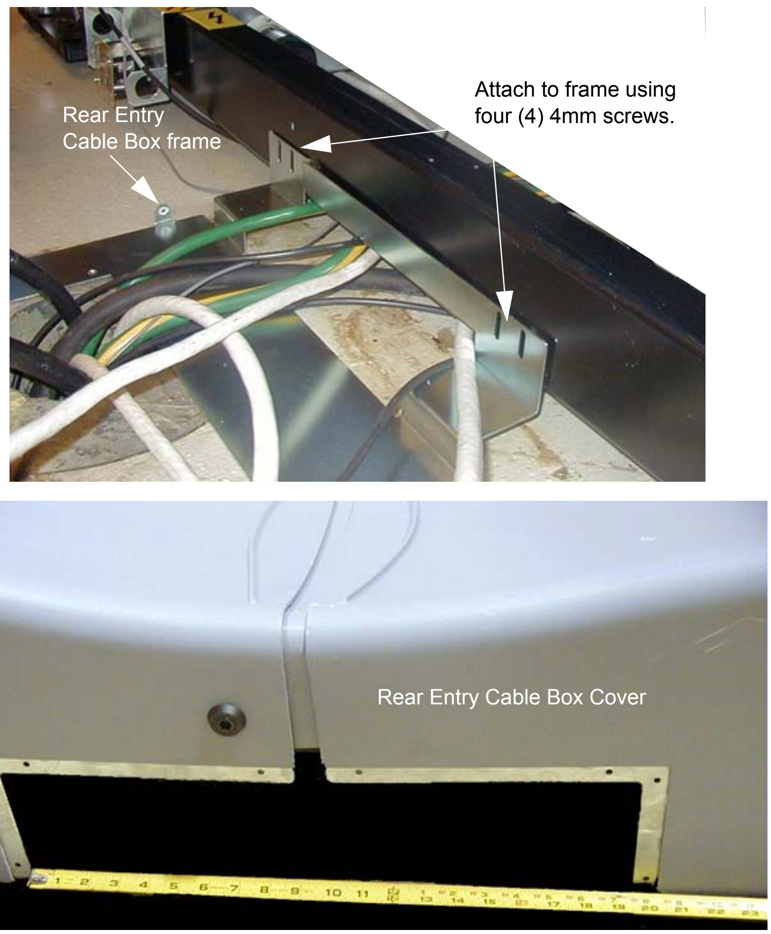
8.) Reassemble the dolly.

Section 5.0
Rear Entry Cable Box

A rear entry cable box (B7850RC) is used when the cables to the gantry cannot be brought up inside the gantry base.The box is not supplied with the system and must be ordered separately.

1.) Attach the rear entry cable box frame to the gantry base using four (4) screws that are shipped with the kit. See Figure 1-46. The assembly can be made to fit floor entrance conduit or surface floor duct.

Figure 1-46 Rear Entry Cable Box



2.) There are three pairs of spacers shipped with this cover. Select the pair that is most appropriate for this site, based on the hardware.

- Solid metal
- Precut L-shaped metal
- Solid plastic - Can be cut

Section 6.0
Install Table Footswitch Assembly

6.1 Requirements

6.1.1 Tools Required

- Standard Install Tool Kit

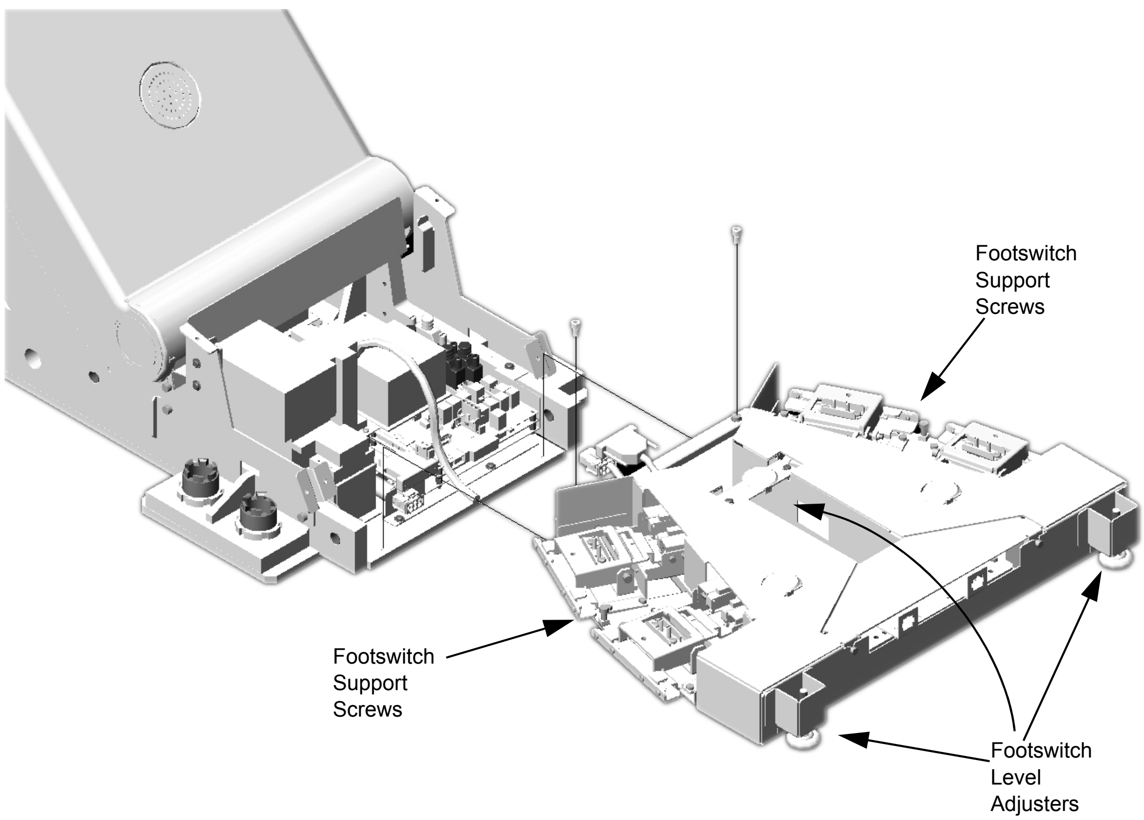
6.1.2 Time and Personnel

- 1 hour labor on site
- 1 Engineer

6.2 Procedure

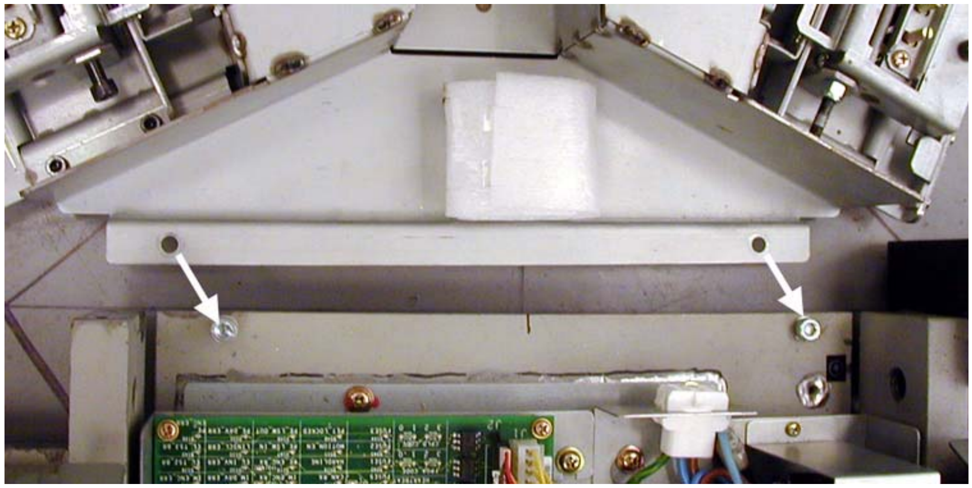
After Table positioning is completed and anchors installed, install the footswitch assembly as shown in Figure 1-47 following the steps below.

Figure 1-47 Footswitch Assembly Installation



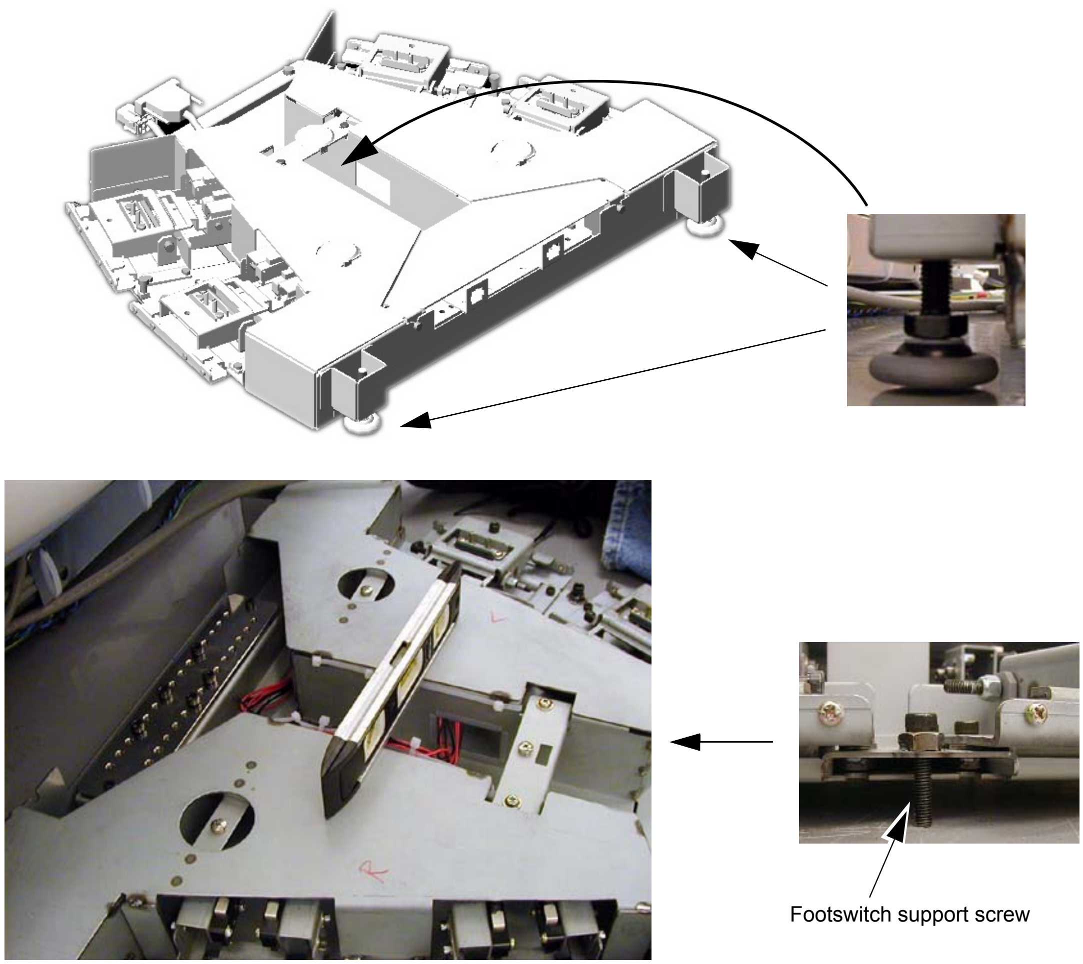
1.) Using two (2) M6 bolts, attach the footswitch assembly to the Table base.

Figure 1-48 Attach Footswitch



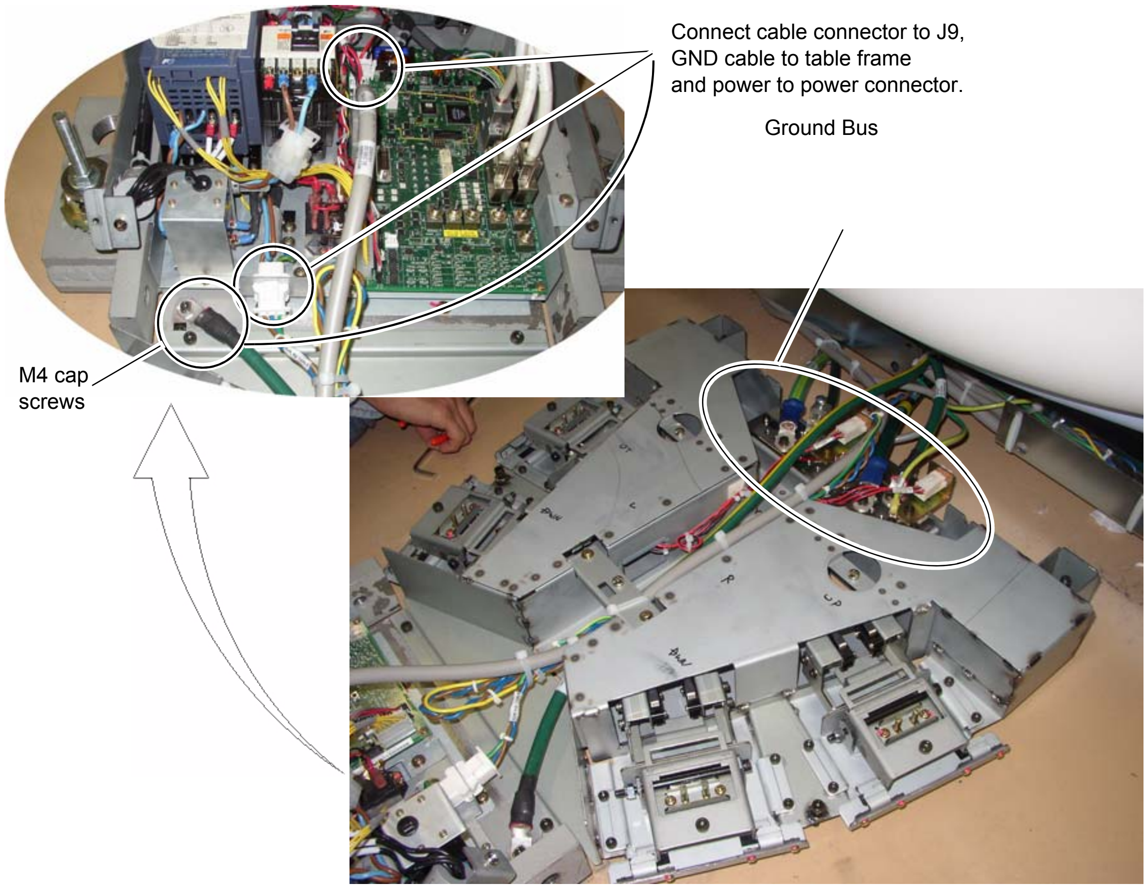
2.) Level the footswitch assembly using the three (3) level adjusters. Two are on the gantry side and one is in the middle. Use a 9" level to check the level in all directions.

Figure 1-49 Level Footswitch



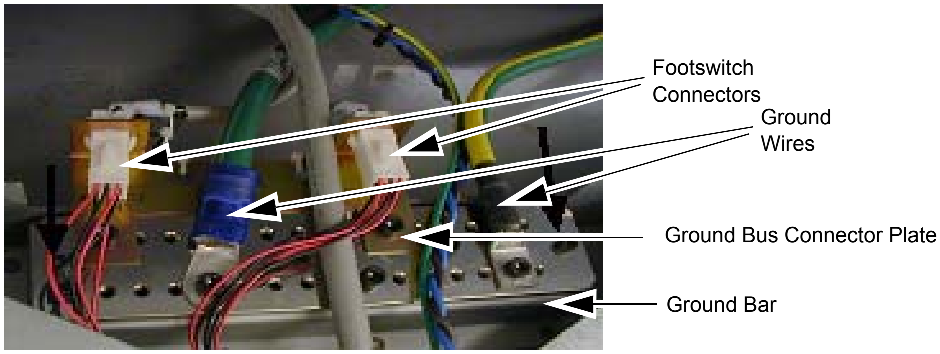
3.) Route the power cables from the gantry as shown in Figure 1-50.

Figure 1-50 Footswitch Assembly Cable Wiring



4.) Connect the ground bus connector plate.

Figure 1-51 Footswitch Ground/Bus Bar



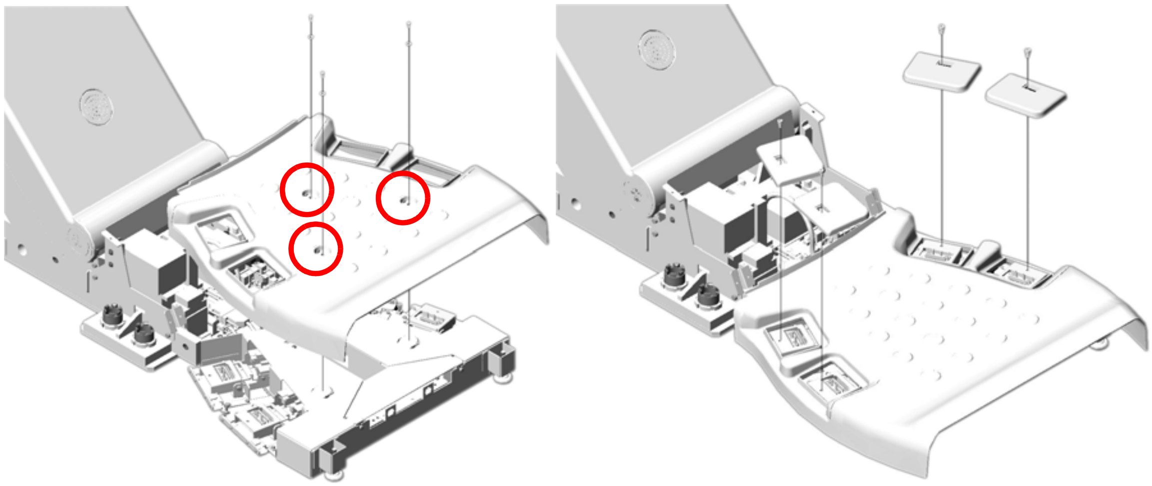
5.) Connect the ground wires (not all shown in Figure 1-51) to the installed ground bus:

Table	#2
Gantry	#1/0
Console	#2
PDU	#1/0
Power Pan	#10

6.) Install the footswitch pedal bracket onto the installed ground bus bar.

7.) Install the footswitch cover using three (3) screws (see Figure 1-52).

Figure 1-52 Footswitch Cover Installation



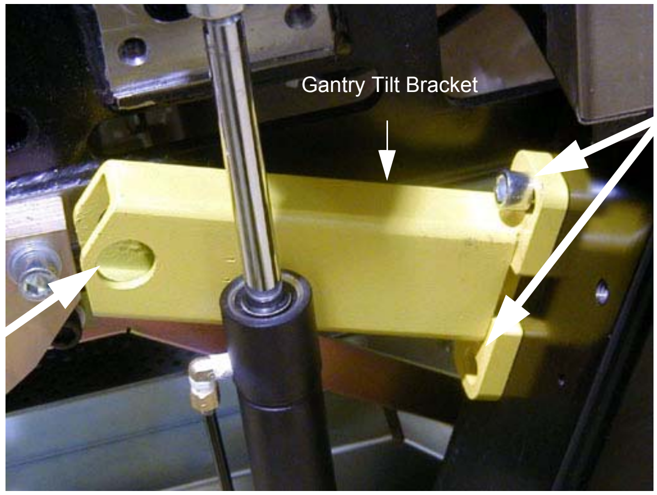
8.) Install cover caps on each pad.

Figure 1-53 Footswitch Pad Caps

9.) Install the four (4) foot pads onto the footswitch assembly.

Section 7.0
Remove Gantry Tilt Bracket

Figure 1-54 Gantry Tilt Bracket Removal



1.) Refer to Figure 1-54. Remove the M12 bolts using a 10 mm Hex wrench.

2.) Loosen the M16 bolt 1-2 turns and check the Gantry tilt bracket, it should be loose to the touch. If loose continue with step 4.

CAUTION

Potential for personal injury.

If there is a load on the tilt bracket, removal may cause the gantry to suddenly tilt all the way back due to a possible lack of hydraulic pressure.

If tilt bracket is not loose, stop and put the M12 bolts back in and tighten tilt bracket back in place.

3.) Check the hydraulic connections for leaks or lack of fluid. You will have to wait until the system can be energized to use the tilt controls to relieve the load on the tilt bracket prior to removal. Do not use force to remove the bracket.

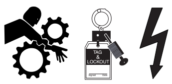
4.) If the bracket feels loose, remove the M16 bolt using a 14 mm Hex wrench.

5.) Remove the bracket.

6.) Reinstall the gantry rear covers and the scan window. (Refer to Appendix A Removal & Installation of Covers, on page 123.)

7.) Store brackets in the service cabinet.

Section 8.0
Position the Power Distribution Unit



WARNING **LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK. USE THE SUPPLIED LOTO KIT.**

1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6") between the PDU and back wall to allow cooling air to circulate.

Note: **Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer's electrical contractor.**

Table 1-6 Contractor Connections

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
TS1	#1	PDB-A	TS1-1	
	#1	PDB-B	TS2-2	
	#1	PDB-C	TS3-3	
	#1/0	GND	N/G **(Do NOT connect anything to neutral point.)**	

2.) Run the main input power conductors and ground though flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.

Figure 1-55 Flexible Conduit for PDU Power

3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.

4.) Cut the three phase and 1/0 ground wire to size.

5.) Attach cables as shown in Figure 1-55 and Figure 1-56.

Figure 1-56 PDU Area Locations

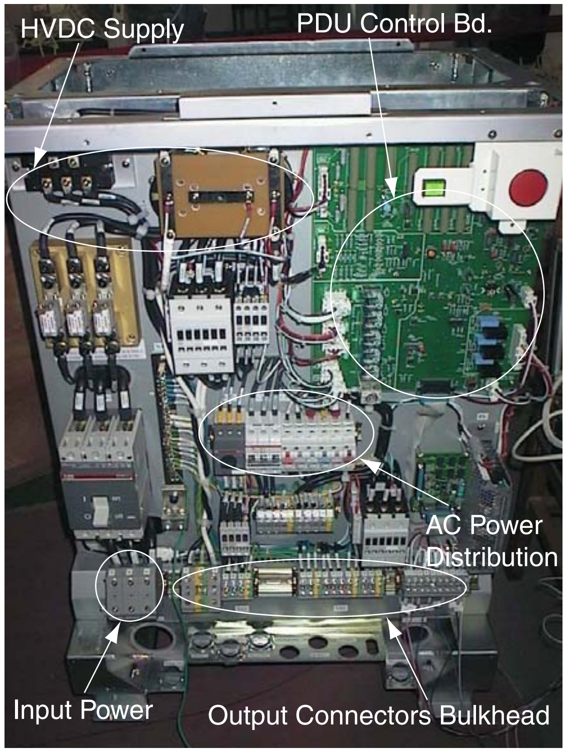
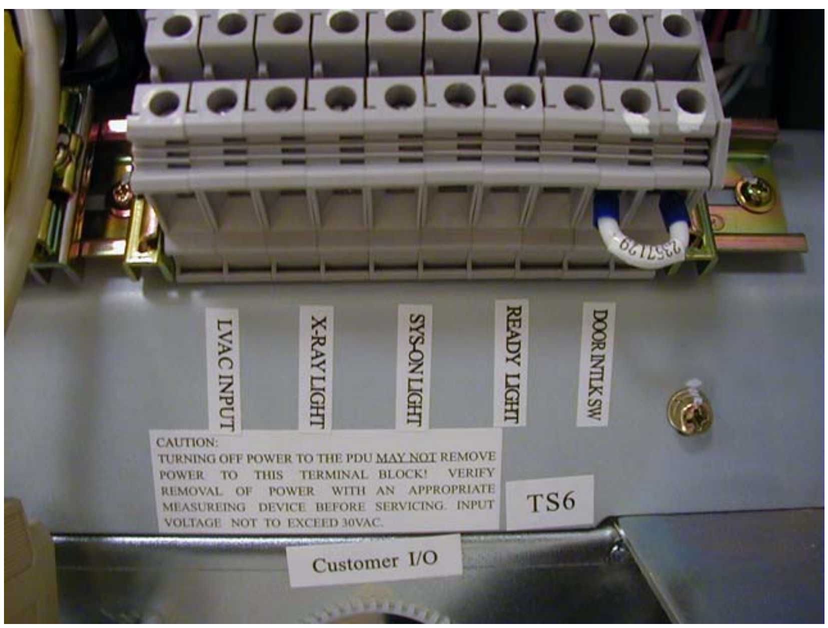


Figure 1-57 PDU Power Connections



Section 9.0
Install Operator Console

9.1 Unpack Console

Figure 1-58 Console boxed on skid

1.) Remove all items from the console.
2.) Remove all packing materials and discard.
3.) Place the step-board under the front edge of the skid and step on it to raise the front edge of the skid as in Figure 1-59.

Figure 1-59 Step-board used to raise front edge of skid

4.) Remove the two front cushions from the bottom of the skid. Refer to Figure 1-60

Figure 1-60 Cushion on bottom of skid

5.) Remove the Seismic Brackets from each side of the console. Refer to Figure 1-61. Save brackets for use later if you need to mount the console to the floor.

Figure 1-61 Removing the Seismic Brackets

6.) Lift up on the strap on the front of the step-board (Figure 1-59) to lower the skid. Remove the step-board.

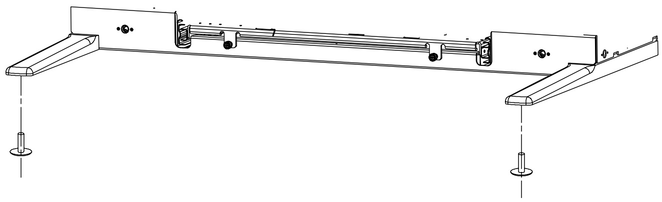
7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This will allow enough clearance to smoothly roll the console down the ramps. Refer to Figure 1-62.

Figure 1-62 Console ready to be unloaded

8.) Move console to installation location.

9.) Attach the two pads to the console stabilizers. Refer to Figure 1-63.

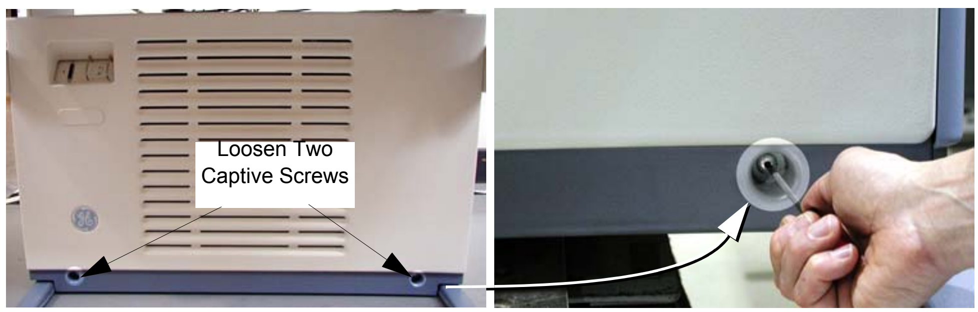
Figure 1-63 Console Stabilizer Pads



10.) Screw the pads all the way into the legs

9.2 Remove Console Covers

Figure 1-64 Front Cover Removal - Global Consoles



1.) From the service desktop, shut down the system
2.) Turn off console power.
3.) Loosen two captive screws at bottom of console.
4.) Rotate bottom of cover outward and upward until it may be lifted free of the console at the top.

9.3 Adjust Table Top Height and Position Console

1.) Refer to Figure 1-65. The console normally arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom.

Figure 1-65 Global Console tabletop adjustments

2.) For optimum operator comfort, align the bottom hole of the keyboard table top with the fifth hole from the bottom on the console.

Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.

- Always select a table top height that permits the operator to see the patient on the table.
- Keep the one bolt-hole relationship between the monitor top and the keyboard table top. Fasten the keyboard table top one (1) hole lower than the console table top (Figure 1-65).

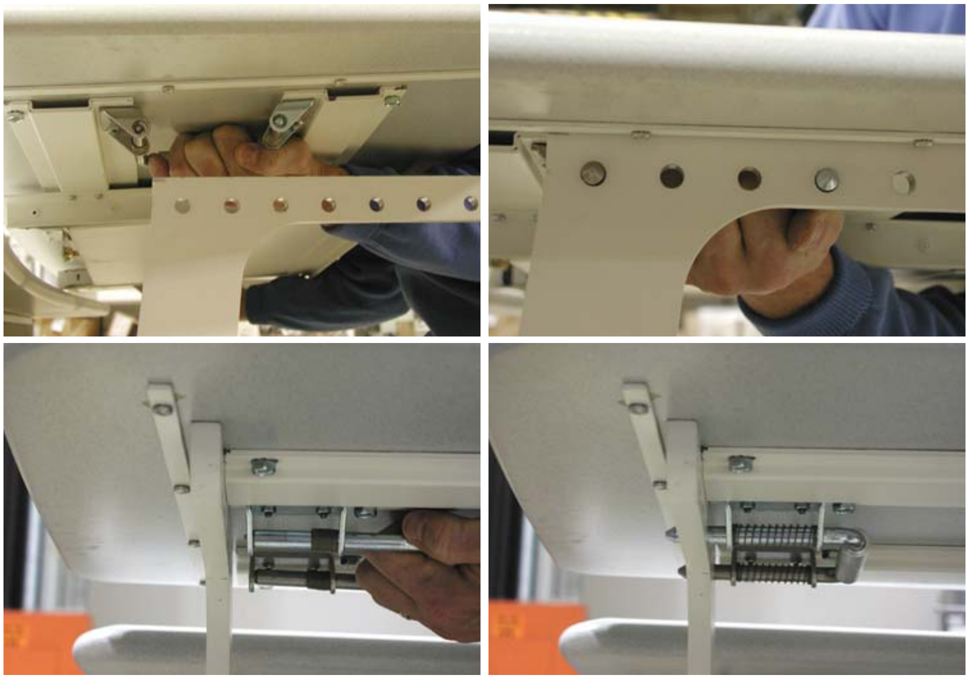
3.) Install the console side covers.

Side covers are left and right specific.

4.) Move the console into its final position, maintaining a 6" (15 cm) minimum distance between the console and the wall.

9.4 Attach Keyboard Table Top

Figure 1-66 Keyboard Table Top Placement



1.) Locate the spring loaded latches at either end of the underside of the keyboard table top.
2.) Pull latches inward (toward center of table top).
3.) Position table on brackets so that latches align with holes.
4.) Release latches, making certain that the pins fully engage the bracket holes (pins should protrude beyond outside edge of bracket).

9.5 Monitor and SCSI Tower Placement

1.) Locate and unpack the two monitors.
2.) Carefully place the monitors on the console desktop.
3.) Locate and unpack the SCSI drive tower.
4.) Place the SCSI drive tower on the console desktop.

Figure 1-67 Global Console tabletop component placement

Section 10.0
Seismic Mounting

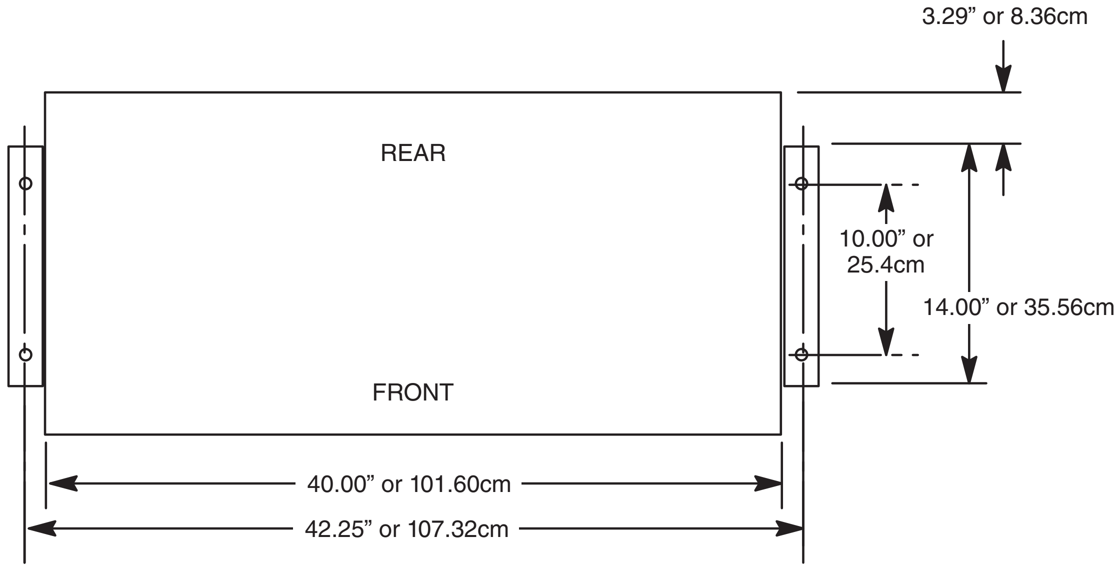
Before proceeding with seismic mounting for any of the components in this section, be sure to allow sufficient space to unbolt and move the component form its mounted location for service.

- All four mounting bolts may need to be removed.
- If removing the component requires lifting, use an appropriate-sized pry bar to lift each corner of the component.
- Two installers may be required to safely complete the task.

10.1 Console

If site specifications require seismic mounting to the floor, use 1/2 in bolts, or those supplied in the seismic mounting kit, to mount the seismic brackets that shipped with the console. Refer to Figure 1-68 for mounting hole locations, and mount the console so it can be easily removed for service.

Figure 1-68 Seismic Console Mounting Hole Locations

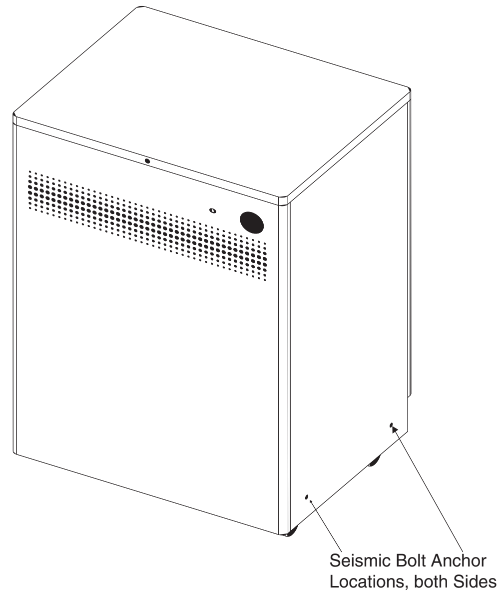


10.2 NGPDU

CAUTION The PDU is very heavy and may present a crush hazard if proper precaution and tools are not used.

If site specifications require seismic mounting, use 1/2 in. bolts, or those supplied in the seismic mounting kit, and the seismic brackets and anchors that shipped with the NGPDU. Refer to Figure 1-69 for mounting hole locations, and mount the PDU so it can be easily removed for service.

Figure 1-69 Seismic PDU Mounting Hole Locations



10.3 Uninterruptible Power Supply (UPS)

CAUTION The UPS is very heavy and may present a crush hazard if proper precaution and tools are not used.

If the site has a UPS (shown in Figure 1-70), refer to the Discovery ST/LightSpeed (7.X) VCT & Pro32 3 Phase UPS Installation manual, Direction 5116426-100. If site specifications require seismic mounting, use 1/2 in. bolts, or those supplied in the seismic mounting kit, and the seismic brackets that shipped with the UPS. Refer to the manufacturer's installation instructions to locate the seismic mounting holes, and mount the UPS so it is easily removed for service.

Figure 1-70 UPS

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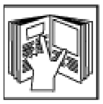
Chapter 2
Power, Ground & Interconnect Cables

NOTICE



Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

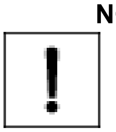
- **Record data collected from procedures in this chapter into Form F4879 when directed, located in Chapter 6 of this book.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

Section 1.0
Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout determines the system cable sequence. If your site has floor or wall ducts that will interfere with placement of the table/gantry, it may be important to have the movers unload the cable boxes (8 & 9) first and run those cables while others unload the subsystems.

- Try to run the system cables after the contractor completes the contractor supplied wiring.
- All ground wires and other contractor wiring should be complete to the point of equipment placement.

NOTICE

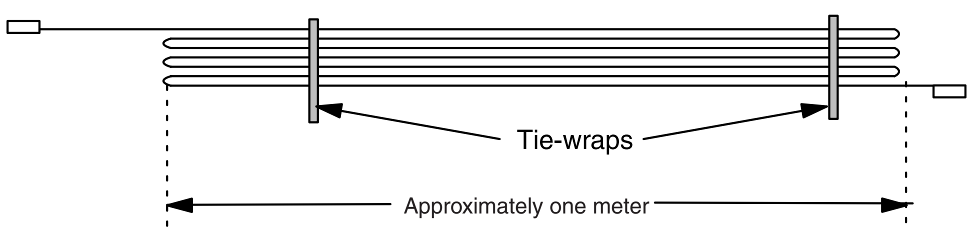


Potential for Equipment Damage

Do not store excess cable in the bottom of the PDU or Gantry.

Do not store excess cable behind or under the installed components (table, PDU, gantry or console). Check with the site electrical contractor, before hiding excess in conduits or cable ducts.

Figure 2-1 Excess Cable Storage Configuration



- Keep signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.
- Check all connections for tightness.
  - Use suitable tools and judgment.
  - Check all visible connections, especially ground connections.
- Check for reasonable cable routing.
  - Take into consideration necessary take-up distances for equipment maintenance, etc.
  - Try to complete as neat a job as possible.

1.1 System Component Identification

Identify all system cables by the system component designators listed in Table 2-1 on page 86. Each end of a system cable has a label, and may have a color near the connector, (refer to Table 2-2 on page 86) to indicate the component and the jack identifier of the component.

Table 2-1 System Component Identifiers

DESIGNATOR	SYSTEM COMPONENT
CT2	Gantry
CT1	Patient Table
PM	Power Distribution Unit
OC	Operator Console (Console Computer)
WL	X-Ray ON Warning Light

1.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. Table 2-2 on page 86 lists the subcomponent, and corresponding color.

Table 2-2 Cable Color Identifiers

SUBCOMPONENT	COLOR
Gantry	Blue
Table	Yellow
PDU	Red
Console Computer	Orange

Table 2-3 LightSpeed 7.X System Interconnect Cables

RUN NO.	DESCRIPTION	PART NUMBER	
		LONG CABLES (KIT 2281840-4)	SHORT CABLES (KIT 2281840-5)
1	Facility MDP to Room Disconnect (A1)	cust. supplied	cust. supplied
2	Room Disconnect (A1) to PDU	cust. supplied	cust. supplied
3	Room Disconnect (A1) to System E-Off	cust. supplied	cust. supplied
4	PDU to Room Warning Light(s)	cust. supplied	cust. supplied
5	PDU to Scan Room Door Switch	cust. supplied	cust. supplied
50	HVDC Power Cable - PDU to Gantry	2343529	2343529-2
51	HVAC Power Cable - PDU to Gantry	2343530	2343530-2
52	LVAC Power Cable - PDU to Gantry	2343528	2343528-2
53	LVAC Power Cable - PDU To Operator's Console	2343531	2343531-2
54	LVAC Power Cable - Gantry to Table	n/a	n/a
55	Ground, PDU to Raceway	2371450	2371450-2
56	Ground, Raceway to Console	2371450-3	2371450-4

Table 2-3 LightSpeed 7.X System Interconnect Cables (Continued)

RUN NO.	DESCRIPTION	PART NUMBER	
		LONG CABLES (KIT 2281840-4)	SHORT CABLES (KIT 2281840-5)
60	LVAC Power Cable - PDU to Optional UPS	-	-
61	LVAC Power Cable - UPS Disconnect Panel to PDU	-	-
90	LVAC Power Cable - PDU to PET	-	-
100	Signal Cable - Gantry to PDU	2333152	2333152-2
101	Signal Cable - Gantry to Console	2333150	2333150-2
102	Signal Cable (Ethernet) - Gantry to Console	2352714-2	2352714-3
103	Data Cable (Fiber Optic) - Gantry to Console	2117848-2	2117848-7
104	Signal Cable - Gantry to Table	n/a	n/a
110	Signal Cable - UPS Control to Room Disconnect (A1)	-	-
111	Signal Cable - UPS Control to UPS Disconnect Panel	-	-
112	Cardiac Monitor Cat 5 - Signal cable between console and Gantry Option Board (GOB)	2352714-2	-

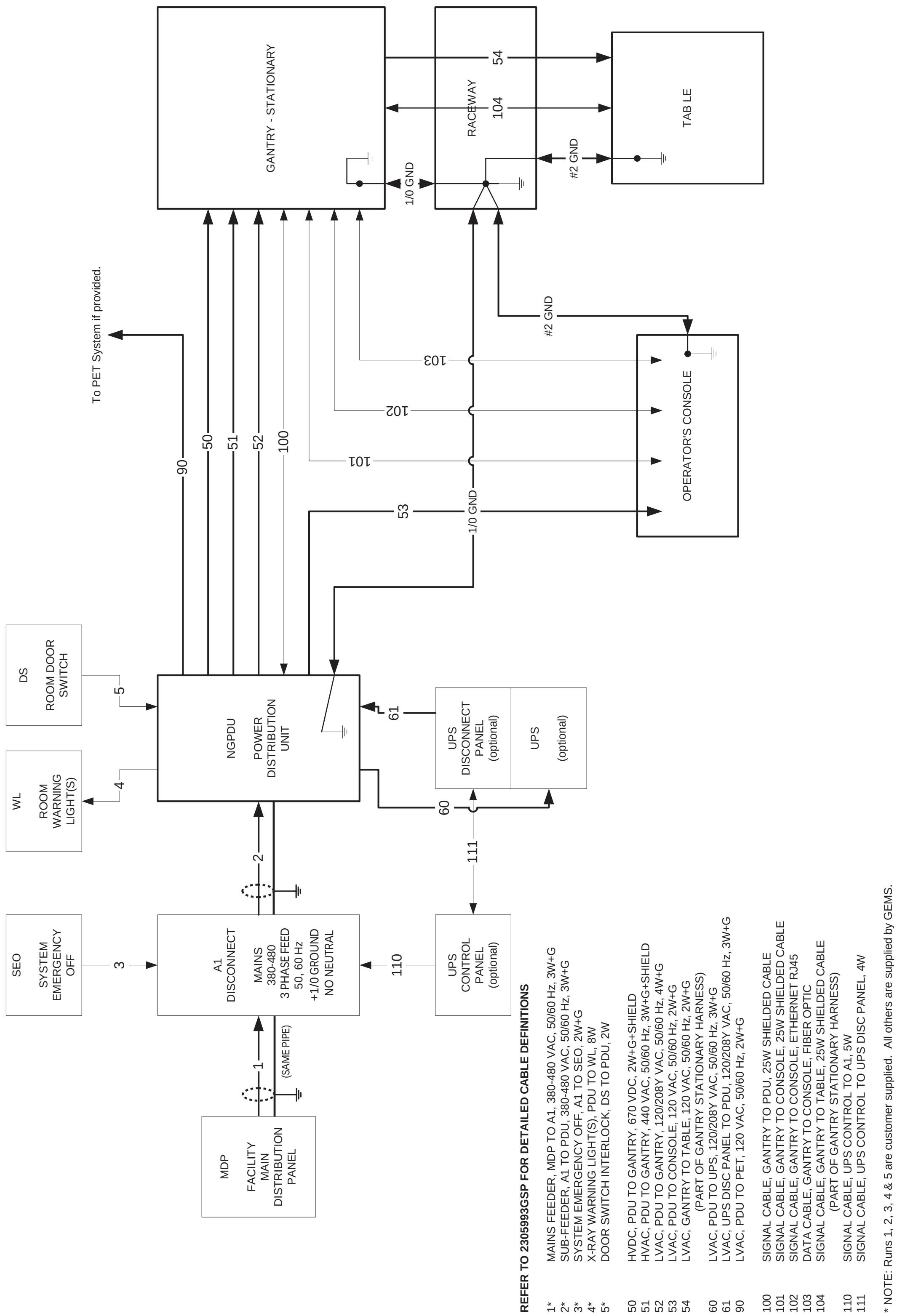
NOTICE **Shortening Power Cables is NOT recommended; especially if the appropriate crimping tool and ferrules are not employed.**

If longer or shorter cables are required, order the correct set.

NOTICE **Excess cables can not be stored under or behind the PDU, Gantry or Console.**

Section 2.0
System Interconnect Diagram

Figure 2-2 System Interconnect Diagram



Section 3.0
Contractor Connections

Table 2-4 Contractor PDU Connections

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1	Load - T1	TS-1 L1	
	#1	Load - T2	TS-1 L2	
	#1	Load - T3	TS-1 L3	
	#1/0	GND	TS-1 GND **(Do NOT connect anything to neutral point.)**	
WL (Warning light) *See Figure 2-30, on page 110.*	#14	LV Source -1	TS6 1	
	#14	LV Source -2	TS6 2	
	#14	X-Ray ON Light -1	TS6 3	
	#14	X-Ray ON Light -2	TS6 4	
	#14	Sys-ON Light -1	TS6 5	
	#14	Sys-ON Light -2	TS6 6	
	#14	Ready Light -1	TS6 7	
	#14	Ready Light -2	TS6 8	
DS (Scan Room Door Switch)	#14	Door SW-1	TS6 9	
	#14	Door SW-2	TS6 10	

Note:
Add #2 ground wire.

IMPORTANT: Add AWG #2 ground wire from table frame to table/gantry raceway ground bar (as shown in Figure 2-2).

WARNING

WORK WITH THE ELECTRICAL CONTRACTOR TO BE SURE EXTERNAL POWER SOURCE IS TURNED OFF.

☐ Check box when complete

Section 4.0
Console Connections

4.1 SCIM, Keyboard, Trackball & Mouse Installation

1.) Route the keyboard cable under the SCIM, as shown in Figure 2-3.

Figure 2-3 SCIM control with keyboard cable routed through SCIM

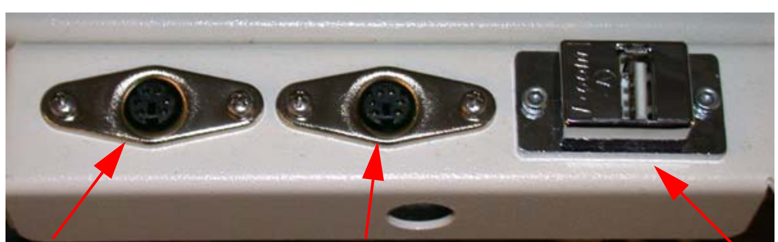
NOTICE

Potential for equipment damage

Never connect a mouse or keyboard with the host computer powered "ON". Doing so can destroy components within the host computer.

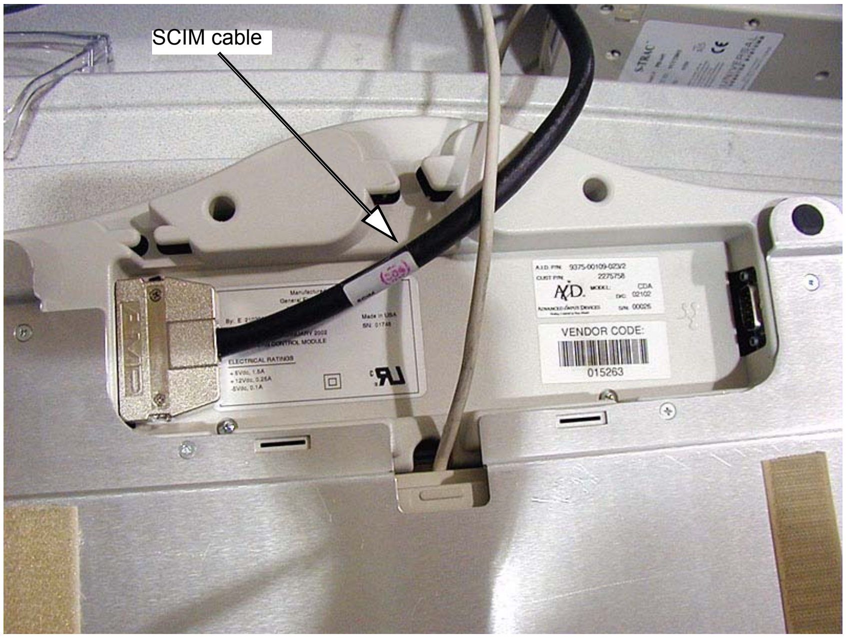
2.) Connect the keyboard and mouse to the ports on the top of the console front bulkhead. The keyboard connects to the port on the right; the mouse connects to the left port (Figure 2-4).

Figure 2-4 Global Console front bulkhead, showing keyboard and mouse connections



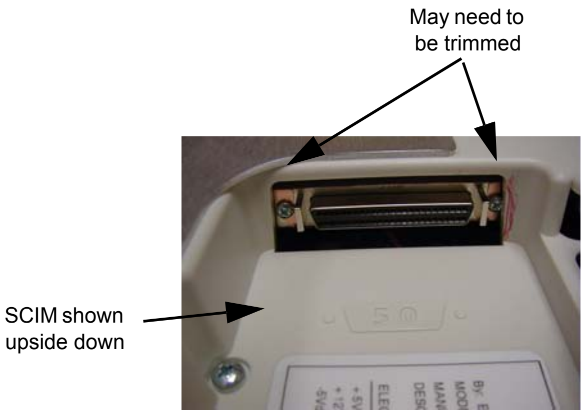
3.) Connect the SCIM cable to the SCIM as shown in Figure 2-5. (Note the cable routing.)

Figure 2-5 SCIM bottom, showing cables and keyboard mounting bracket



4.) Make sure the SCIM connector fits snugly. Some plastic molding may need to be removed to allow the cable to fit properly.

Figure 2-6 SCIM Connector Area



5.) Select and install the proper keyboard overlay for your system: (1) with tilt or (2) without tilt.

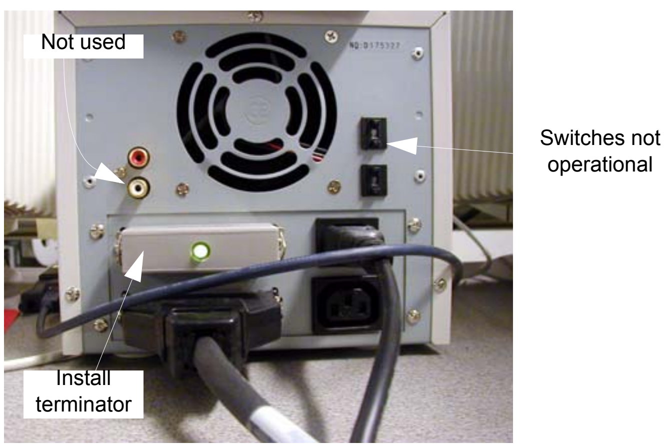
Verify that none of the buttons get caught and stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.

6.) The keyboard should attach to the SCIM using the supplied Velcro strip and fit snugly against the SCIM when finished, as shown in Figure 2-7.

Figure 2-7 SCIM connected to the keyboard with the US English tilt overlay installed

4.2 Connecting the SCSI Tower

Figure 2-8 SCSI tower: Connections (left) and Optional Seismic Mounting Bracket (right)



Check the box when each step is completed:

1.) ❒ Connect the SCSI cable to the rear of the SCSI tower.

Note: The cable is included in the SCSI Cable Kit and is the same as the DASM cable.

2.) ❒ Connect the SCSI terminator to the rear of the SCSI tower.

3.) ❒ Connect the power cable to the rear of the SCSI tower.

4.3 Connecting the LCD Monitor

With the monitor and computer switched off, connect the video signal cable to the monitor's video input.

NOTICE **Equipment Damage Possible**

Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, otherwise the pins might bend.

Connect the following cables:

SCIM:

- SCSI Cable
- Template
- Keyboard

Scan Monitor:

- Video Cable
- Power Cable
- Cable Keeper

Display Monitor:

- Video Cable
- Power Cable
- Cable Keeper

Note: A DARC video cable is shipped with the system. This is only to be connected when troubleshooting. Do not connect during normal installation.

4.4 Power Panel Connections

NOTICE **Console power is two-phase power. Outlet assignments are critical.**

1.) Connect the console power cable to the console power panel.

2.) Connect console component power cords as listed in Table 2-5. Plug LAN cables into the rear bulkhead on the console.

Table 2-5 Console Component Connections

VCT M3 LAN CABLES	SIGNAL CABLES	POWER CABLES
• Hospital Network	• J20 Cable	• 208 VAC Power
• Gantry Base LAN	• Dual Fiber DIP	• #2 Ground
• VCT Cardiac		

Note: The DIP signal cable is keyed to only fit one way.

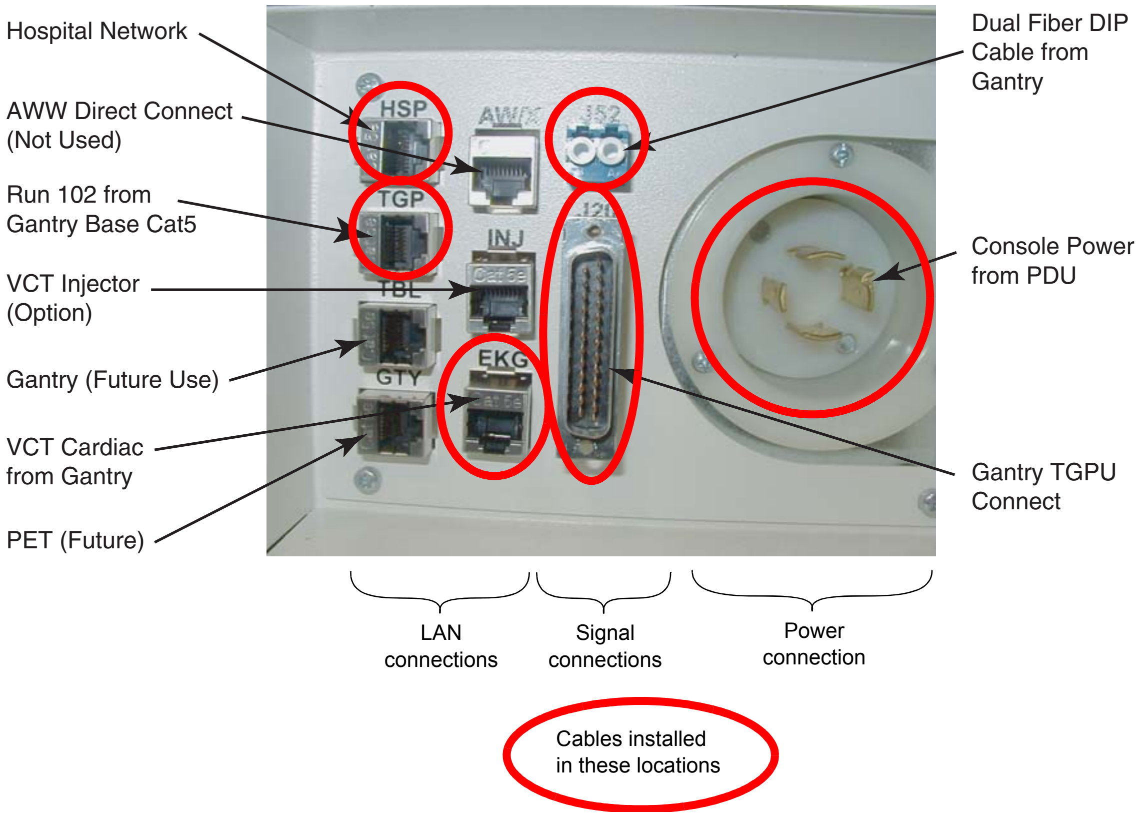
"J" numbers increment from top to bottom and left to right.

The console is shipped with all internal power cables connected to their assigned power slots.

4.5 Console Connections

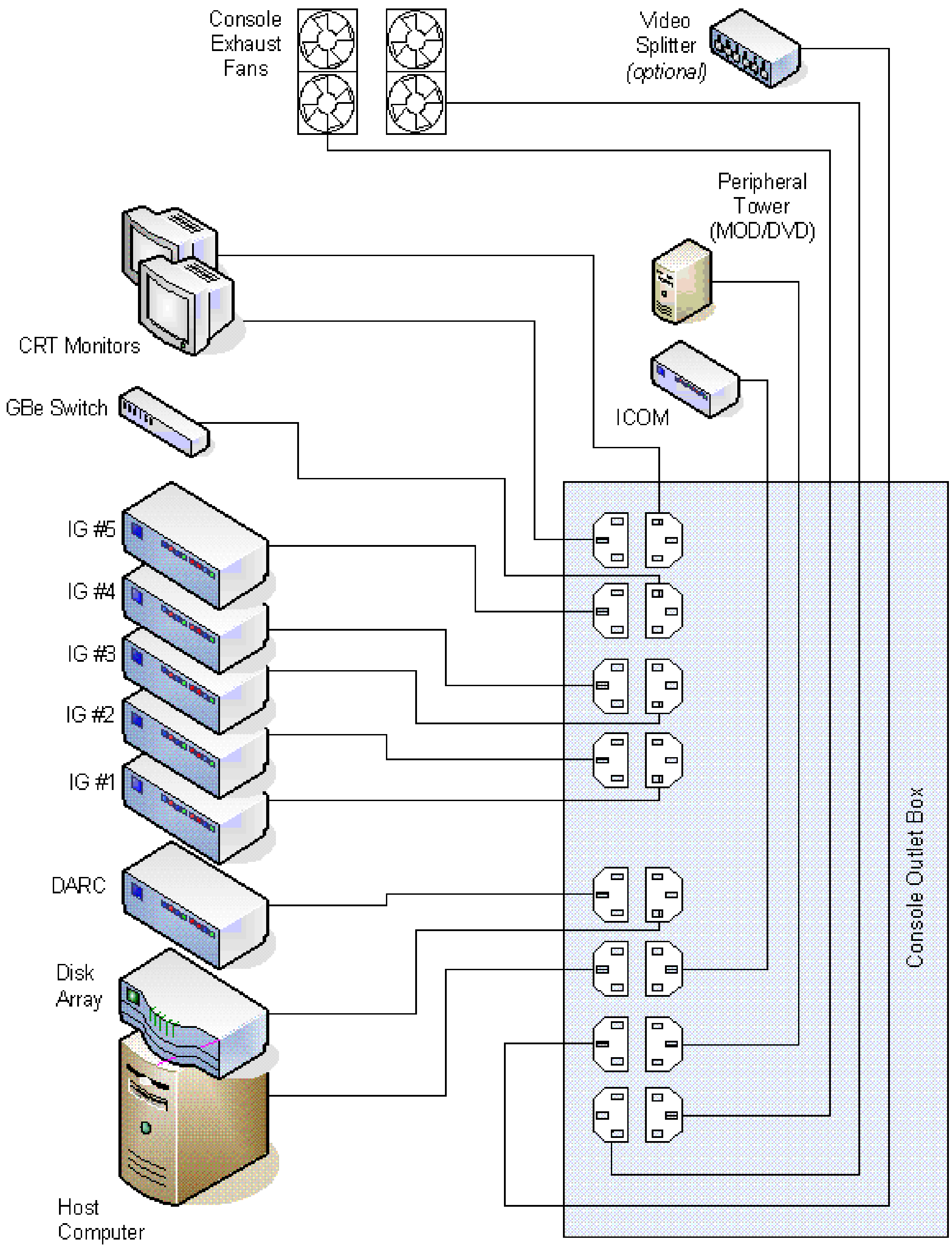
Plug cables into rear bulkhead on console.

Figure 2-9 Console Rear Bulkhead



3.) System cable connections are located on the rear of the console. Connect as shown in Figure 2-10 below.

Figure 2-10 Console Power Connections

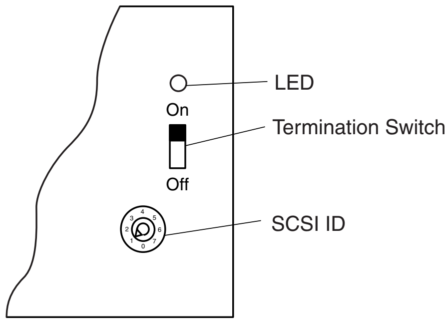


4.6 Installing a DASM (Hardware Option)

If you have a digital DASM to install, complete the following steps to connect it to the computer.

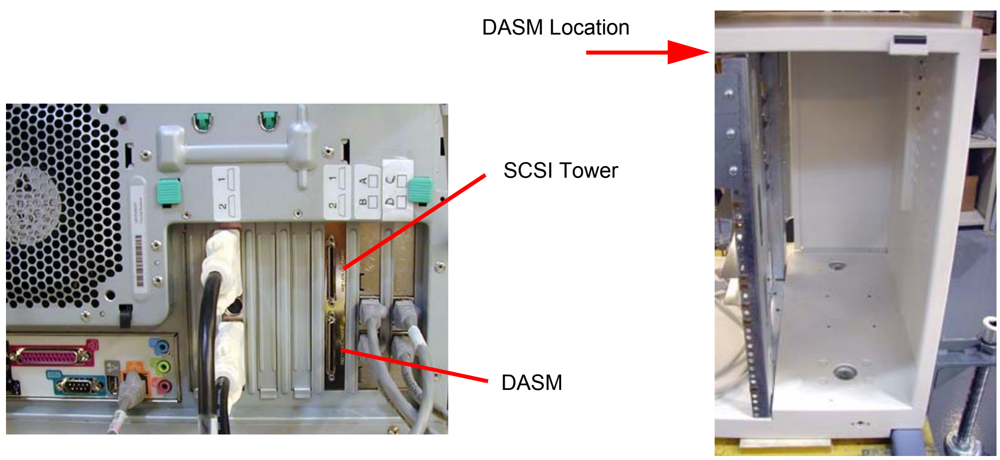
1.) Turn the DASM over so you are looking at the bottom.
2.) Set the DASM SCSI ID to 1.
3.) Set the DASM termination to ON.

Figure 2-11 DASM Bottom



4.) Set the RS232/RS422 switch to the RS232 position, for a digital DASM.
5.) Refer to Figure 2-12. Attach the SCSI cable supplied with DASM kit to the back of the DASM.

Figure 2-12 Rear view of HP xw800, showing DASM SCSI cable connection



6.) Attach the free end of the SCSI cable to the rear of the Host Computer (see Figure 2-12).
7.) Plug DASM power cord into J10 (or any open slot).
8.) Attach remaining cable(s) from camera to DASM and install the DASM to the right of the slide-out tray.

Note: See your *Software Load Procedures* manual for instructions on configuring cameras.

4.7 Modem Option

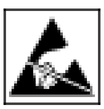
Note: A signed deviation is required to install the modem option.

If you have a modem to install, do it now. Place the modem on top of the intercom box and attach it with Velcro to the power supply.

Hook up the power, phone, and serial line as shown in the interconnect drawing. The drawing is located in the back cover of the console.

Figure 2-13 Modem

4.8 Install Drives

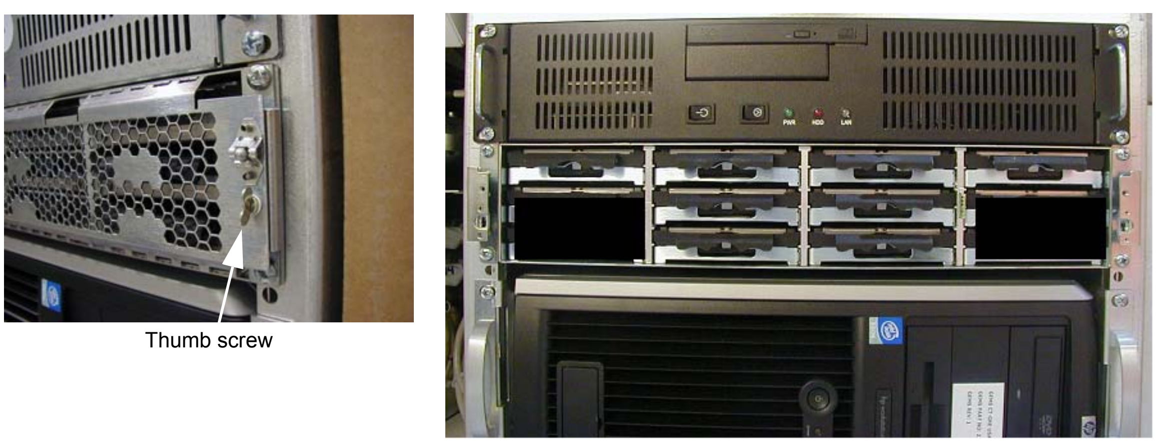


NOTICE **ESD can damage hard drives. Use proper ESD precautions when handling hard drives.**

Install drives in the eight (8) slots shown in Figure 2-14 as follows.

1.) Using the thumb screws, remove the front cover. Use caution so you do not damage the tabs on the top edge or the small plates on the bottom edge.
2.) Remove the cardboard shipping box with slots for eight drives on the right side of the console.
3.) Install eight drives in the open slots. They snap into place.
4.) Replace the box with the installed drives.
5.) Using the thumb screws, replace the front cover.

Figure 2-14 Hard Drive Installation



Section 5.0
Install Options

5.1 Install Optional Bar Code Reader

Follow the installation procedures in the Bar Code Reader box.

When finished neatly dress all cables.

5.2 Install Optional Remote Monitor

Follow the installation procedures in the Remote Monitor box.

5.3 Install Cardiac Respiratory Gating Option

Follow the instructions shipped with the option. The options board should be only mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

5.4 Install Respiratory Gating Option

Follow the instructions shipped with the option. The option is attached to the GOB, and should be only mounted on the non-motor side of the gantry. Neatly dress all cables along the gantry base so that the base covers fit properly.

5.5 Install Injector Option

Follow the instructions shipped with the option.

5.6 Install The IVY Monitor and Stand

Follow the instructions shipped with the Monitor and stand kit. Review the instructions carefully before assembling the stand and accessory basket to avoid repeated steps.

5.7 Customer Accessories (Head Holders and Extender)

Open the boxes and installed the appropriate language warning labels.

The head holders are shipped with shims installed to assure proper fit. Check that shims are included and a pair is installed. The holder should fit snugly.

5.8 Install Service Cabinet

The service cabinet you receive may ship disassembled. Assembly takes about 1.5 hours.

1.) Assemble the cabinet following the instructions located in the cabinet's shipping box.
2.) When you complete assembly, place the cabinet in the location shown on the site print.
3.) Place all service materials shipped with the system in the service cabinet.

5.9 Install UPS

Follow the instructions shipped with the UPS option. The option ships with two sets of instructions, a GE set and a Powerware set.

The GE set instructs you to install the connections between the UPS and the PDU and between the UPS and the A1.

Note: A GE A1 Disconnect with UPS controls is required for this option.

The Powerware set instructs you to internally connect the batteries and do a power-up check. Refer to both manuals for additional guidance.

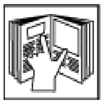
WARNING **LOCKOUT/TAGOUT IS REQUIRED WHEN WORKING IN THE A1 DISCONNECT.**

Section 6.0
Gantry Cable Connections

1.) Connect the gantry CAT 5 cable to the gantry base connector.

2.) Rear cable entry is standard for LightSpeed 7.X VCT. Install the rear cable entry box (B7850RC) now, before routing cables to gantry.

NOTICE



CABLE CONNECTION NOTICE: Potential for equipment damage.

- **Observe correct polarity when connecting the high voltage DC power. Reversing these leads will result in serious equipment damage.**
- **The HVDC positive conductors have red insulation and are labeled “ONE.” The HVDC negative conductors have black insulation and are labeled “TWO.” Lead “ONE” must be connected to lead “ONE,” and lead “TWO” must be connected to lead “TWO.”**
- **Observe correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected top to bottom.**

3.) Install the cables to the gantry power pan. The power pan is located on the rear of the gantry at its base. See Figure 2-15, and Figure 2-16 for connections.

Note: The gantry 120VAC cable may not fit under the gantry frame. Install this cable before gantry placement—or remove the power plug—to route it under the gantry. In doing so, be careful not to damage the cable or power plug.

Leave approximately 10’ of the dual channel fiber and gantry Cat 5 cables under the gantry frame for troubleshooting.

Figure 2-15 Gantry Power Pan Connections

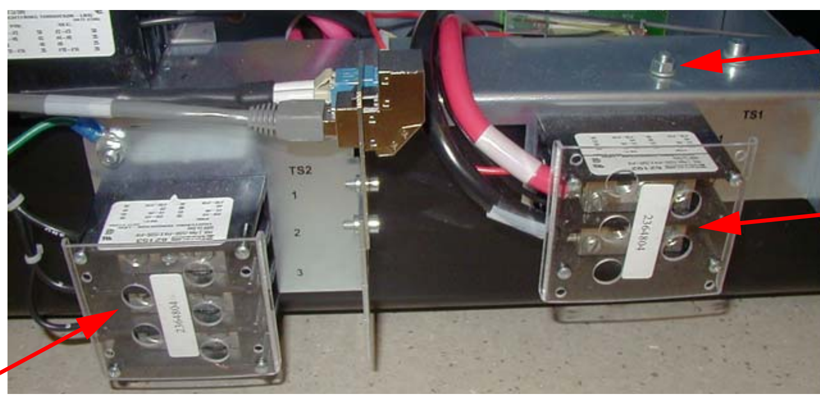
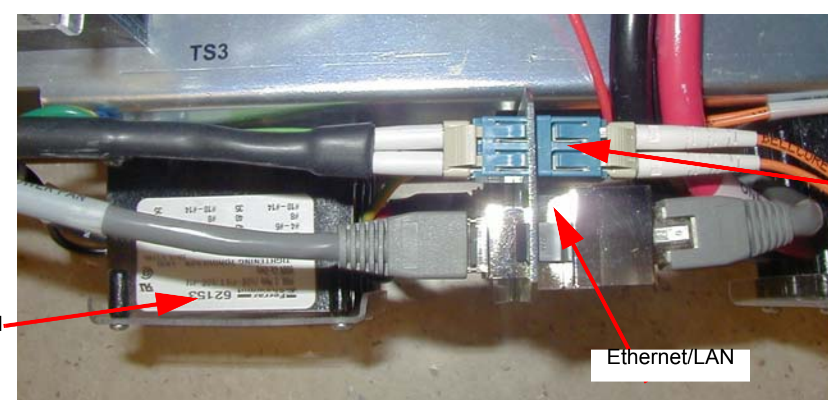


Figure 2-16 Close-up of Power Pan Bulkhead

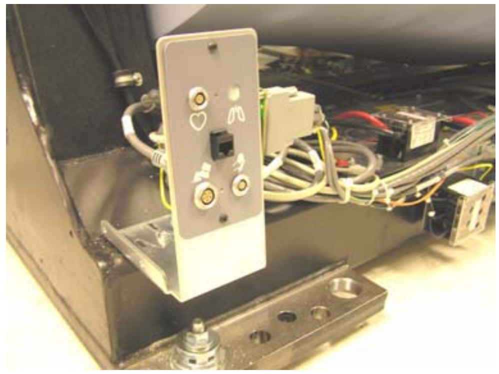


4.) If the system has a second CAT5 LAN cable for the cardiac option, run this cable between the console and the gantry.

5.) Connect this cable to the gantry base cardiac monitor base plate (GOB board). Different plates are included with the cardiac option kit.

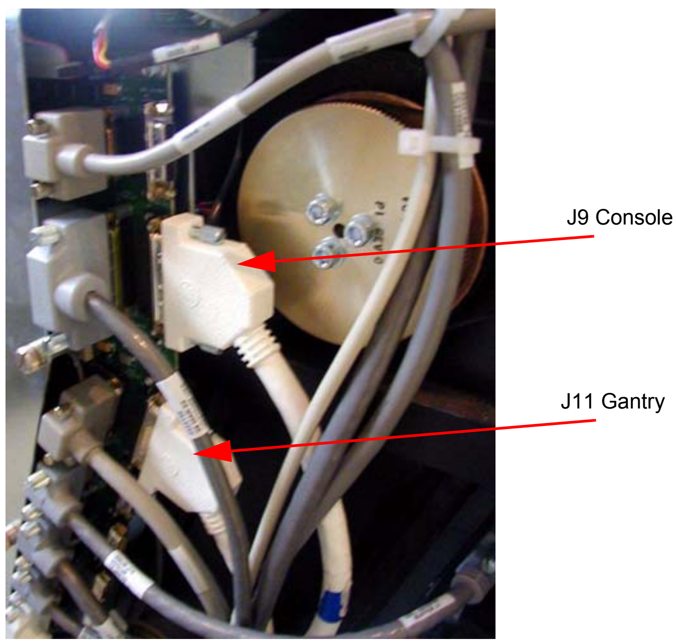
NOTICE **The GOB connectors are not soldered to the board. DO NOT use the cable to pull on the board.**

Figure 2-17 Cardiac Option Board



6.) Install cables to the gantry TGPU.

Figure 2-18 TGPU Connections



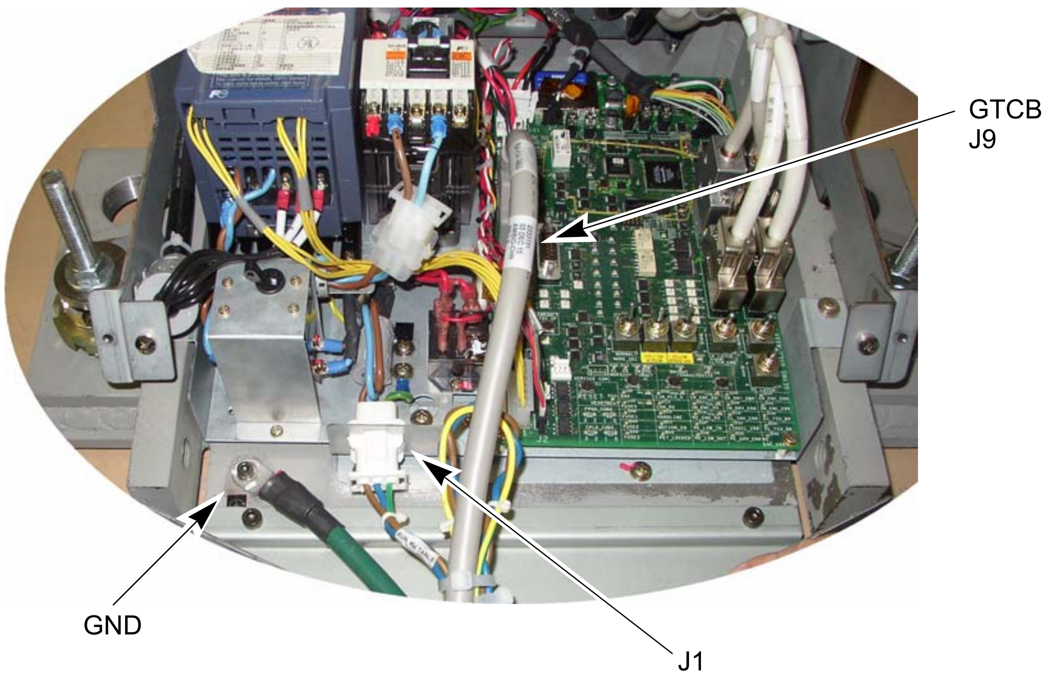
Section 7.0
Table Connections

Pull and connect the following cables:

Table 2-6 Table Cable Connections

TABLE	FROM	CABLE DESCRIPTION
J1 table power	Gantry	120 VAC
J9 table control	Gantry	Signal Cable
Table ground	Gantry	Table ground

Figure 2-19 Table Connections



Note: You may need to add the table ground cable.

☐ Check box when complete.

Section 8.0
PDU Cable Connections & Configuration

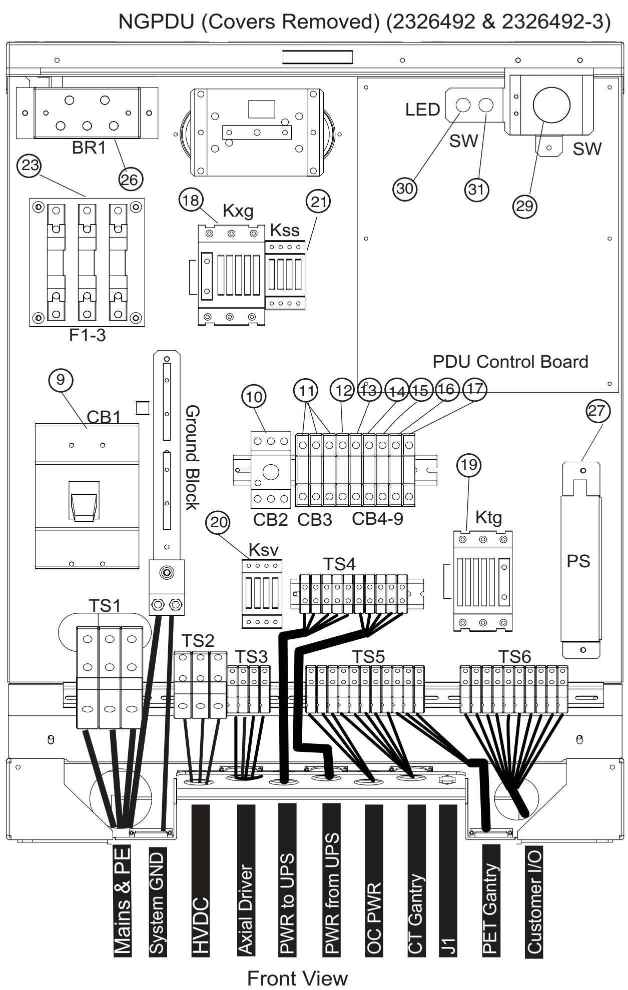
CAUTION **Do not work in an energized PDU. When working on the PDU, follow this simple rule: Always tag and lock out power to the PDU at the "main" disconnect. Failure to due so can result in electrocution or death.**

Do NOT apply power to the PDU until all work has been completed and all PDU covers are in their proper place.

8.1 Introduction to NGPDU

As seen in Figure 2-20, a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use Figure 2-20 for reference. The PDU has been designed to have cables routed into the PDU from behind and/or beneath it.

Figure 2-20 PDU Cable Connections - Front

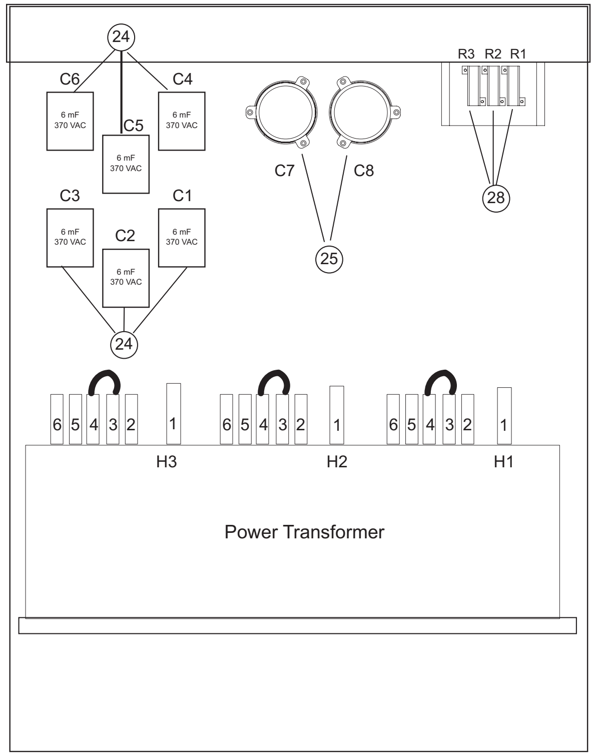


Note: Wire colors may vary

HVDC	TS2-1	Red
	TS2-2	Grn/Yellow
	TS2-3	Black
Axial	TS3-1	Black
	TS3-2	Red
	TS3-3	Orange
	TS3-4	Grn/Yellow
Console Power	TS5-0	Black
	TS5-1	Brown
	TS5-2	Blue
	TS5-3	Grn/Yellow
GT Gantry	TS5-4	Black
	TS5-5	Red
	TS5-6	Orange
	TS5-7	Blue
	TS5-8	Grn/Yellow

Figure 2-21 PDU Cable Connections - rear

NGPDU (COVERS REMOVED) (2326492 & 2326492-2)

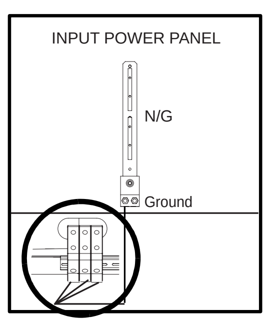


Rear View

8.2 Panel - 380 - 480VAC Mains "TS1" Input Power Connection

1.) Remove the TS3 panel front cover.
2.) Strip the wires to fit securely on the power block.
3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
4.) Insert "vault" ground into PDU "vault" ground lug.
5.) Tighten all fasteners securely and replace the TS3 front panel.

Figure 2-22 Input Power Panel Connections



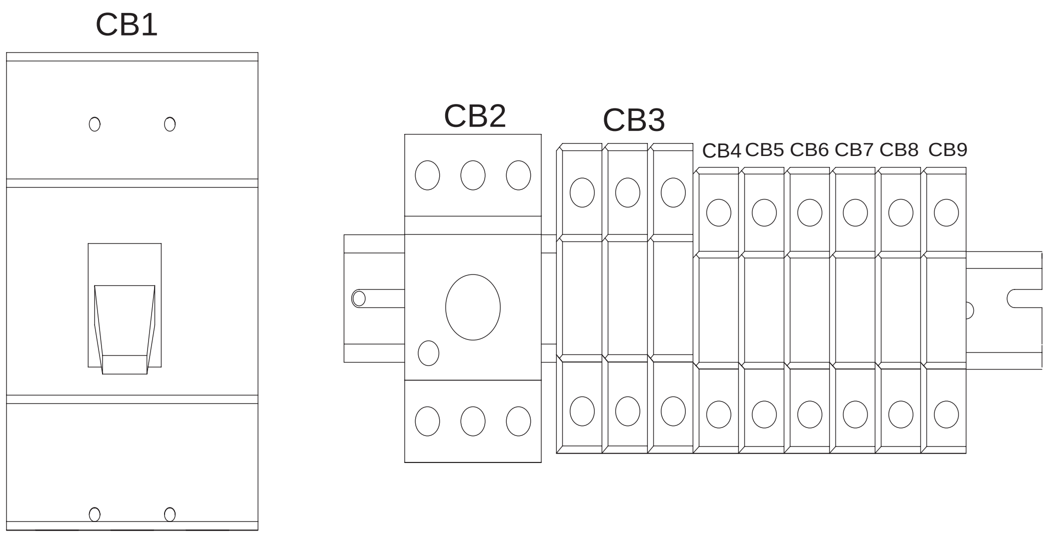
When Mains power is available to the PDU, the "TS1" power light will be illuminated. (See Figure 2-20.)

☐ Check box when complete.

8.3 Panel - Circuit Breakers

Place the circuit breakers in the "off/down" position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the "ON" positions.

Figure 2-23 Circuit Breaker Panel



By design, when CB3 is in the "OFF" position, circuit breakers 4, 5, 6, and 7 are switched "OFF". CB3 is essentially in series with these breakers.

Table 2-7 Panel Circuit Breaker Descriptions

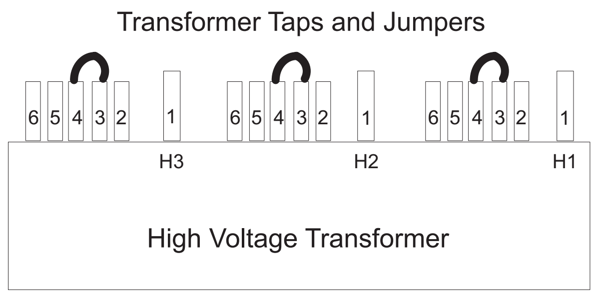
CIRCUIT BREAKER	DESCRIPTION
CB3	Fully Winding Protection (Master power of CB 4, 5, 6, and 7)
CB4	CT Gantry Service Outlets
CB5	CT Gantry rotating loads
CB6	Table & CT Gantry Station Loads
CB7	Operator Console
CB8	PET Gantry
CB9	NGPDU Control Power Supply

☐ Check box when complete.

8.4 Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory.The taps should be set as shown in Figure 2-24.

Figure 2-24 PDU Tap Positions for 480 volt operation

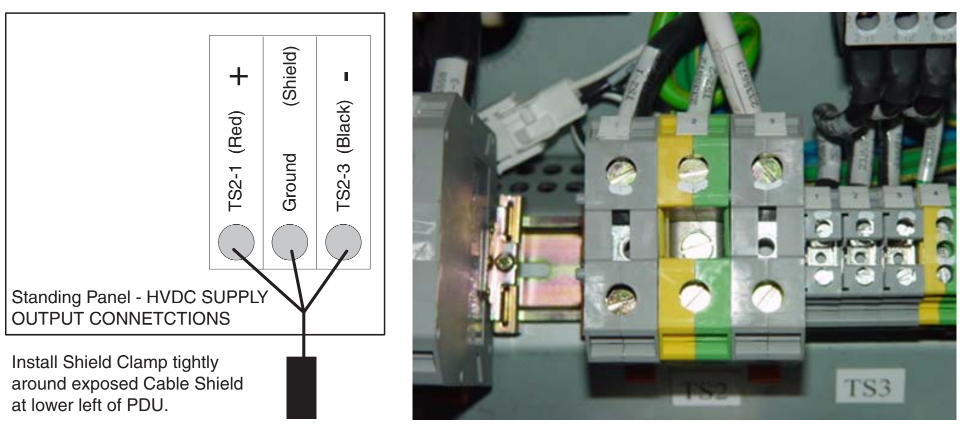


☐ Check box when complete.

8.5 HVDC Connection

Connect the internally shielded HVDC cable to TS2 on the standing panel. See Figure 2-20 for the location of the connector and Figure 2-25 for details. Observe polarities and grounds.

Figure 2-25 HVDC Connection

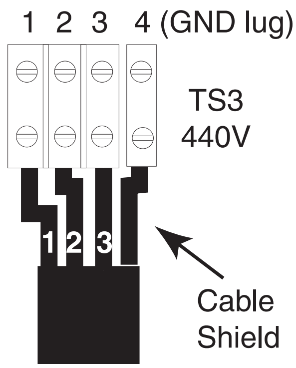


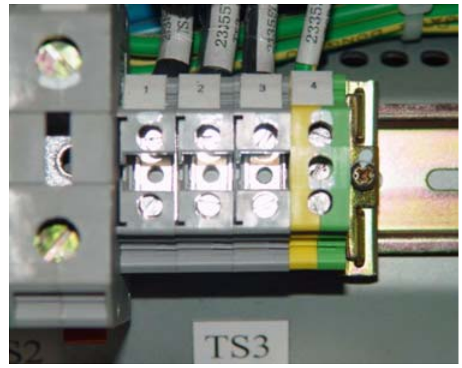
☐ Check box when complete.

8.6 440V Connection

Connect the internally shielded 440V cable from the gantry to TS3 on the panel. See Figure 2-20 for the location of the connector and Figure 2-26 for details. Observe the labels on the cable leads for proper identification and orientation.

Figure 2-26 440VAC Connection



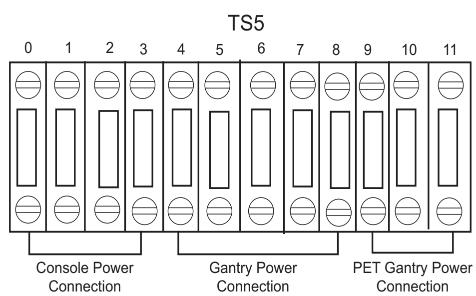


☐ Check box when complete.

8.7 Gantry & Console Power Connections

Plug the Gantry power cable wires into "J4" and the Console power cable wires into "J5" as shown in Figure 2-27.

Figure 2-27 Gantry & Console Power Connections



☐ Check box when complete.

8.8 Console Power Cable Re-termination

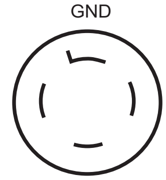
Power cable re-termination should be used as a last resort. Short and long cables are available.

1.) Carefully remove the power plug and **record the color of the wires** in Table 2-8. The terminals are labelled X, Y, W and G on the plug.

Table 2-8 Console Power Cable Termination

TERMINAL	X	Y	W	G
Description	Hot	Hot	Neutral	Ground
Color (may vary)	Black	Brown	White	Green
PDU Connections	TS5-0	TS5-1	TS5-2	TS5-3

Figure 2-28 Console Power Plug



2.) Cut the cable to desired length and dress ends.

3.) Re-install the power plug, according to the orientations recorded in Table 2-9.

4.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 2-9 Resistance Verification Points

FROM PDU PLUG	TO CONSOLE PLUG END	
CB1 -11 (J5 Phase X)	Phase X Hot	☐ Check box when complete
N/G-Neutral TS5-2	Phase Y Hot	☐ Check box when complete
A3 Neutral Bus Bar (J5 -13 Phase W)	Phase W Neutral	☐ Check box when complete
A3 Ground Bus Bar (J5 -22 Ground Green)	Ground (Green) Ground Green screw	☐ Check box when complete

☐ Check box when complete.

8.9 PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into “J1” on the “A4” panel. Secure it by using the fasteners intergrated into cable’s connector shell.

☐ Check box when complete.

8.10 System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See Figure 2-20, on page 103, and Figure 2-29, below.

Figure 2-29 PDU System Ground Connection

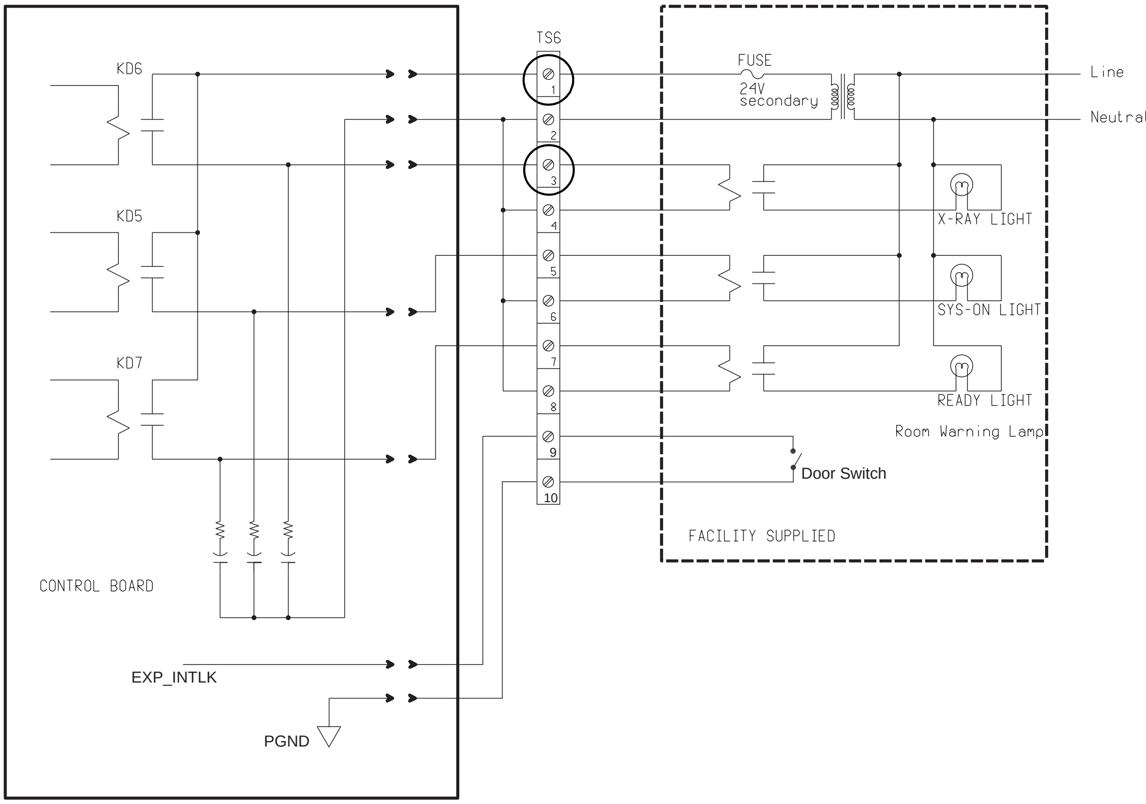
☐ Check box when complete.

8.11 Warning Light & Door Interlock Connections

8.11.1 Warning Light Configuration & Connection

1.) Warning Light is controlled by signals from the system.
2.) This step is site specific. The PDU by default is configured for “no” external warning light connection. If you have external warning lights, see Figure 2-30 for proper connection.

Figure 2-30 Typical TS6 Warning Light & Door Interlock Connections



It is recommended that you use the four (4) wire method of adding an X-ray warning light to a room, as shown in Figure 2-30. When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

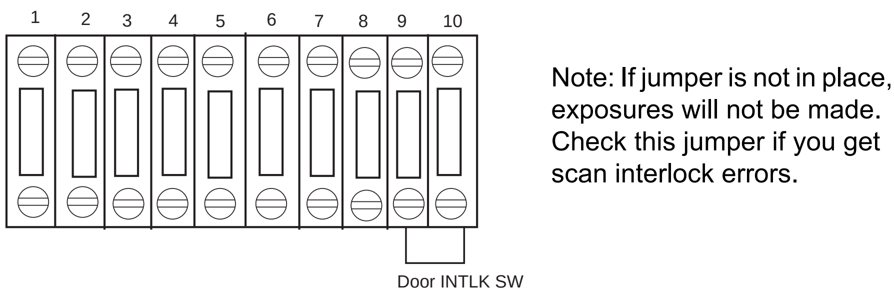
☐ Check box when complete.

8.11.2 Door Interlock Connections

Door interlocks are used to prevent X-Rays from being generated when the scan room door is open.

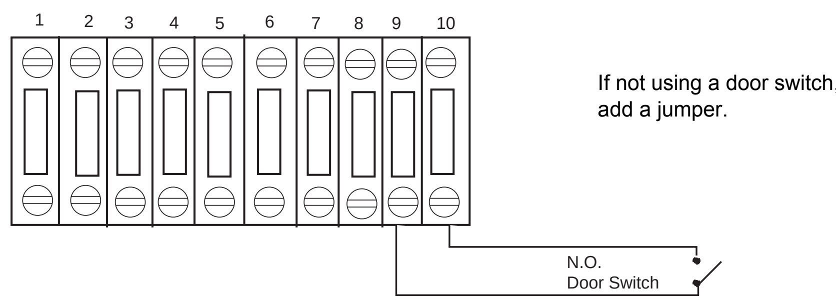
The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 9 & 10 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See Figure 2-31.

Figure 2-31 Without a Door Interlock



To use the system with a a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

Figure 2-32 With a Door Interlock

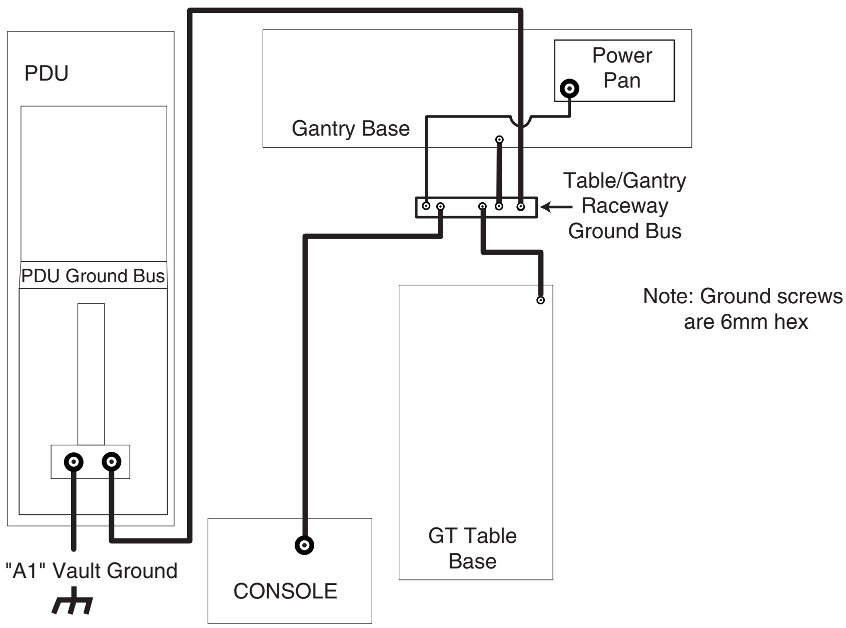


☐ Check box when complete.

Section 9.0 System Ground Connections

As seen in Figure 2-33, the Table/Gantry raceway ground bus is used to centralize all system grounding. The system ground is tied to vault ground at the PDU, through its chassis.

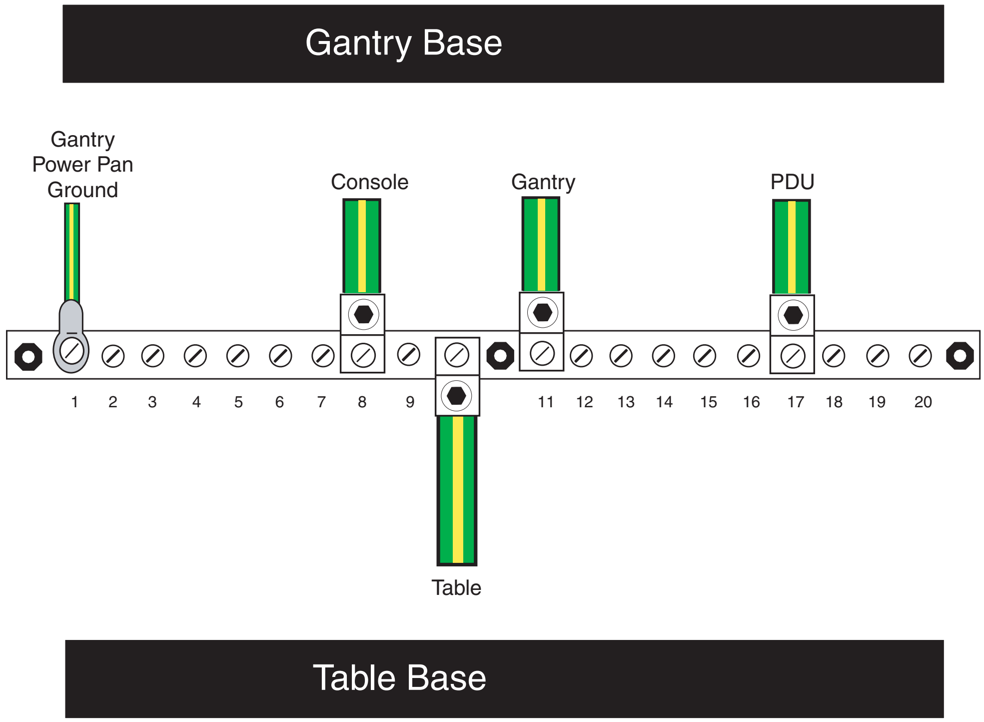
Figure 2-33 CT System Ground Connections



The gantry is tied to system ground at a number of points. It is important that all of these ground connections are securely made. See Figure 2-34.

Check that the terminals will not be loosened, by moving or swinging the cables rather strongly by hand.

Figure 2-34 Table/Gantry Raceway Bus - Grounds



Various types and sizes of wire are used to ground the system. Please use the type and sizes specified in Table 2-10.

Table 2-10 System Ground Connections

AWG #	CONNECTION TO	CONNECTION TO
#1/0	PDU	Power Main
#1/0	Gantry (Power Pan)	Raceway
#2	Console	Raceway
#1/0	Gantry	Raceway
#2	Table (frame)	Raceway
#1/0	PDU	Raceway

☐ Check box when complete.

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Chapter 3
System Continuity & Ground Checks

NOTICE

Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- **Record data collected from procedures in this chapter into Form F4879 when directed, located in Chapter 6 of this book.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

Section 1.0
System Continuity & Ground Checks (Mechanical Contractor)

Use this section to check cable and ground connections.

1.1 Tools Required

- Digital VOM
- 30 ft of #18 wire
- 600 VAC meter leads

1.2 Procedure

Reference Figure 3-1: Front View of NGPDU, with Covers Removed on page 117 and Figure 3-2: Gantry Power Pan on page 117.

WARNING

USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
2.) Put the UPS in the Service Position.
3.) Remove the PDU A3 input power panel cover.
4.) Verify, with a voltmeter, that mains power is disconnected.
5.) Verify that less than 1 ohm of resistance exists between the following ground connections:

Table 3-1 Mains Connections to PDU

FROM	TO	
Wall ground connection	PDU Cabinet	☐ Check box when complete

6.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 3-2 Resistance Verification Points

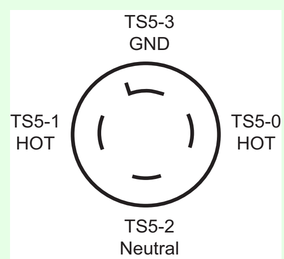
FROM	SIGNAL NAME (COLOR)	TO	
PDU TS2-1	+HVDC (Red)	Gantry HV Power Pan TS1-1	☐ Check box when complete
PDU TS2-2	HVDC Ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU TS2-3	-HVDC (Black)	Gantry HV Power Pan TS1-2	☐ Check box when complete
PDU Ground Bus	HVDC shield	Gantry HVDC cable shield	☐ Check box when complete
PDU TS3-1	Axial drive 440vac (Black)	Gantry HV Power Pan TS2-1	☐ Check box when complete
PDU TS3-2	Axial drive 440vac (Red)	Gantry HV Power Pan TS2-2	☐ Check box when complete
PDU TS3-3	Axial drive 440vac (Orange)	Gantry HV Power Pan TS2-3	☐ Check box when complete
PDU TS3-4	Axial drive ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU Ground Bus	Axial drive shield	Gantry 440 VAC cable shield	☐ Check box when complete
PDU TS5-0	120vac Phase B (Black)	Console Power Plug: 	☐ Check box when complete
PDU TS5-1	120vac Phase A (Brown)		☐ Check box when complete
PDU TS5-2	120vac Neutral (Light Blue)		☐ Check box when complete
PDU TS5-3	Ground (Green/Yellow)		☐ Check box when complete
PDU TS5-4	120vac Phase A (Black)	Gantry LV Power Pan A1J1 FIlter - L1 (by power plug)	☐ Check box when complete
PDU TS5-5	120vac Phase B (Red)	Gantry LV Power Pan A1J1 FIlter - L2 (by power plug)	☐ Check box when complete
PDU TS5-6	120vac Phase C (Orange)	Gantry LV Power Pan A1J1 FIlter - L3 (by power plug)	☐ Check box when complete
PDU TS5-7	120vac Neutral (Light Blue)	Gantry LV Power Pan A1J1 FIlter - N (by power plug)	☐ Check box when complete
PDU TS5-8	Ground (Green/Yellow)	Gantry Power Pan Chassis A1J1 Filter Ground Stud	☐ Check box when complete

Figure 3-1 Front View of NGPDU, with Covers Removed

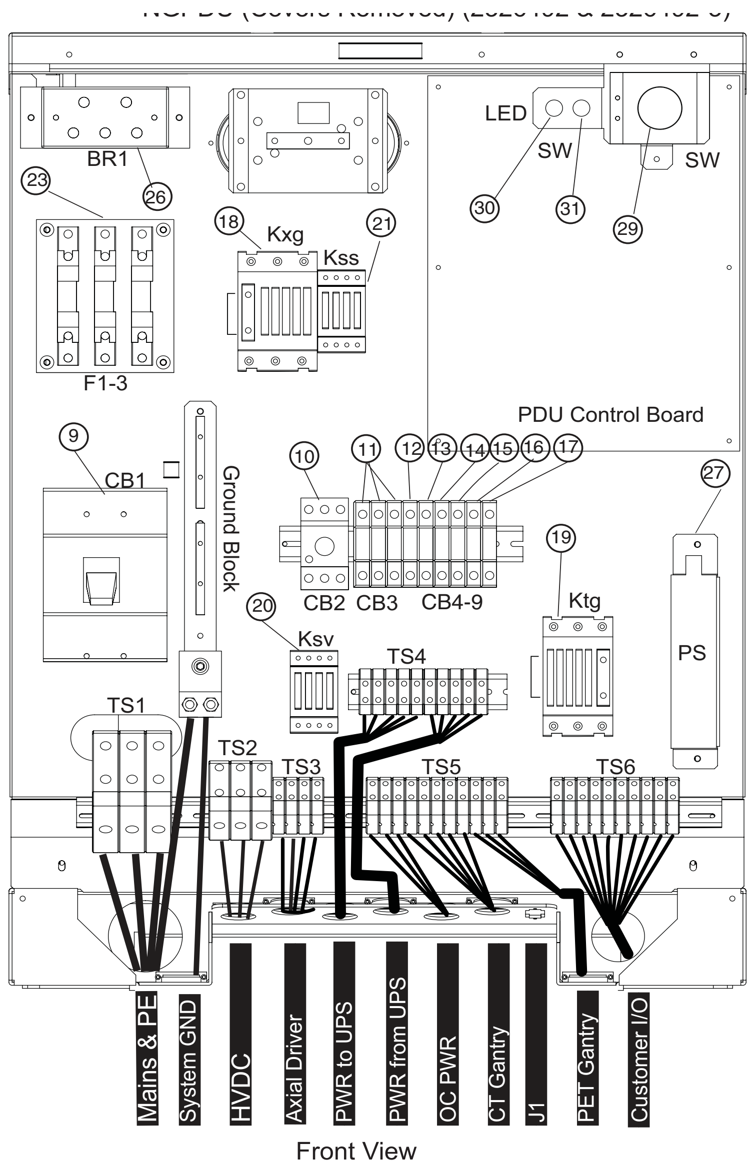
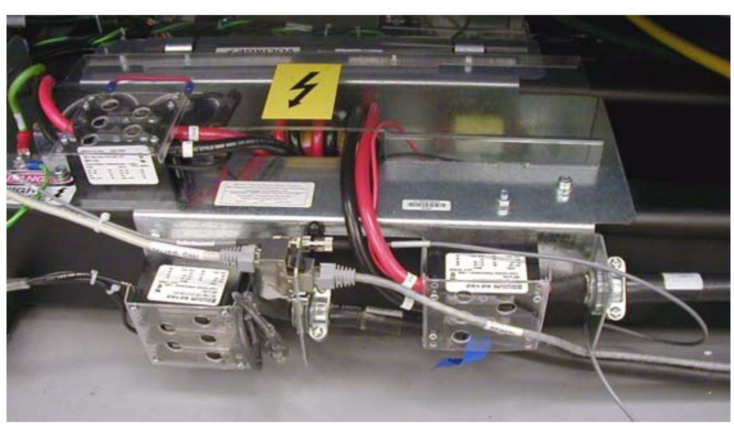


Figure 3-2 Gantry Power Pan



WARNING **TURN OFF ALL PDU CIRCUIT BREAKERS.**

7.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

Table 3-3 No Continuity Verification Points

FROM PDU	TO A1 BREAKER BOX	
TS2-1 (+HVDC) (Red)	vault ground	☐ Check box when complete
TS2-3 (-HVDC) (Black)	vault ground	☐ Check box when complete
TS3-1 (440vac output) (Black)	vault ground	☐ Check box when complete
TS3-2 (440vac output) (Red)	vault ground	☐ Check box when complete
TS3-3 (440vac output) (Orange)	vault ground	☐ Check box when complete

8.) Leave the metal cover off the PDU A3 input power panel until you complete the checks in the next section.

Section 2.0
Site Ground Continuity Check

1.) Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

Table 3-4 Resistance Verification - Site Ground

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

2.) Install remaining covers on:
- Gantry
- Table
- Raceway
- Console
- PDU

Section 3.0
Axial Head Holder Shim Installation

3.1 Overview

This procedure applies to the following Table Types:

GT 2000	All head holders
GT 1700	All head holders
HP 1600	All head holders
PET Tables	All head holders

3.2 Requirements

Table 3-5 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
1	10 mins	15 mins	5 mins

Table 3-6 Tools and Test Equipment

ITEM	QTY	EFFECTIVITY	PART#	MANUFACTURER
Standard FE Tool Kit	1	-	-	-

Table 3-7 Replacement Parts

ITEM	QTY	EFFECTIVITY	PART#	MANUFACTURER
Shim Kit	1	-	-	-

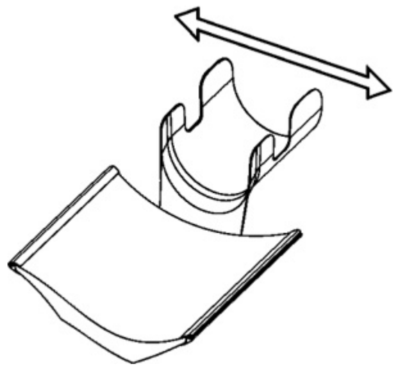
NOTICE **Understand and Follow All General Table Safety Procedures.**

3.3 Required Conditions

Check head holder for a tight fit. If the head holder fit is loose, follow this procedure.

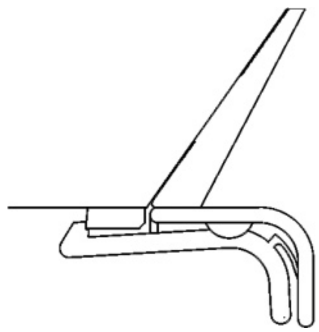
Introduction:

- Some Axial Head Holders have a large free-play in the horizontal direction which could potentially lead to motion and therefore image artifacts.
- Installation of the 2327335 rubber shim kit can minimize this motion.

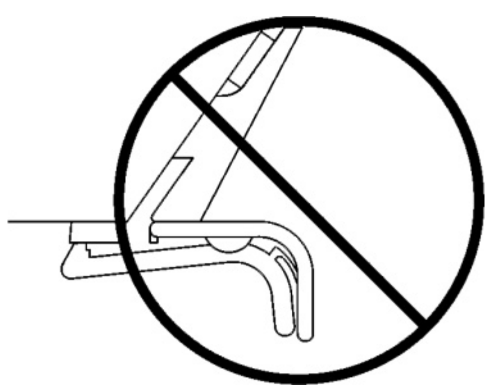


Notes before Selecting Shim Thickness:

- While selecting the best shim size, do not attach the rubber shim yet using the adhesive on the back. It is best to use a piece of tape to hold on the shim in order to see if the size is correct.
- Selecting a shim size that is too thick may result in:
    - Difficulty latching the head holder properly. The head holder must latch so that a patient is not injured.
    - Damage to the plastic latch or the plastic screws that secure it.



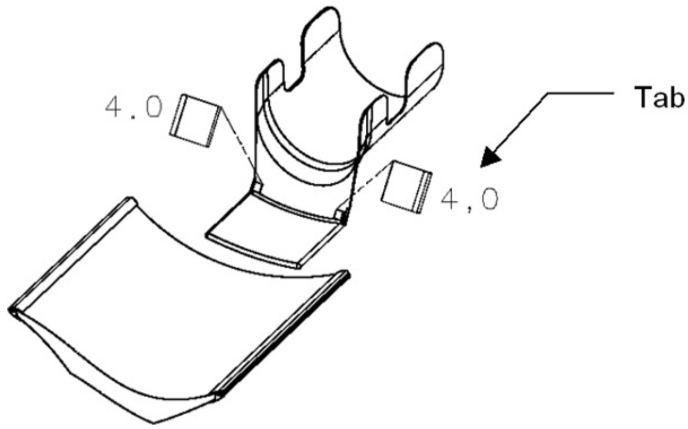
Correct - Head Holder is latched onto first step of plastic latch mechanism (The head holder does not need to be latched onto the second step)



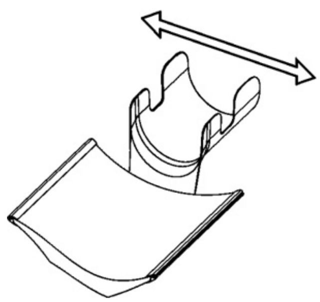
Wrong - Head Holder is NOT latched after installing shims

3.4 Procedure

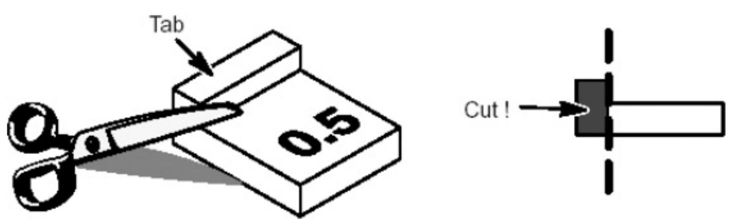
1.) First place the two 4.0mm shims (thickest size) onto both edges of the head holder as shown (use a piece of tape to temporarily secure them)
   - The shim must be placed with the tab facing out
   - The thickness is printed on the shim



2.) Insert the head holder into the cradle
3.) Check if the head holder is latched onto the cradle at the first step of the plastic latch mechanism. (The head holder does not need to be latched onto the second step)
4.) Check if the head holder has a small free-play in the horizontal direction



5.) If the rubber is too thick, repeat steps 1-4 using a thinner shim (3.5, 3.0…0.5mm) until the head holder is latched (without excessive force) and fits securely in the cradle.
If the thinnest shim (0.5mm) is too tight, the tab can be cut off to reduce the thickness



6.) Clean off the surfaces where the shims will mount using alcohol.
7.) Peel off the paper from the back of the selected shims and attach with the tabs facing out. Hold each shim with your fingers for a few seconds to attach it to the head holder.

3.5 Finalization

Review latching of head holder with customer after installation

Section 4.0
Mechanical Installation Completion Checklist

Complete the entire Mechanical Installation Completion checklist and return it to GE. The electronic file for the checklist is found on the LightSpeed 7.X Service Information CD-ROM.

- From the PDF, it may be accessed by clicking on the following link: Mechanical Checklist
- The checklist (`Mechanical Vendor Installation List.dot`) is also located in the `forms` directory of the CD-ROM.

NOTICE **In Chapter 6, Section 3.0: Mechanical Installation Completion Checklist on page 114, there is a copy of the electronic form. This should be used as a guide for *completion of actions* only. The electronic form contains additional fields. It MUST be completed in its entirety and returned to GE. (Return instructions are on the form.)**

Appendix A
Removal & Installation of Covers

Section 1.0
Gantry Scan Window

1.1 Remove Scan Window

1.) Grip the window by the metal band at the top and pull firmly but carefully downward.

2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.

Figure A-1 Scan Window Removal

3.) Hold the scan window as shown in Figure A-2 and remove from between the front and rear covers.

Figure A-2 Removing Gantry Scan (Mylar) Window

4.) To prevent damage, place the scan window in a secure place.

1.2 Install Scan Window

Note: The front and rear covers must be installed before installing the scan window.

5.) Shape the scan window as shown in Figure A-3, and nest the scan window at the bottom of the opening between the front and rear covers, (Figure A-4) with the rivets in the 6 o'clock position. Remember the rivets must be in the 12 o'clock position when the mylar window is fully installed.

6.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.

7.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

Figure A-3 Install Scan Window

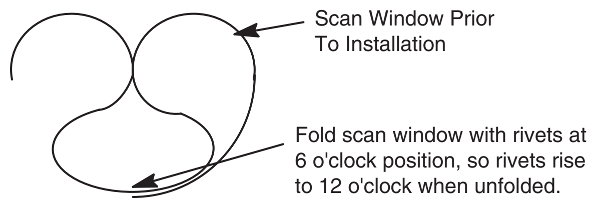
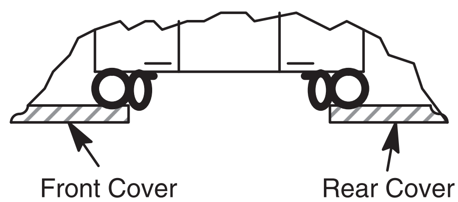


Figure A-4 Scan Window Nested Between Front and Rear Cover



Section 2.0
Gantry Side Covers

2.1 Side Cover Removal

1.) Lower table to home (lowest) position.

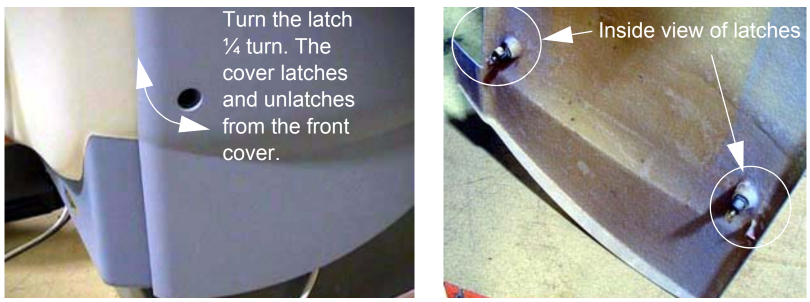
CAUTION

Potential for injury if covers removed and power is left "ON".

Always remove the right side cover first, and turn OFF power at the Service Switch Panel.

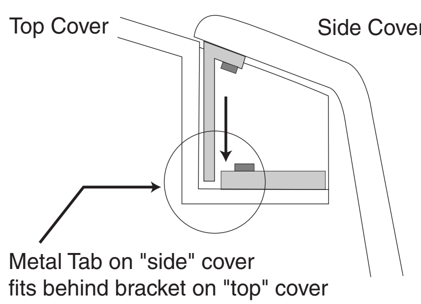
2.) Use an 8mm Hex wrench to unlatch the side cover from the front cover. See Figure A-5.

Figure A-5 Side Cover Latches



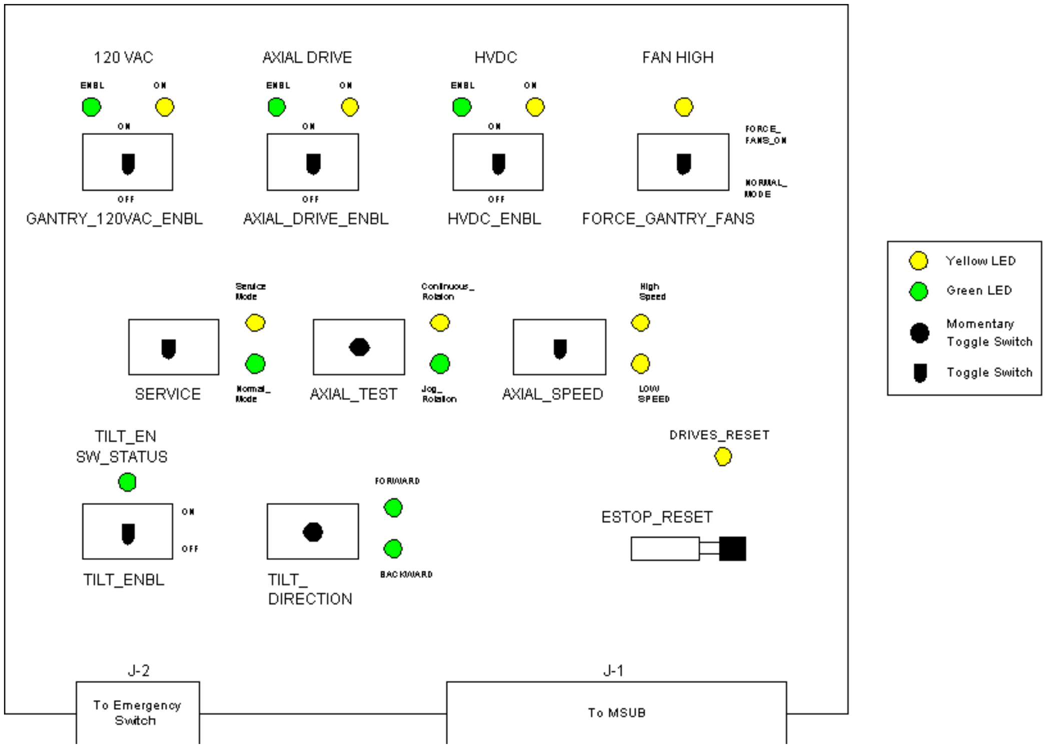
3.) Remove the right side cover by lifting it upward to release the two (2) latches, located on the top edge of the cover. Once removed, the MSUB/TGP left side should be exposed.

Figure A-6 Side and Top Cover Clasp



4.) Turn OFF the **120 VAC**, **AXIAL DRIVE** and **HVDC** power switches on the gantry service switch panel (see Figure A-7).

Figure A-7 Gantry Service Switches



5.) Repeat steps 1-3 for the left side cover.

2.2 Side Cover Installation

1.) To install a side cover, place it over the top cover and let the two (2) side cover latches slide behind the metal tabs, located on the top cover. See Figure A-6.

2.) Use Hex wrench to secure the side cover to front cover by turning the bolts a quarter turn. See Figure A-5.

Section 3.0
Gantry Top Covers

CAUTION **Potential for Electric Shock.**

Gantry Power Is Present.

Before you remove top covers, always make sure the three (3) power switches have been turned off. (See Figure A-7.)

CAUTION **Potential for Personal Injury.**

Top Cover is Heavy.

Follow the Removal Procedure.

3.1 Top Cover Removal

3.1.1 Additional Tools

- Front and rear cover dollies

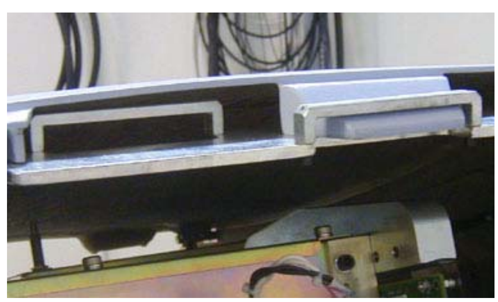
3.1.2 Procedure

1.) Remove any associated side covers if installed. On a new installation, the top covers are shipped tie-wrapped in place. Use a pair of side cutters to remove, and discard.

2.) The top cover fans are connected to the gantry using a Molex connector. Locate and disconnect the fan power plug.

3.) Take the end of the top cover closest to you and tilt upwards. Grip the cover on both sides. You will notice that the top covers have edge guides that center the cover on the raised ridges of the front and rear covers. This allows the cover's tab to disengage from the mounting bracket. See Figure A-8. Slide the cover down using the gantry front and rear covers as support until the cover is at chest level.

Figure A-8 Top cover tabs and bracket



4.) Lift the cover clear, then place the flat end on the floor.

5.) Repeat steps 1 - 4 for the other cover.

3.2 Top Cover Installation

The top cover consists of two (2) pieces. Install the front and rear gantry covers, if not already installed. See Section 5.0 on page 131, and Section 4.0 on page 129.

1.) Lift the top cover so that the cover is chest high. Fit the cover guides on the top of the cover into the rail slots on the gantry covers.
2.) Slide the top cover up the front and rear covers onto the top cover ramp. The cover should snap into place in the top cover bracket.
3.) Insert the cover-locking tabs (at the bottom of the top cover) into the slots on the gantry covers.
4.) Check cover alignment. Adjust as needed.
5.) Reconnect the fan power.
6.) Repeat above steps 1 - 5 for the other cover.

Section 4.0
Gantry Rear Cover

4.1 Removal

1.) Assemble the rear cover dolly.

a.) Tighten the two (2) shoulder bolts to the rear cover.

Figure A-9 One side of the Rear cover dolly

b.) Fit side dolly through the shoulder bolts and secure assembly with two (2) wing nuts. See Figure A-9.

c.) Repeat steps a and b for the other side dolly.

CAUTION

Potential for injury if covers removed and power is left "ON".

2.) Disconnect cables on the right side of the rear cover.

3.) Remove rear cover.

a.) Disengage upper and lower cantrell brackets on both sides of the rear cover.

1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See Figure A-13.

2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.

b.) Disengage the rubber retaining straps on both sides.

4.2 Installation

1.) Position cover in back of gantry

2.) Attach the rear cover

   a.) Align the studs on both sides of the rear cover with the receivers located on the gantry frame.

   b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps.Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

Note: You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

3.) Reattach upper and lower cantrell brackets on both sides.

   a.) Remove upper cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides.

   b.) Remove lower cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place.

Note: Adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not re-quire adjust during normal use.

4.) Remove dolly, disassemble and store safely away.

5.) Re-attach cables to cover.

6.) Re-install the mylar (scan) window according to Section 1.0 on page 123.

Section 5.0
Gantry Front Cover

NOTICE **Potential for cover damage.**

Front and rear cover removal and installation can be safely accomplished by one (1) person using the dollies provided with the system. Failure to use these dollies will significantly increase the likelihood of damage to the covers. Do not lean covers against walls.

5.1 Front Cover Dolly Setup

DANGER **DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.**

CAUTION **Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.**

1.) Arrange Dolly sections for assembly. The base and post can be assembled only one way. Refer to Figure A-10 and Figure A-11.
   - The base uses two (2) palm screws to clamp the four (4) legs in the open or usage mode.
   - The base also uses the same palm screws to prevent the legs from falling in storage mode.
   - The top post can be inserted in either base and is keyed for proper engagement.
   - The top post locking pin prevents the sections from separating during usage.

Figure A-10 Front Cover Dolly in Storage Mode

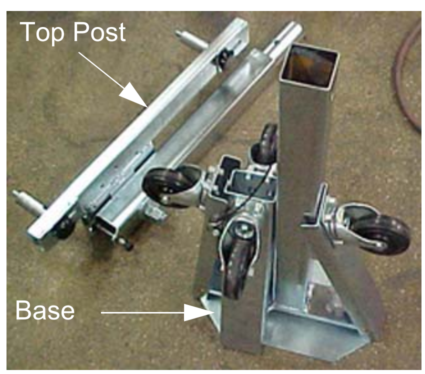
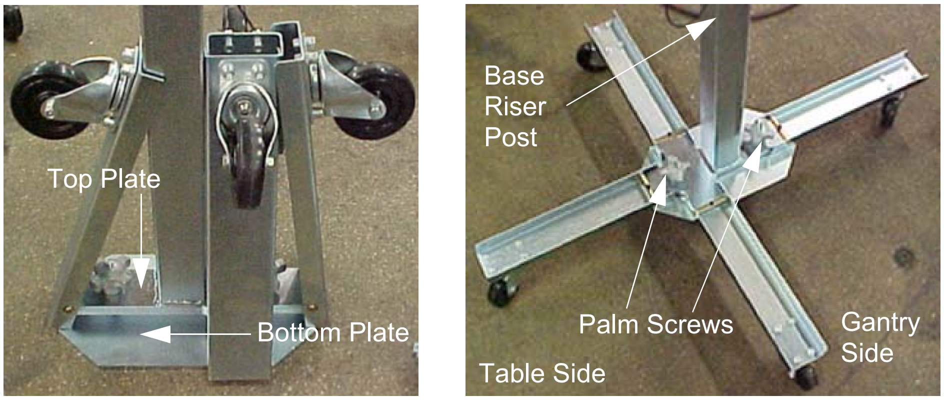


Figure A-11 Front Cover Dolly Base Assembly



2.) Unfold the base legs by loosening both palm screws to the top of their travel.

3.) Carefully unfold the legs so that the castors touch the floor.

4.) Tighten the palm screws to clamp the legs between the base top and bottom plates.

Note: Lifting the base by the riser post while leaving the castors on the floor will ease palm screw tightening. Reference Figure A-11.

WARNING **ENSURE BOTH PALM SCREWS ARE TIGHTENED SECURELY AND THE LEGS ARE CLAMPED TIGHTLY BETWEEN THE BASE TOP AND BOTTOM PLATES. FAILURE TO DO SO WILL RESULT IN INSTABILITY DURING FRONT COVER HANDLING.**

5.) Insert top post into the base riser post. Align the key for complete engagement.

6.) Insert top post locking pin to secure both top and bottom sections.

7.) Reverse above steps to disassemble.

Note: For base storage only one (1) palm screw needs to be tightened. This will engage the bottom base plate and the leg ends preventing the legs from unfolding during transport and storage.

5.2 Removal

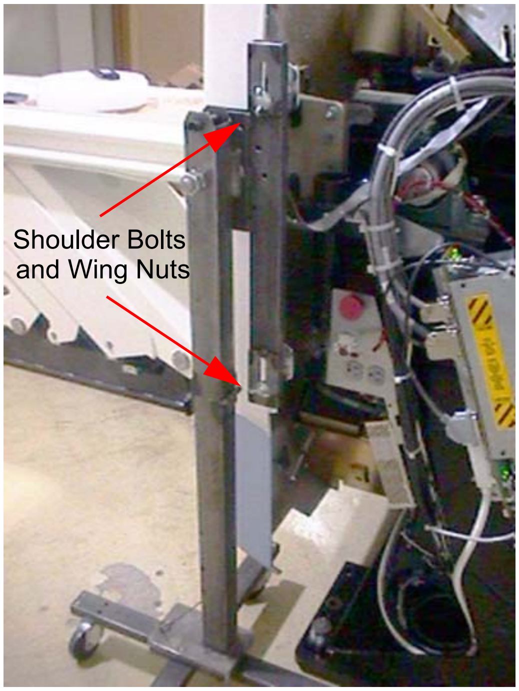
1.) Position the table at its lowest position.

2.) Remove gantry side and top covers, if you have not already done so. See Section 2.0 on page 125. Make sure that the three (3) power switches have been turned off. See Figure A-7.

3.) Assemble the front cover dolly.

a.) Tighten the two (2) shoulder bolts to the gantry securely. This will make cover installation easier. See Figure A-12.

Figure A-12 Front Side Dolly



b.) Attach side dolly to the shoulder bolts and secure assembly with two (2) wing nuts.

c.) Repeat steps a and b to assemble the other side dolly.

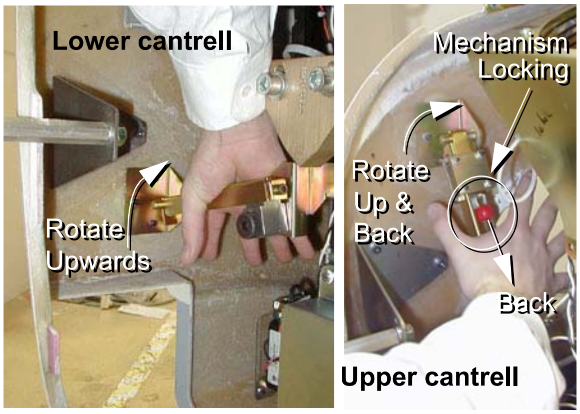
4.) Detach front cover J3 and J2 and front cover BKHD J1 cables.

5.) Remove front cover

a.) Disengage upper and lower cantrell brackets on both sides of the cover.

1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See Figure A-13.

Figure A-13 Releasing cover brackets

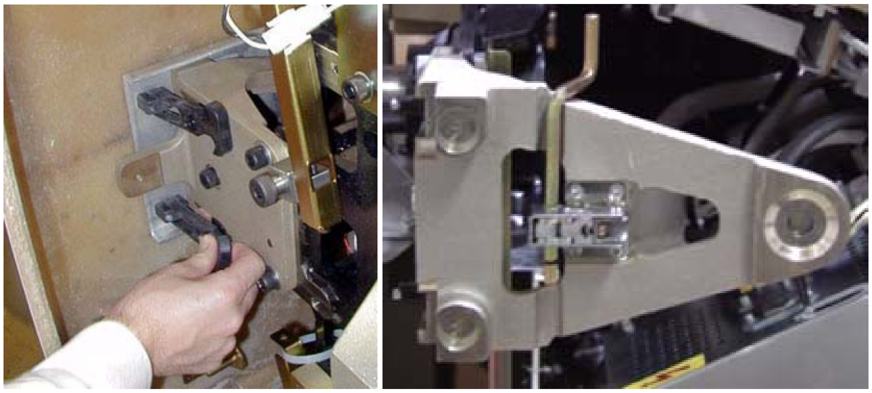


2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.

b.) Disengage the rubber retaining straps on both sides. You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

c.) Also lift and rotate cover locking arm to unlocked position.

Figure A-14 Rubber retaining straps and Cover Locking mechanism



6.) Rotate front cover away from gantry.

a.) Move front cover away from gantry giving enough space (about 5 feet) between front cover and gantry.

b.) Pull the locking pin and rotate front cover away from gantry. Place locking pin in one of the side dolly perforations. See Figure A-15.

Figure A-15 releasing Front Cover Dolly Hinge

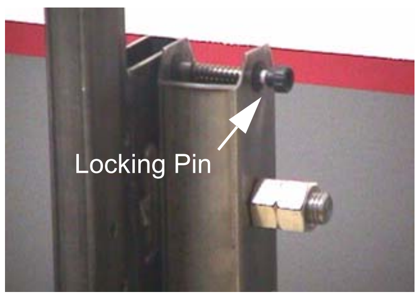
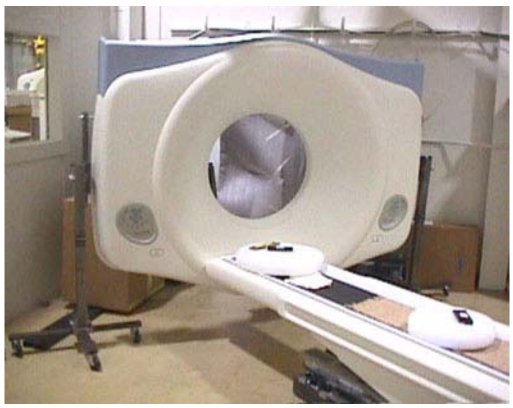


Figure A-16 Front Cover Removal Sequence



1. Upper Left>Rotate cover to make room for step 2.
2. Upper Right>Rotate cover to clear the table. Roll the cover to foot end of table.
3. Lower Left> Rotate the cover upside down to provide clear work area.
   Note: Remove Gantry Display and Control Panel in position 2.

7.) Rotate the cover horizontally and move it back and over the table to a safe location. Once in a safe location, you may over-rotate the cover full vertically but upside down.

5.3 Installation

1.) Remove the gantry display and control assembly from their service positions and re-attach them to the gantry cover.
- a.) Disconnect cables from Display and Gantry Control Panels.
- b.) Install Gantry Display in front cover. Secure the 5 thumbscrews. With a flat-blade screw driver gently tighten past finger tight.
- c.) Install the gantry control panel making sure the ball studs are secure within the receivers.
- d.) Re-attach cables.

2.) Rotate gantry back to its vertical position.

NOTICE **Potential for front cover damage.**

When you put rotate the gantry back to its vertical position, make sure not to scratch the front cover with the edge of the table cradle.

3.) Attach the front cover.
- a.) Align the studs on both sides of the front cover with each associated receiver. Receiver is located on the gantry frame.

Figure A-17 Cover stud and Mounting bracket receiver

b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps.Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

You may find it helpful to lift "up" on the cover to align the stud while attaching the rubber retaining straps.

4.) Re-attach upper and lower cantrell brackets on both sides.

a.) Remove upper Cantrell brackets from service position and rotate them into position over their associated retaining pins. See Figure A-18.

Figure A-18 Service position of upper and lower cantrell brackets

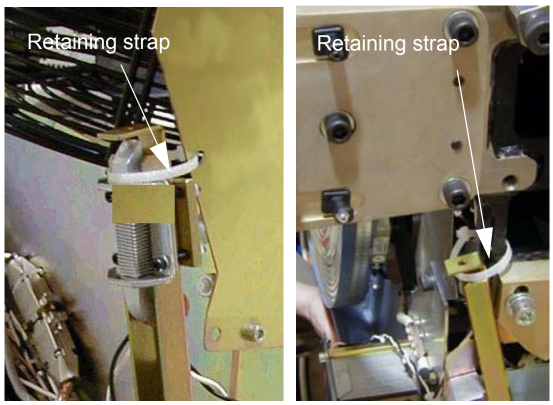
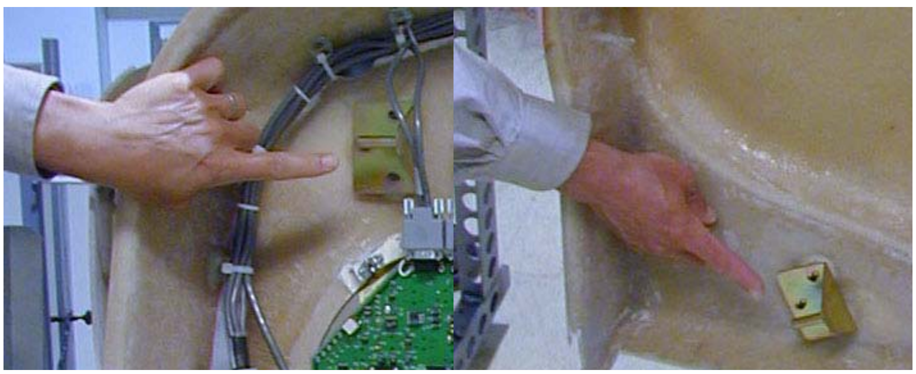


Figure A-19 Cover retaining pins (top and bottom)



Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides. See Figure A-20.

Figure A-20 Locking the cover brackets into place

b.) Remove lower cantrell brackets from service position (see Figure A-18), and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. See Figure A-20.

Note: Mis-adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.

5.) Remove dolly, disassemble and store safely away for later use.

6.) Re-attach cables to cover.

Section 6.0
Gantry Cover Alignment Adjustment Guide

6.1 Overview

This section explains the adjustment capability available for the Gantry covers and guidelines for performing adjustments.

6.2 Tools Required

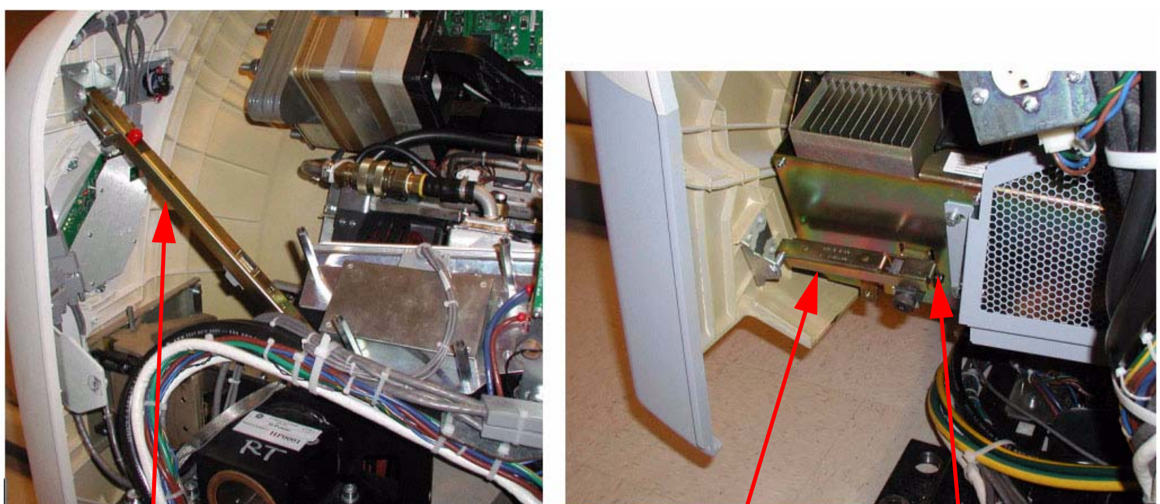
3mm, 5mm, 8mm, Hex wrenches

6.3 Adjustment Guide

6.3.1 Front/Rear Cover adjustment

The front and rear covers are held in a fixed vertical position by the cover locking mechanisms at the vertical middle of each cover on each side. The only adjustment capability is found on each of the cover cantrells (see Figure A-21, one near each corner of the front and rear covers. Adjustments are best done at zero tilt.

Figure A-21 Cantrells



1.) Remove the Gantry side covers using a 8mm hex wrench to release the bottom latches. Refer to Section 2.0 on page 125, for further details.
2.) Disable gantry rotation, 120VAC and HVDC via the service switch panel.
3.) Disconnect the fan power via the white molex connector (one for each top cover). Remove the top covers by lifting up and pulling to the side of the gantry.
4.) Remove the scan window to avoid creases or kinks during cover movements.

5.) Using a 5mm hex wrench, adjust the bolt on the gantry end of the cover cantrells (see Figure A-21) to lengthen or shorten the cantrell arm that will push or pull the cover corner. The distance between the inside edges of the front and rear covers will be typically 27" + 1/8” (689mm). Distance should remain the same as measured along the inside edges from top to bottom with both covers hanging level as seen with a level on the inside vertical edge.

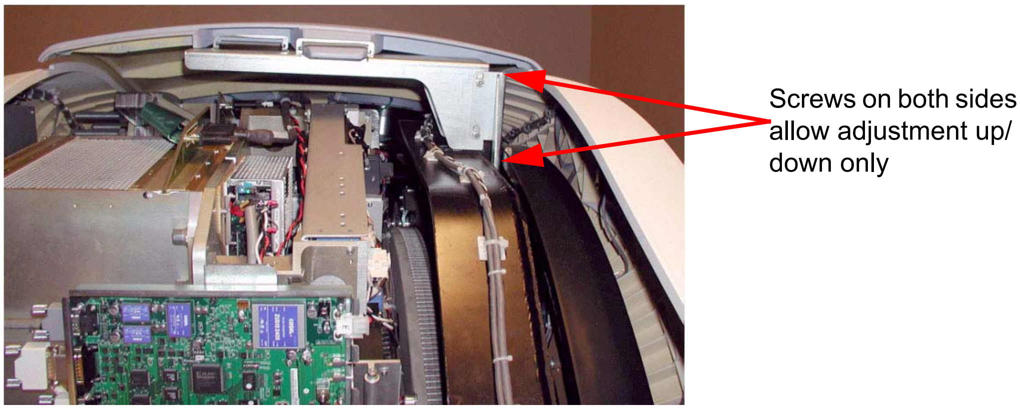
6.3.2 Scan Window

Position the scan window in the opening to verify that it seats all the way around with no gaps. The scan window should seat easily if the front and rear covers have been adjusted properly. If not, go back and readjust such that the scan window seats properly. There is no easy solution other than adjusting and looking for fit.

6.3.3 Top Cover Adjustment

The top covers rest in a top bracket that is mounted to the stationary gantry frame and subsequently just rest on top of the front/rear covers. There are also two alignment fingers on the outer edge of the top covers that need to sit in brackets on the front/rear covers.

Figure A-22 Top Cover Bracket

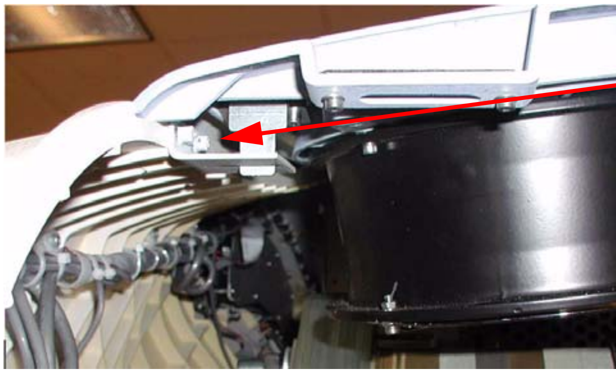


1.) Set the top covers back on the gantry and look for proper fit. The top covers should rest on the front/rear covers all along the edges. If they do then skip the rest of this section.

2.) If the top covers do not rest on the gantry at the top then lift them back off and use a 5mm hex wrench to loosen the bolts that hold the top cover bracket for adjusting vertical alignment to allow the top covers to rest on the front and rear covers.

3.) The brackets on the front/rear covers for the fingers on the outer edge of the top covers are adjustable using a 3mm hex wrench and sliding the bracket till it lines up with the top cover finger. (See Figure A-23)

Figure A-23 Top Cover Fingers



4.) Verify the top cover tapered edges fit over the lip along the top edge of the front/rear covers. If not, go back to Section 6.3.1 and readjust the cantrells as necessary.

5.) When done adjusting the top covers, reconnect the fan power molex connections.

6.3.4 Side Covers

The side covers just hang from the top covers by two fingers that are not adjustable. When set into the top covers, the side covers should hang straight down with the tapered edges fitting over top of the lip of the front/rear covers.

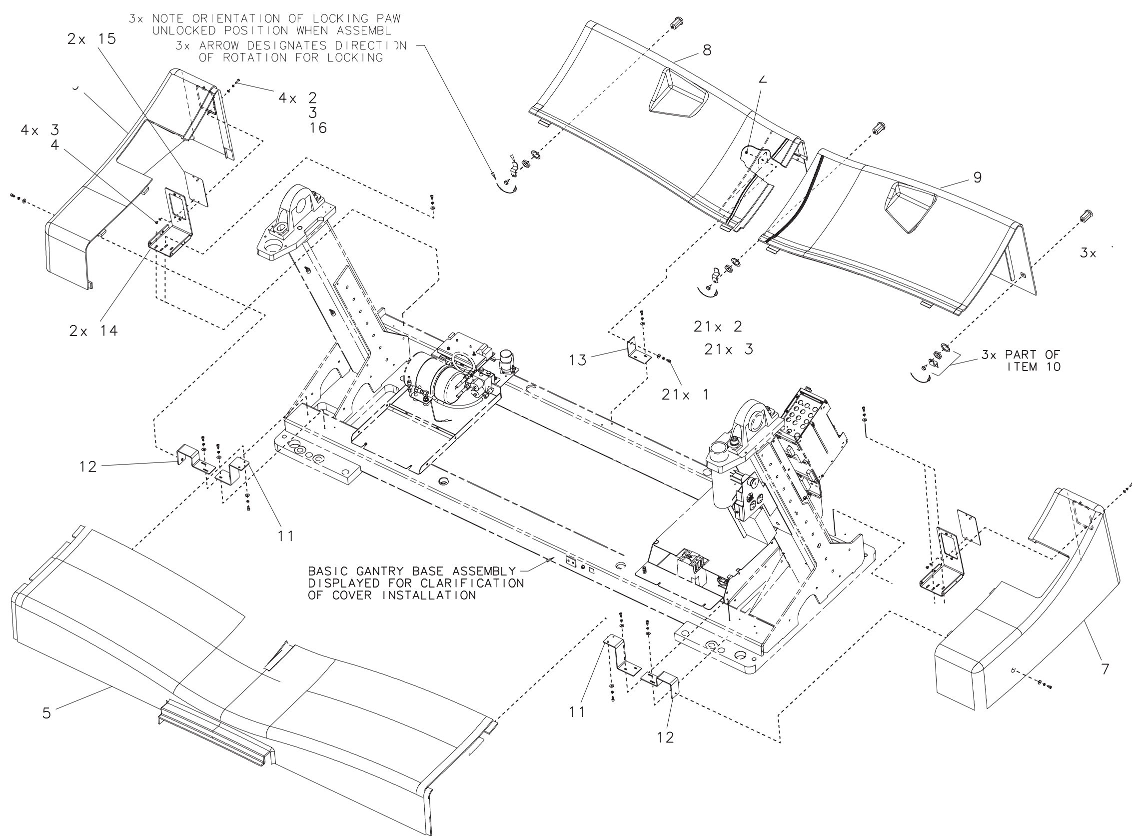
Section 7.0
Gantry Base Covers

Refer to Figure A-24 for the following assembly sequence.

Note: Tighten means torque to 2.3 Nm

1.) Assemble two (2) brackets (item 11) to front cover (item 5), using four (4) hardware items 1, 2 and 3. Align leg of brackets parallel to front edge of cover and tighten. Position cover on gantry base, with bracket slots aligned to gantry holes. Center cover left to right, and attach with four (4) hardware items 1, 2 and 3, as shown, and tighten.
2.) Assemble two (2) brackets (item 12) and two (2) brackets (item 13) to gantry base, using eight (8) hardware items 1, 2 and 3. Finger tighten hardware with brackets moved outward to end of slots. Install side covers (items 6 & 7) on base, pushing brackets (items 12 & 13) inward until properly aligned with front cover. Remove side covers, tighten fasteners and replace side covers, using two (2) hardware items 1, 2 and 3 and one item 4 on each cover, and tighten.
3.) Assemble last bracket (item 13) loosely to gantry base with two (2) hardware items 1, 2 and 3. Assemble latch (item 10) to rear cover (item 8), oriented as shown. Install rear cover (item 8) to base, properly aligned to side cover (item 6). Attach rear cover to bracket with hardware items 1, 2, 3 and 4, tightening all fasteners. Lock latch (item 10).
4.) Assemble two (2) latches (item 10) to rear cover (item 9), oriented as shown. Place cover on gantry base, aligned to covers 7 and 8. Lock both latches.

Figure A-24 Gantry Base Covers



Section 8.0
CT UMI Tilting Dolly

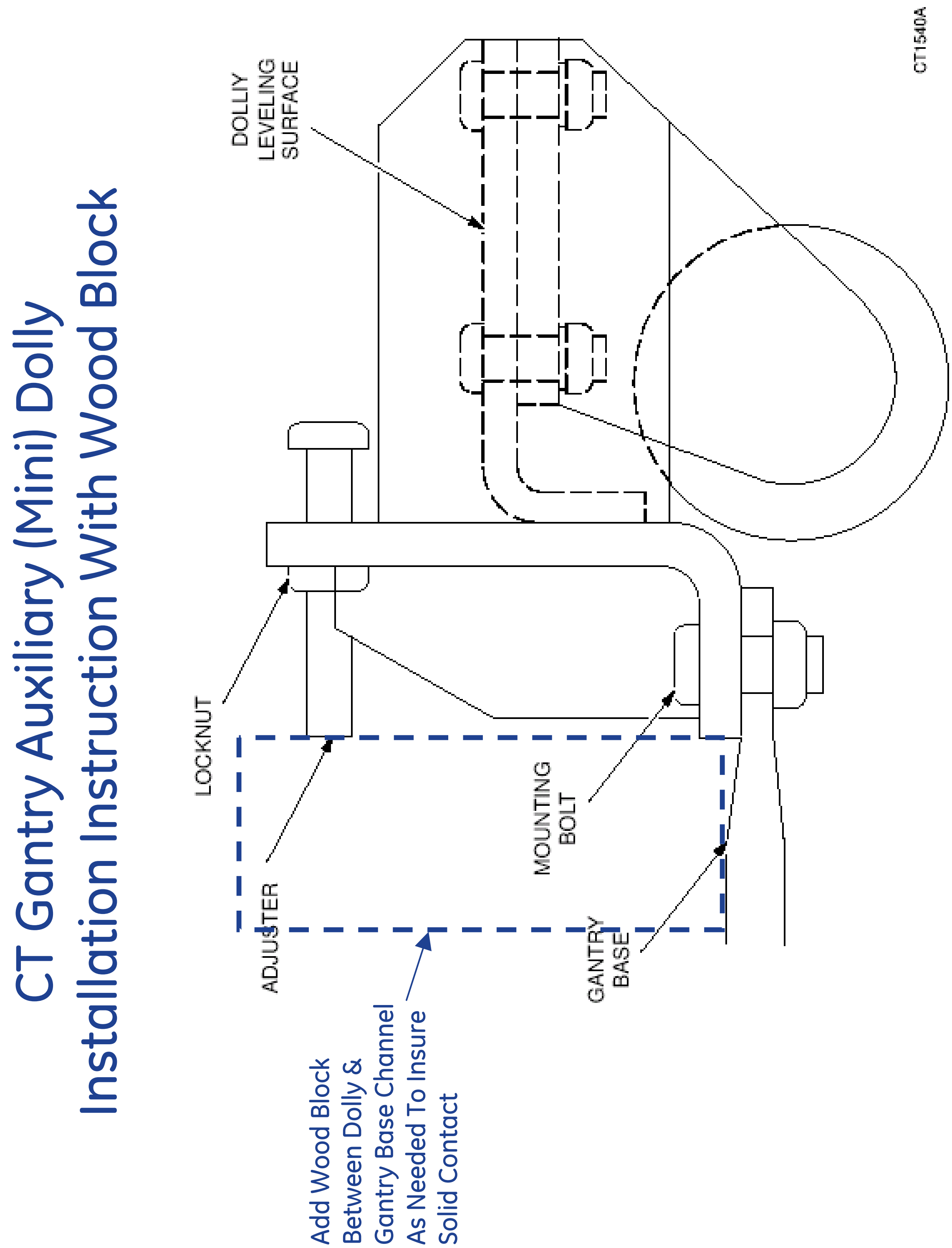
8.1 Installation Procedure

1.) Remove dolly side rails.
2.) Remove two inboard caster towers, and install and pin them at gantry end of dolly.
3.) Elevate all caster towers until their casters contact the floor.
4.) Remove gantry end HSA dolly end, replace dolly pins in the frame receptacles, retain HSA dolly end.
5.) Retrieve and attach pickers to gantry end of table with 16 millimeter (mm) by 40 mm cap screws.
6.) Remove and retain spacers and nuts from 16mm x 240 mm SHCS. Retain for return.
7.) Remove foot end HSA dolly end, replace dolly pins in the frame receptacles, retain HSA dolly end.
8.) Remove and retain 5/16 inch diameter pivot lock pin. Note that this pin is slightly smaller than all other pins used on the dolly, and the only pin that will work well at the pivot.
9.) Remove and retain the sheet metal component at each side of the base weldment and both screws.
10.) Pivot frame open and surround table, maintaining approximate equal clearance between the frame and table on each side. The caster towers may require adjustment to allow ease of movement.
11.) Pivot frame back to approximately square.
12.) Attach and pin lifting arms to pickers.
13.) Carefully and squarely align the 16 mm x 240 mm SHCS to the table. Caster tower adjustments may be required.
14.) Thread the 16 mm x 240 mm SHCS into table until their threads extend inboard of the base weldment.
15.) Remove side cover bracket.
16.) Attach and pin arm separator in place.
17.) Insert pivot lock pin.
18.) Attach and pin HSA dolly ends. Some height adjustments may be required. Insure dolly pins are fully seated.
19.) Remove, stow, and pin caster towers. They may need adjustments to ease pin insertion.
20.) Attach and pin side rails. Be alert for pinch points of fingers between the side-rail and lifting arm. Best results are often achieved by pinning one side-rail before inserting the second.
21.) Manually rotate jack screw to take up play at pickers.
22.) Elevate HSA dolly ends for transport.

8.2 Tilting the Dolly

1.) Verify the blue handle/foot end is 14.5 inches away from the rectangular tunnel in the leg. If not:
    a.) Remove the shroud on the patient's left side
    b.) Detach the transit arm travel limiter
    c.) Verify the table transit lock is positioned correctly
    d.) Reattach the shroud on the patient's left side
2.) Lower HSA dolly ends until table is in contact with floor.
3.) Unpin, remove, and retain side rails, replace pins place pins in a convenient location on HSA dolly end.
4.) Attach and pin all caster towers in active positions. Height adjustments may be required to ease pin insertion.
5.) Elevate caster towers until their casters are in contact with floor.
6.) Remove HSA dolly ends and replace dolly pins in receptacles in frame.
7.) Elevate caster towers to appropriate height. A good target is to elevate until the dolly pins clear the floor, and are supported by their rings.
8.) Rotate jackscrew to tilt table.
    a.) Regularly observe the clearance between the table and the floor; adjust caster towers to maintain an effective distance of approximately one inch, depending on the surfaces and thresholds to be traversed.
    b.) Regularly observe the clearance between the pivot pin of the frame and the table. Excessive rotation of the table may cause the rounded “sweep” of the table to contact the frame, and subsequently damage the table.

Section 9.0
Gantry Auxiliary (Mini) Dolly Installation With Wood Block



Appendix B
Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in Section 2.5, on page 30.

Table B-1 Required Tools

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Adapter		Sears Industrial: $^{3}/_{8}$” to $^{1}/_{2}$” (9-4258)
Ball-Peen Hammer		Sears Industrial: 1lb/2lb (9-38465)
Canned Air		Miller Stephenson: Aero Duster (MS-222N)
Clamp on Amp Meter		Sears Industrial: 9-WTAD105
Combination Wrench Set		Sears Industrial: U.S. Standard & Metric (9-44048)
Cordless Screwdriver		Sears Industrial: 9-MU65401
Deep Well Socket		Sears Industrial: $^{3}/_{4}$” X $^{3}/_{8}$” (included with 9-34496)
Dental Pick		
Diagonal Cutting Pliers		Sears Industrial: Small (9-45077)

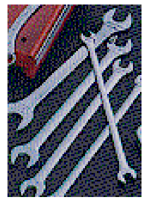
* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Drill		Sears Industrial: $^3/_8$" or $^1/_2$" (9-27859)
Drill Adapter		Sears Industrial: 3" X $^3/_8$" (9-APSZ24)
Drill Bit Set		Sears Industrial: U.S. Standard (9-66084)
DVM		Sears Industrial: 9-82028 Sears Industrial: 9-FL873
Extension for Ratchet Wrench		Sears Industrial: 3" X ½" (9-44133)
Gloves		Sears Industrial: Large (9-40502)
Hammer Drill		Sears Industrial: ½" (9-27205)
Hex Bit Set		Sears Industrial: ¼" (9-SK45508)
Hex Key (Allen Wrench) Set		Sears Industrial: U.S. Standard (9-46284)

* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Level		Sears Industrial: 4’ (9-39856)
Masonry Bit		
Open-End Wrench (Thin or Standard Tappet)		Snap-on: 10mm (SRSM10) & 21mm (LTAM2124)
Pozi Screwdriver		
Ratchet Wrench		Sears Industrial: $^{3}/_{8}$" (9-43175)
Reciprocating Saw with Blades		Sears Industrial: 9-MU650921
Safety Glasses		Sears Industrial: 9-18650
Safety Shoes		
Screwdriver Set		Sears Industrial: Phillips & Straight (9-41505)
Socket Set		Sears Industrial: Standard $^{3}/_{8}$" (9-34496)

* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Sockets		Sears Industrial: 1 $^1/_8$" X $^1/_2$" (9-47516)
Step Ladder		Sears Industrial: 6’ (9-WN6006)
Tongue & Groove Pliers		Sears Industrial: Large (9-CL440)
Torpedo Level		Sears Industrial: 9" (9-39829)
Torque Wrench		Sears Industrial: $^3/_8$" (9-WR3470)
Universal Joint		Sears Industrial: $^3/_8$" (9-4435)
Vacuum Cleaner		Sears Industrial: 8 Gal (9-17780)
* Part Numbers given for reference only. GE Healthcare does not endorse any tool brand name.		

Appendix C
Option Cables

Run all option cables that connect to the Table, Gantry or Console.
For example the remote monitor (through the splitter) and SmartScore.
For more information, refer to “LightSpeed Linux CRT & LCD Monitor Option”

Figure C-1 Remote Monitor Splitter

Figure C-2 SmartScore

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AppendixD
Regulatory Clearance Quick Reference Guide

Section 1.0
Regulatory Code Description

Egress: 29 CFR 1910 Subpart E (OSHA) and NFPA 101 (Life Safety Code) define the minimum requirements for means of egress. The requirement most applicable to equipment installation and room layout is minimum width of exit access. Under OSHA 1910.37(f)(6), the minimum width of exit access shall in no case be less than 711 mm (28 in) from any potentially occupied point in the room.

Under NFPA 101 (2006 edition) 7.3.4.1, the minimum width of any means of egress is 914 mm (36 in). However, NFPA allows this to be reduced to 711 mm (28 in) around furniture or equipment, provided that a 914 mm (36 in) clearance would otherwise be available without moving permanent walls.

Electrical Clearance: 29 CFR 1910 Subpart S (OSHA) and NFPA 70E (Standard for Electrical Safety in the Workplace) define minimum clearance requirements for the workspace around electrical equipment. Under both OSHA 1910.303(g)(1) and NFPA 70E (2004 edition) 400.15, a minimum clear space of 914 mm (36 in) depth (with minimum 762mm (30 in) width and 1981 mm (78 in) height) must be provided in front of electrical equipment with parts operating at 600 volts or below and likely to require examination, adjustment, servicing, or maintenance while energized.

This safety clearance requirement applies to all GEHC equipment. Although 914 mm (36 in) is the minimum clearance for most installations, the standards require an increased minimum clearance distance where parts operate above 150 volts (but still below 600 volts) under the following circumstances:

- If the wall or surface directly facing the electrical equipment is grounded (e.g. brick, concrete, or tile) or includes grounded protrusions (such as medical gas ports, metal door or window frames, water sources and metallic sink structures, metallic cabinetry, electrical disconnects or emergency off panels, air conditioners or vents), then a 1067mm (42 in) clearance depth is required.
- If the possibility exists of exposed and unguarded live parts on both sides of the workspace (for example if a power distribution unit were positioned on the wall directly facing the GEHC equipment), then a 1219 mm (48 in) clearance depth is required.

Section 2.0
Terms and Definitions

Egress: The path of exit from within any room. U.S. regulations require a minimum of 711 mm (28 in) of continuous and unobstructed space, including trip hazards along the path of exit.

Workspace: The dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. The U.S. regulation minimum is 914 mm (36 in), but additional conditions can increase the minimum dimension requirement. GEHC defines this as the envelope of the component superstructure with the external covers in place.

Service Access Width: The width of the workspace in front of the equipment. A minimum of 762 mm (30 in), or the width of the equipment, whichever is greater.

Head Clearance: The height dimension of the workspace. The height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). 1981 mm (78 in), or the height of the equipment, whichever is greater.

Grounded Wall: Any wall that can be electrically conductive to earth ground. Masonry, concrete, and tile are considered conductive. Additional commonly found aspects of a wall should also be considered grounded. This is not an all-inclusive list:

- Medical gas ports and plates
- Metal doors and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinetry
- A1 main disconnect panel
- Equipment Emergency Off panels
- Industrial equipment (such as air conditioners and vents)
- Expansion joints
- Surface raceway
- Exposed wall conduits
- Floor outlets boxes

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks
- Ceiling tile grids

Section 3.0 Regulatory Minimum Working Clearance by Major Subsystem

Requirements apply to equipment operating at 600V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.

Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.

Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance.

WORKSPACE REQUIREMENT	MINIMUM CLEAR SPACE	ADDITIONAL CONDITIONS
Direction of service access: Front of console	914mm (36 in.)	No exposed live part hazards with cover in place. If the console is placed under a counter, the front edge of the console must be even with the vertical edge of the console workspace. Note: This component is typically serviced from the front with access to the rear.
Service access width: Left-Right of workspace	762 mm (30 in.)	This is the width of the workspace in front of the equipment. 762 mm (30 in.) minimum or the width of the equipment, whichever is greater.
Head clearance	1981 mm (78 in.)	The height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). 1981 mm (78 in.) or height of equipment, which ever is greater.
Note: Distances are measured to the finished covers.		

Table 3-8 Console Subsystem

WORKSPACE REQUIREMENT	MINIMUM CLEAR SPACE	ADDITIONAL CONDITIONS
Direction of service access: Front of PDU	914 mm (36 in.)*	No exposed live part hazards with cover in place. This component is typically serviced from the front with access to the rear. *1219 mm (48 in.), if exposed live parts of 151 - 600V are present on both sides of workspace, with the operator between. *1067 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600V are present.
Service access width: Left-Right of workspace	762 mm (30 in.)	This is the width of the workspace in front of the equipment. 762 mm (30 in.) minimum or the width of the equipment, whichever is greater.
Head clearance	1981 mm (78 in.)	The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s). 1981 mm (78 in.) or the height of equipment, which ever is greater
Note: Distances are measured to the finished covers.		

Table 3-9 PDU Subsystem

WORKSPACE REQUIREMENT	MINIMUM CLEAR SPACE	ADDITIONAL CONDITIONS
Direction of service access: All sides	914 mm (36 in.)*	No exposed live part hazards with cover in place. This component is typically serviced from all four sides. *1219 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. *1067 mm (42 in.), if the opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service access width: Left-Right of workspace	762 mm (30 in.)	This is the width of the workspace on each side of the equipment. 762 mm (30 in.) minimum or the width of the equipment, whichever is greater.
Note: Distances are measured to the finished covers.		

Table 3-10 Gantry Subsystem

WORKSPACE REQUIREMENT	MINIMUM CLEAR SPACE	ADDITIONAL CONDITIONS
Direction of service access: Table head or foot	914 mm (36 in.)*	No exposed live part hazards with cover in place. This component is typically serviced from all 4 sides This is the width of the workspace on each side of the equipment. 762 mm (30 in.) min or the width of the equipment, whichever is greater.
Direction of service access: Table sides	914 mm (36 in.)*	
Direction of service access: Table foot [CT ONLY]	711 mm (28 in.)*	*457 mm (18 in.) minimum for Front Gantry Cover removal only if unobstructed egress space of 711 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service access width: Left-Right of workspace	914 mm (36 in.)	This is the width of the workspace in front of the equipment. 914 mm (36 in.) minimum or the width of the equipment, whichever is greater.
Note: Distances are measured to the finished covers.		

Table 3-11 Table Subsystem

Section 4.0
Minimum Room Size (Limited Access)

The CT Gantry Left Side Limited Access Initiative provides the capability to reduce the minimum room size for CT Systems while still meeting all installation requirements and specifications. This adds left side flexibility, allowing the CT system to be sited in rooms with widths 559 mm (22 in.) smaller than the current minimum room width.

In the minimum room configuration, left-side access and egress may be restricted for sites with less than 711 mm (28 in.) of left side clearance. Refer to your site's installation print for your room's detail.

4.1 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout as shown on your site print shall meet all regulatory requirements as described in the installation manual. Additional room components such as cabinets reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

For minimum room size sites, where the gantry distance between the wall and side cover is between 357 mm – 686 mm (14 in. – 27 in.) the left side cannot be considered or used for egress. In this configuration, egress around the left side of the gantry will be restricted. Unobstructed egress route shall be provided around the table foot end or behind the back of the gantry. If millwork is required or will be installed later, consider room placement locations that allows for regulatory compliance and clearances.

4.2 Operational Caution

In this minimum room layout (357 mm – 686 mm [14 in. to 27 in.]) the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall. Detailed customer installation tasks are detailed in the product PIM. Chapters 1-4.

4.3 Suggested Room Size

This room configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add millwork, while meeting all regulatory requirements. This room would be compatible with most two-step future installations.

4.4 Recommended Room Size

This room configuration allows for some future upgrades. It has sufficient workspace but limited space to add millwork and meet all regulatory requirements. This room may be compatible with some two-step future installations.

4.5 Minimum Room Size

This room configuration allows for no future upgrades. It has limited workspace and no in-room millwork, but meets all regulatory requirements. This room is not compatible with two-step future installations.

Section 5.0
System Specifications (Pro32 & VCT 64 Systems)

5.1 Suggested Room Size

VCT with GT1700 = 4 m x 6.71 m (14 ft., 0 in. x 22 ft.)

VCT with GT2000 = 4 m x 7.32 m (14 ft., 0 in. x 24 ft.)

Same Regulatory requirements apply

5.2 Recommended Room Size

VCT with GT1700 = 4.11 m x 6.1 m (13 ft., 6 in. x 20 ft.)

VCT with GT2000 = 4.11 m x 6.71 m (13 ft., 6 in. x 22 ft.)

Same Regulatory requirements apply

5.3 Minimum Room Size

VCT with GT1700 = 3.58 m x 6.1 m (11 ft., 8 in. x 20 ft.)

VCT with GT2000 = 3.58 m x 6.71 m (11 ft., 8 in. x 22 ft.)

Same Regulatory requirements apply, with the addition of no energized left side service.

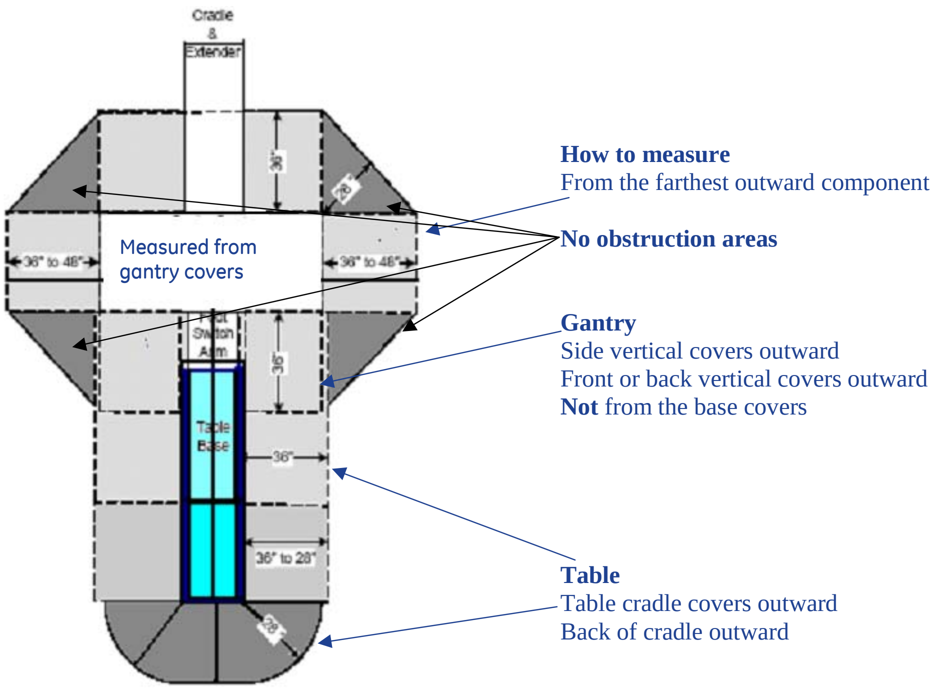
5.4 Rooms w/ Less Than 711 mm (28 in.) Egress Clearance around Table Foot End

Egress requires a clear unobstructed route out of the room, either around the back of the gantry or around the back of the table. If your egress route is not around the back of the table, maintain 18 in. of clearance between the back of the table, with a continuous width of 126 in. on each side to any obstruction so that the front cover can be removed. Refer to the Pre-Installation manual for more details on service clearances.

Exceptions

Rooms smaller than 3.58 m x 6.1 m (11 ft., 8 in. x 20 ft.,) or 3.58 m x 6.71 m (11 ft., 8 in. x 22 ft.), will require construction to meet the minimum

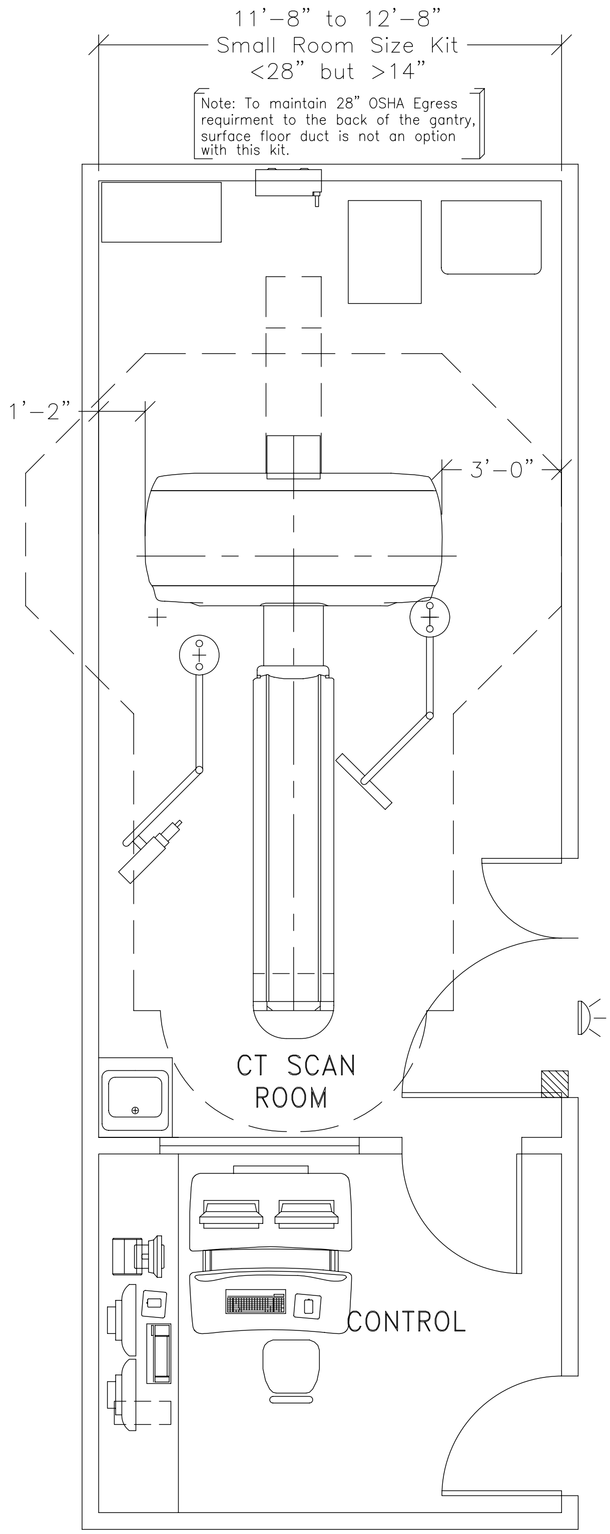
Section 6.0
How to Measure



Section 7.0
Minimum Room Size & Requirement Layouts

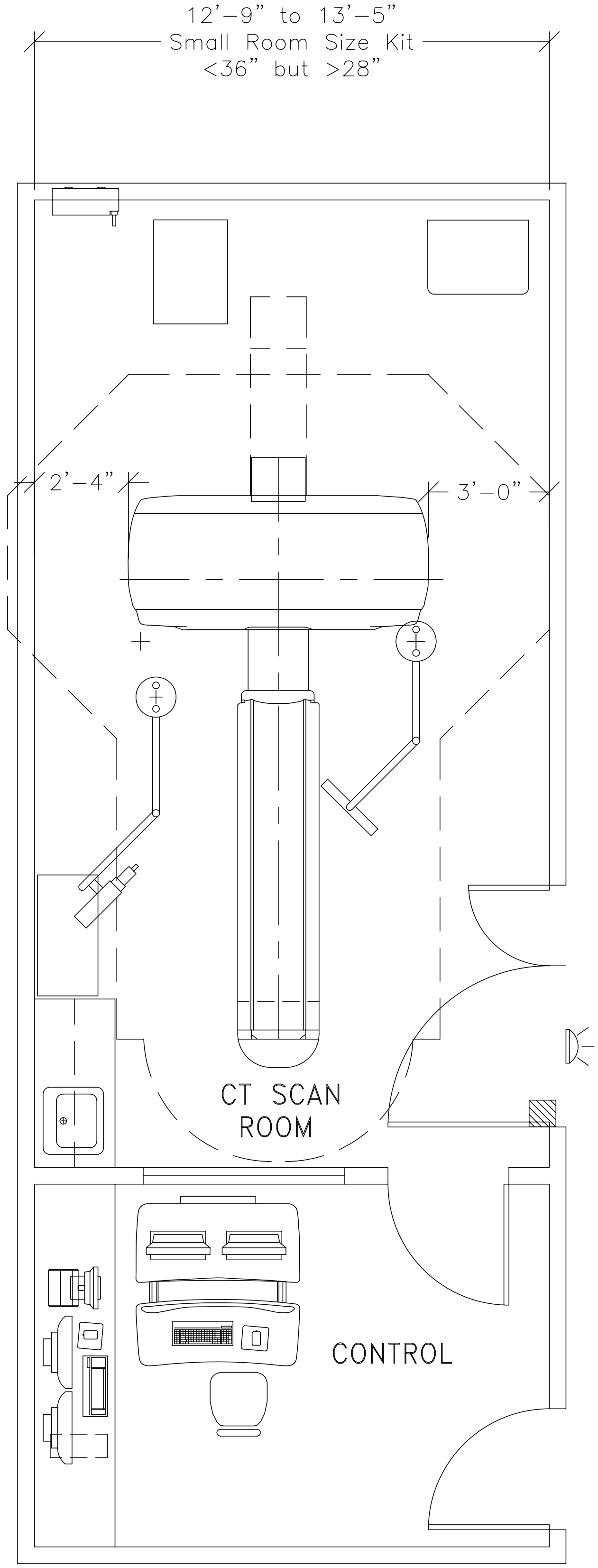
Room A - Less than 711 mm (28 in.) but greater than 356 mm (14 in.) measured from the covers to the left sidewall.

In this configuration service, egress and workspace are compromised around the gantry.

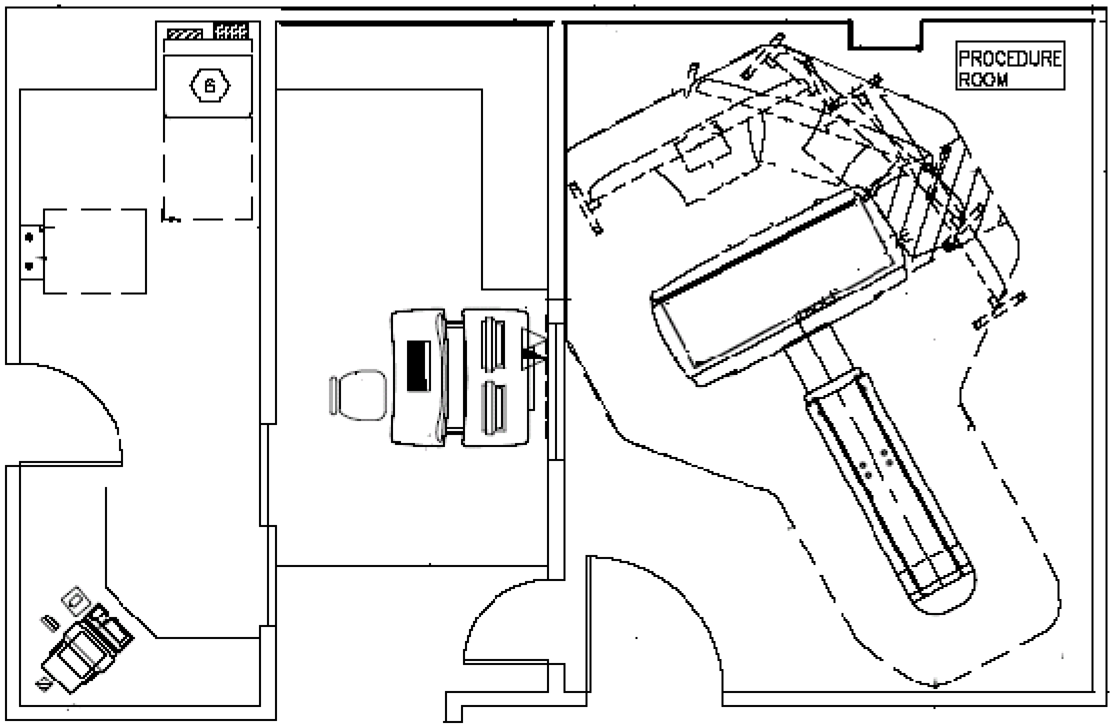


Room B - Less than 914 mm (36 in.) but greater than 711 mm (28 in.) measured from the covers to the left sidewall.

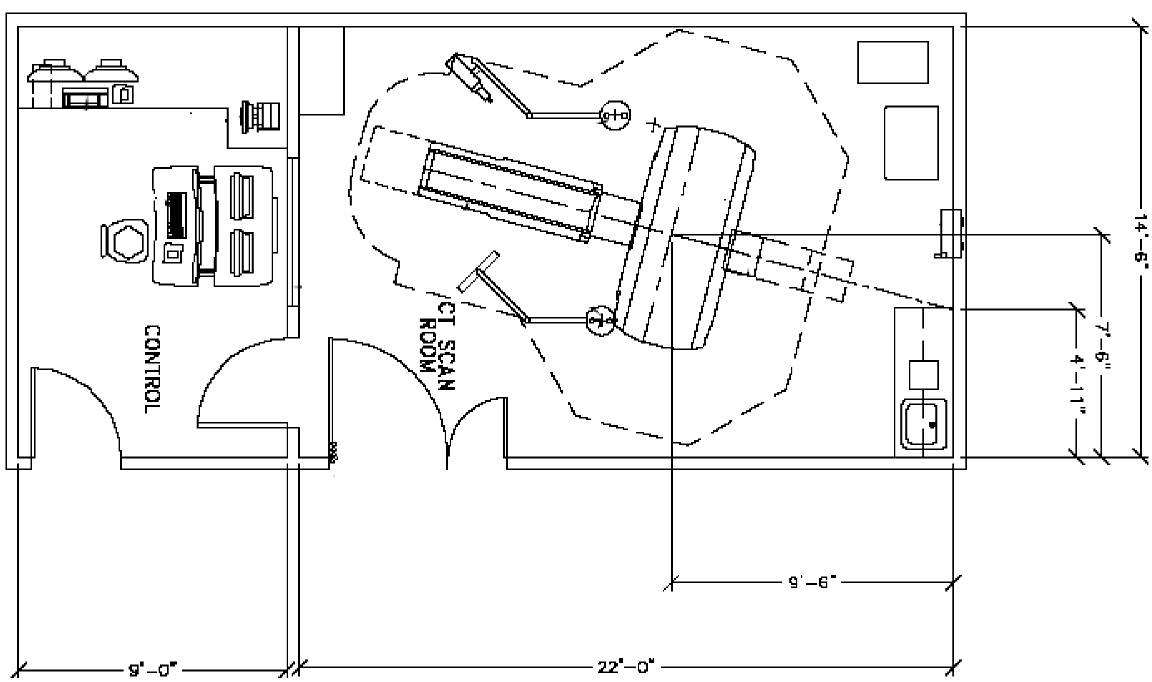
In this configuration service, egress and workspace are acceptable around the gantry.



Section 8.0
Recommended Room Size & Requirement Layouts



Note: Your room layout may meet the recommended or suggested room requirements but look different than the room shown above. Contact your sales person to have a detail room layout completed for your site.



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